BASE24®

SAA AS2805 Interface Manual

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Preface

The BASE24 SAA AS2805 Interface Manual describes the SAA AS2805 Interface process, which is used to establish a communications link between a BASE24 network and a co-network. This co-network may be another BASE24 network, but may also be any transaction processing network which conforms to the standards set forth in the Australian Standards 2805 Manual for message formats and processing.

This manual describes the SAA AS2805 Interface process, which uses the Australian Standards 2805.2 - 1986 message formats.

Audience

This manual is intended as a reference for those persons responsible for the interface between BASE24 and a co-network. It provides a comprehensive description of all the functions performed by the SAA AS2805 Interface process involving incoming and outgoing message traffic for both acquirer and issuer processors.

Additional Documentation

The BASE24 documentation set is arranged so that each BASE24 manual presents a topic or group of related topics in detail. When one BASE24 manual presents a topic that has already been covered in detail in another BASE24 manual, the topic is summarised and the reader is directed to the other manual for additional information. Information has been arranged in this manner to be more efficient for readers that do not need the additional detail and at the same time provide the source for readers that require the additional information. This manual contains references to the following BASE24 publications:

- The ***BASE24 Logical Network Configuration File Manual*** discusses in detail the assigns and params used in configuring the SAA AS2805 Interface.

- The ***BASE24 Log Message Reference Manual*** discusses in detail how to electronically access log message documentation generated by the SAA AS2805 Interface.
- The ***BASE24 Network Environment Source File Manual*** discusses in detail the process, station, line, and device declarations used in configuring the SAA AS2805 Interface.

- The ***BASE24 Base Files Maintenance Manual*** documents the BASE24 data entry screens for the External Message File (EMF), the Key Files (KEYF and KEY6), the Enhanced Interchange Configuration File (ICFE), the Acquirer Processing Code File (APCF), the Issuer Processing Code File (IPCF), the

Card Prefix File (CPF), and the Surcharge File (SURF). The ICFE file is used to configure the parameters described in this manual for the SAA AS2805 Interface. The APCF and IPCF are used to configure the transactions allowed to be sent and received by the SAA AS2805 Interface. The CPF and SURF are used to configure surcharging. The EMF and the KEYF or KEY6 are used to configure the messages and parameters described in this manual.

- The ***BASE24 File and Record Structures Reference Manual*** documents how to access the file and record structures for the Interchange Log File (ILF).

- The ***BASE24 Tokens Manual*** documents token processing. It includes information on how to configure the tokens that get logged to the Interchange Log File (ILF), sent in SAA AS2805 messages, and extracted from the ILF.

- The ***BASE24-atm Transaction Processing Manual*** documents the BASE24-atm standard internal message format (STM).

- The ***BASE24-pos Transaction Processing Manual*** documents the BASE24-pos standard internal message format (PSTM).

- The ***BASE24 Transaction Security Manual*** discusses in detail Personal Identification Numbers (PINs), Message Authentication Codes (MACs), and dynamic key management used to set up transaction security for the SAA AS2805 Interface using Atalla and Racal security modules.

- The ***BASE24 ERACOM Security Control Module (SCM) Reference Manual*** discusses in detail Personal Identification Numbers (PINs), Message Authentication Codes (MACs), and dynamic key management used to set up transaction security for the SAA AS2805 Interface using ERACOM ProtectHost White APCA security module.

- The ***BASE24 ERACOM Security Module (ESM) Reference Manual*** discusses in detail Personal Identification Numbers (PINs), Message Authentication Codes (MACs), and dynamic key management used to set up transaction security for the SAA AS2805 Interface using ERACOM ProtectHost White Mark II security module.

- The ***BASE24 Transaction Security Services (TSS) User Guide*** documents the use of TSS as an interface between the security utilities bound into BASE24 processes and hardware security modules.

Software Release

This manual documents standard processing for BASE24-atm and BASE24-pos Release 6.0.

Customer-specific code modifications (i.e., CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Manual Summary

The ***BASE24 SAA AS2805 Interface Manual*** is divided into the following sections:

Section 1, "Introduction," provides basic information about the SAA AS2805 Interface process and provides an overview of SAA AS2805 Interface functionality.

Section 2, "Configuration and Control," provides a description of the steps required to install the SAA AS2805 Interface, as well as guidelines for setting up the required control files.

Section 3, "Network Management and Operations," describes the network management messages supported by the SAA AS2805 Interface and provides message flows that illustrate the processing that occurs when these messages are sent.

Section 4, "Store-and-Forward Processing," describes how the SAA AS2805 Interface performs Store-and-Forward processing. It includes information about what messages can be placed in the SAF and when messages from the SAF are sent. It also provides message flows that illustrate Store-and-Forward processing.

Section 5, "Transaction Handling," describes how the SAA AS2805 Interface handles various types of transactions.

Section 6, "Cutover and Settlement," describes cutover processing and report generation. Message flows describing how the SAA AS2805 Interface handles various types of cutover transactions are also included in this section.

Appendix A, "SAA AS2805 Interface Startup and Control," provides information about the configuration of the SAA AS2805 Interface process, initialisation processing and network control.

Appendix B, "Message Mapping" provides information about the mapping of fields between the BASE24 internal message, and the AS2805 external message.

Appendix C, "Log Messages" provides documentation for the log messages that can be generated by the SAA AS2805 Interchange Interface Process.

Appendix D, "SAA Tokens" describes the SAA message tokens.

Publication Identification

Two entries appearing at the bottom of each page uniquely identify this BASE24 publication. The publication number (e.g. SW-DS770-1 for the ***BASE24 SAA AS2805 Interface Manual***) appears on every page to assist readers in applying updates to the appropriate manual. The publication date (e.g. 5/98 for May, 1998) identifies the last time the manual has been revised. This date can be used to verify that the reader has the most current information and is particularly helpful when dealing with HELP24 to answer operating questions. The print record, located on the page immediately following the title page, is also helpful when verifying that a manual contains the most current information. An entry is added to the print record each time a BASE24 publication is modified in any way. Each print record entry includes the date of the change, a unique version number, and the type of change (new, update, or reissue). Version 1 is assigned to a new manual. Later versions can be updates or reissues, depending on the changes being made. An update typically involves changes on a small number of pages throughout the manual. A reissue occurs when all pages of the manual are replaced because changes affect a large number of pages.

Section 1

Introduction

This section introduces the BASE24 SAA AS2805 Interface and provides an overview of SAA AS2805 Interface functionality. It also identifies the BASE24 files used by the SAA AS2805 Interface and the message types supported.

SAA AS2805 Interface Overview

The SAA AS2805 Interface process, shown in the illustration on the next page, is responsible for the online message transfer between BASE24 and its co-networks. This process performs the following functions:

- Facilitates message exchange between BASE24 and the co-network.

  When BASE24 receives a transaction from the co-network, the SAA AS2805 Interface translates the message into the appropriate internal message format and sends the message to a BASE24 Authorisation process. When BASE24 receives a transaction for the co-network to authorise, the SAA AS2805 Interface translates the message from the internal message format to the external message format and forwards the message to the co-network.

- Provides store-and-forward capabilities.

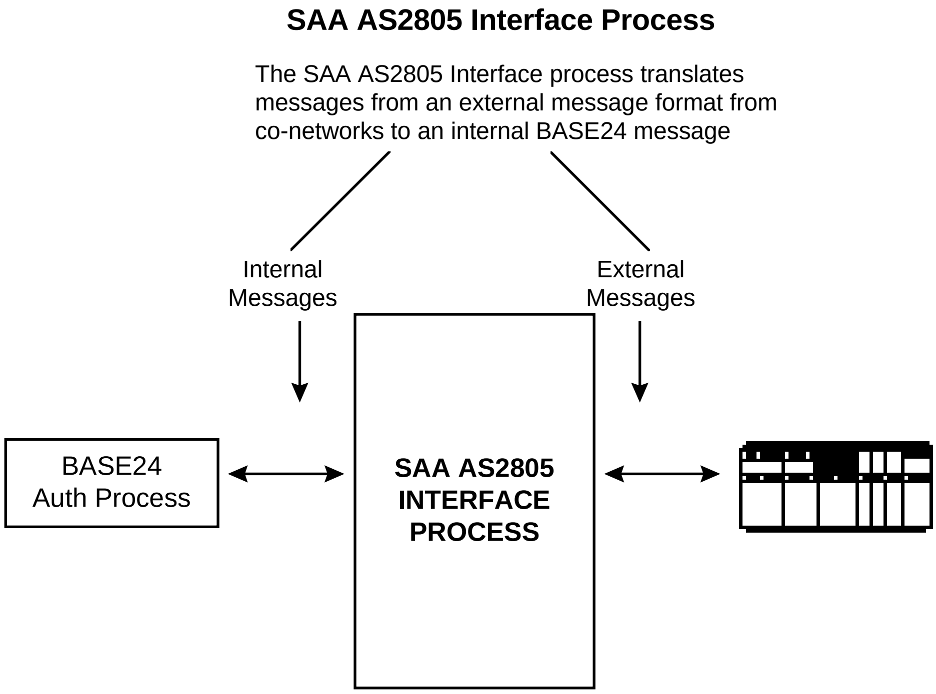
  When the link between BASE24 and the co-network is down, the SAA AS2805 Interface stores certain messages intended for the co-network in its Store-and-Forward File (SAF). When the link is restored, these messages are retrieved from the file and sent to the co-network.

- Facilitates settlement between BASE24 and the co-network.

  The SAA AS2805 Interface produces a set of interchange reports to assist in reconciling transactions between BASE24 and the co-network. These reports allow the BASE24 participants to reconcile the settlement positions of the co-network and BASE24. The SAA AS2805 Interface also handles online exchange of reconciliation totals between BASE24 and the co-network.

- Manages communications between BASE24 and the co-network.

  The SAA AS2805 Interface sends and receives network management messages to determine the status of the link between BASE24 and the co-network. Network management messages include logon, logoff, echo test, dynamic key management and cutover messages.



Co-networks

The BASE24 Interchange or SAA connects a BASE24-atm or BASE24-pos network with a co-network—another BASE24 network or another transaction processing network that conforms to SAA AS2805 standards. These standards include sending and accepting the appropriate form of the SAA AS2805 external message and providing the proper message flows to effect processing.

SAA serves as a link between two networks. It allows a cardholder of one network to initiate a transaction at a terminal belonging to the other network. Although the co-network can be another BASE24 network, for the purposes of this manual the co-network is considered to be unknown.

Stations

Stations provide the means by which communications takes place between BASE24 and a co-network. All messages sent to a co-network are actually sent to the stations defined for the co-network. Likewise, all messages received from a co-network are actually received from stations.

Stations and their lines must be defined to BASE24 via the BASE24 Network Environment Source File (NEFS). Stations and lines are tied to their respective co-networks through the Interchange Configuration File (ICF). At least one station is required for each co-network and a maximum of 2 stations is allowed per co-network. For information on defining stations and lines to BASE24, refer to the ***NET24-XPNET Network Control Command Reference Manual***.

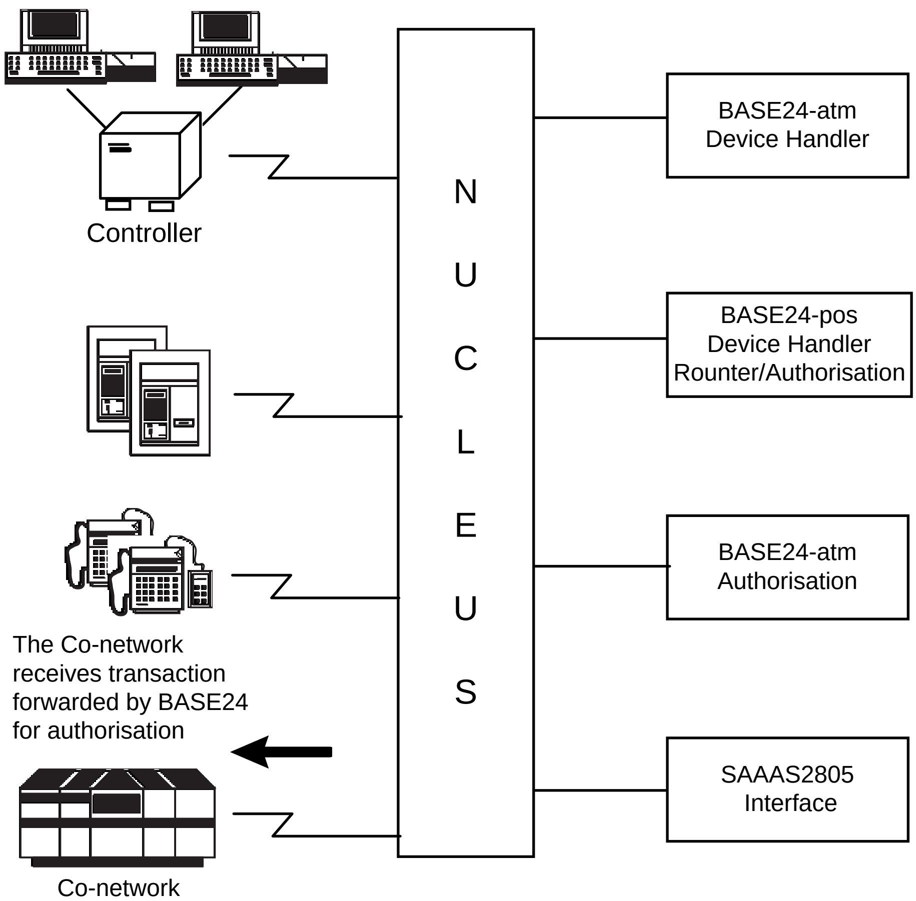
Note: All protocol-level functions between BASE24 and the stations are actually handled by the XPNET process using Tandem's communications control software. The SAA AS2805 Interface process sends and receives messages using the XPNET process. The XPNET process maintains communications links with the stations according to the protocols under which the stations are defined.

Issuer and Acquirer Co-networks

In the financial industry, the terms *issuer* and *acquirer* define the parties on opposite ends of the electronic message exchanges required to authorise and process transactions.

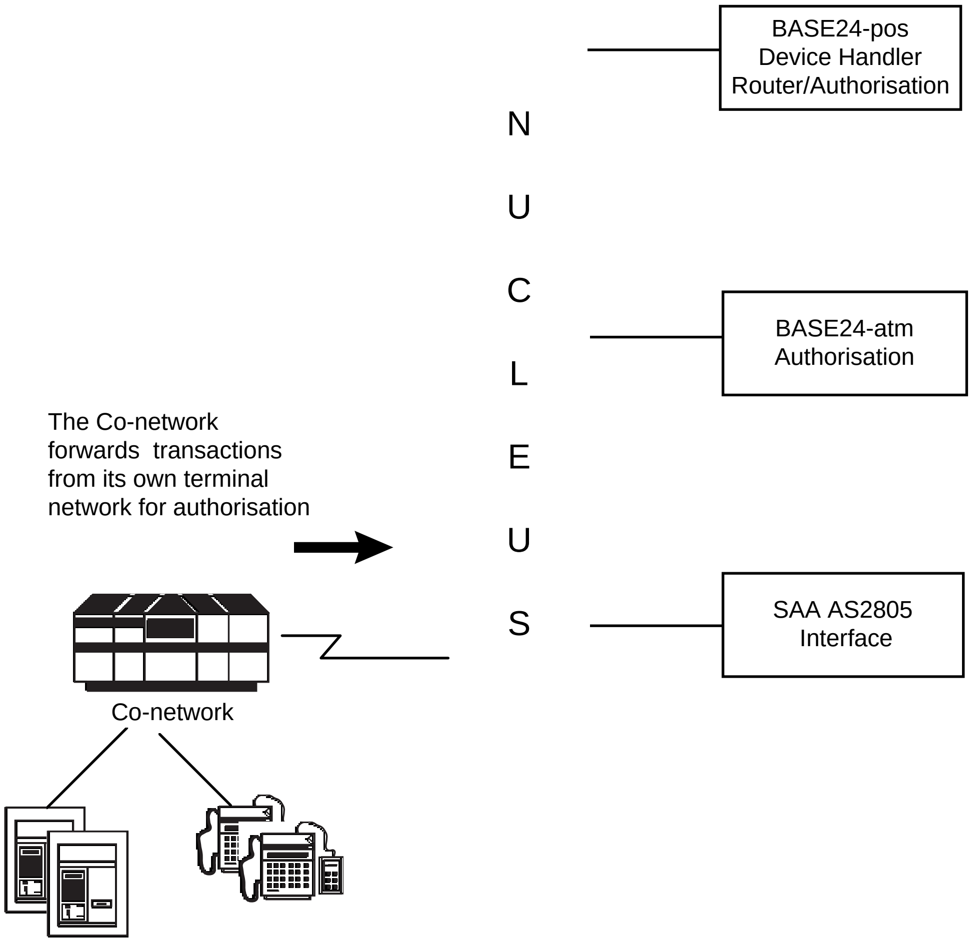
Issuers are card-issuing or account-owning institutions or their agents. These institutions or agents are responsible for authorising cardholder transactions and keeping track of cardholder accounts. Acquirers are the parties initiating transactions on behalf of the card acceptors.

Co-networks acting on behalf of an issuer in a BASE24 message exchange are referred to as *issuer co-networks*. Issuer co-networks authorise and/or post transactions sent to them by BASE24, as shown in the following diagram.



Co-network Acting as an Issuer

Co-networks acting on behalf of an acquirer in a BASE24 message exchange are referred to as *acquirer co-networks*. Acquirer co-networks support ATM or POS networks external to BASE24 and forward transactions to BASE24 for authorisation, as shown in the following diagram.



Co-network Acting as an Acquirer

Note: Because of their general position in the transaction path, issuer co-networks are also referred to as *back-end co-networks* and acquirer hosts are also referred toas *front-end co-network*s.

Issuer and Acquirer Co-network Support

The SAA AS2805 Interface process can function in the role of an issuer or an acquirer. When the SAA AS2805 Interface process sends financial transactions to an issuer co-network for authorisation, the SAA AS2805 Interface process is representing the transaction acquirer, even though the transaction may not have originated at a BASE24-controlled terminal. When financial transactions are being sent to the SAA AS2805 Interface process for authorisation from an acquirer co-network, the SAA AS2805 Interface process is representing the card issuer, even though the SAA AS2805 Interface process may only forward the transaction to an issuer co-network for authorisation.

Co-networks can also function on behalf of both card issuers (or account owners) and transaction acquirers, as would be the case when an institution issues cards and supports its own network of ATM or POS devices.

These options are controlled via the ACQUIRER and ISSUER fields on ICFE screen 3.

Internal Message Format

The internal message format is used whenever information is routed between the SAA AS2805 Interface process and the rest of BASE24. Depending on the BASE24 product receiving the message, the internal message is commonly referred to as the BASE24-atm Standard Internal Message (STM) or the BASE24-pos Standard Internal Message (PSTM).

For more information on the STM or PSTM, see the BASE24-atm or BASE24-pos transaction processing manual.

SAA AS2805 External Message Format

The structure and content of the external message used by the SAA AS2805 Interface processes to communicate with their co-networks is patterned after the standard external message developed by Australian Standards 2805 (AS2805). It is a variable-length, variable-content message that can be configured differently, based on the type of message being sent.

Store-and-Forward Function

A store-and-forward function is incorporated into the SAA AS2805 Interface process to collect transaction messages during periods when a co-network is unavailable.

When a co-network is unavailable, messages required as notification to the co-network are stored in the Store-and-Forward File (SAF). These messages are then transmitted later when communications are restored. For more information on store-and-forward processing, see section 4.

Multiple Product Support

The same SAA AS2805 Interface process can be set up to process co-network messages for multiple BASE24 products.

Message Processing

Due to the variances in processing that must occur in handling messages for different products, the SAA AS2805 Interface processes messages differently based on the message type and product ID for the message. For more information on how messages flow back and forth between the co-network and the SAA AS2805 Interface, please refer to the following sections, based on the type of message:

- Network management message flows, including those for dynamic key management, are documented in section 3.
- Store-and-forward message flows are documented in section 4.
- BASE24 transaction flows are documented in section 5.
- Cutover message flows are documented in section 6.

The remainder of this subsection provides an overview of transaction flows for normal and link-down message processing.

Normal Message Flow

When the SAA AS2805 Interface process receives a message from the co-network, it is translated from the external message into the appropriate internal message format. The internal message is then sent to the appropriate BASE24 Authorisation process. When the message has been processed, the authorising process sends an internal response back to the SAA AS2805 Interface. Using information from the internal message, the SAA AS2805 Interface formats an external response message and sends it back to the co-network.

When the SAA AS2805 Interface process receives a message from a BASE24 Authorisation process, it is translated from the internal message into the external message format. The external message is then sent to the co-network to be authorised. When the message has been processed, the co-network sends an external response back to the SAA AS2805 Interface. Using information from the external message, the SAA AS2805 Interface formats an internal response message and sends it back to the BASE24 Authorisation process.

Link-Down Message Flow

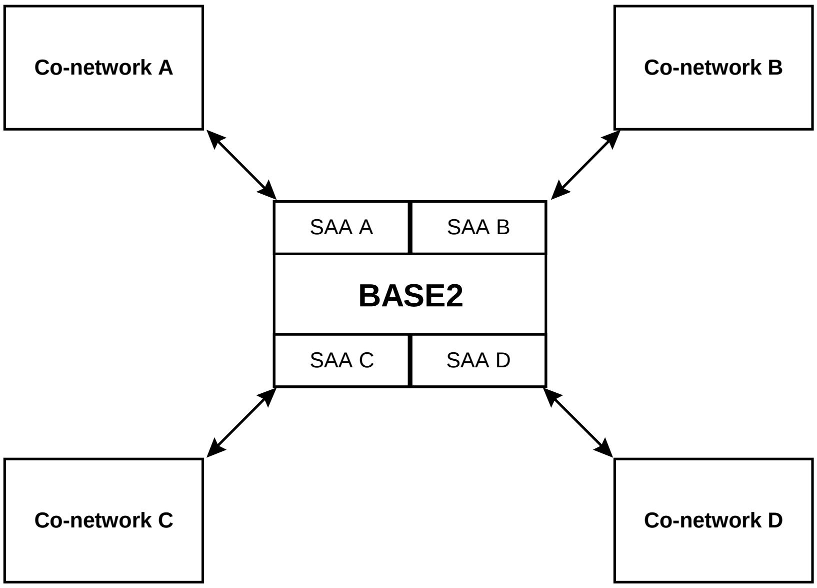
If stand-in authorisation is allowed, the acquirer network has the option of authorising requests when the link to the issuer network is down. The SAA AS2805 Interface returns the request message to the BASE24 Authorisation process as a failed request needing offline authorisation. If the transaction is approved, a force post message is forwarded to the issuer network when the link is re-established.

If stand-in authorisation is not allowed and an alternate destination is available, the link-down situation is handled by using alternate routing to send the transaction to the alternate destination. However, if stand-in authorisation is not allowed and an alternate destination is not available, the SAA AS2805 Interface responds to a link-down situation by denying all transactions until the link is re-established.

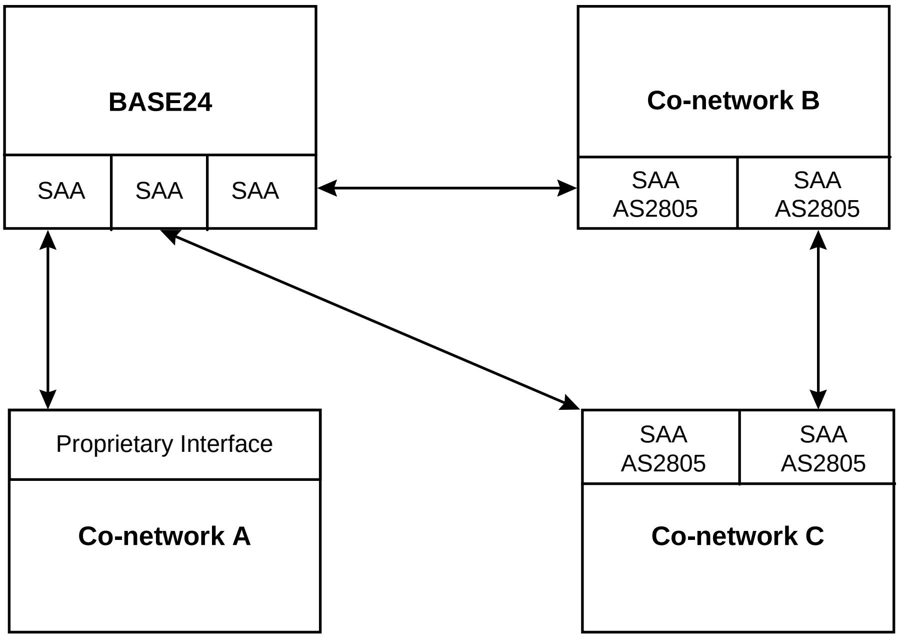
SAA AS2805 Networks

The main purpose of this manual is to describe the connection between a BASE24 system and a single co-network. However, it is possible to connect a BASE24 system to multiple co-networks. This type of a system requires a separate SAA AS2805 Interface process for each co-network to which the BASE24 system is connected.

An SAA AS2805 network can be configured to require that all transactions be routed through a single member of the SAA AS2805 network. For example, a primary BASE24 system configured as the focal point for exchanging transactions within the entire SAA AS2805 network. In this type of arrangement, the primary BASE24 system has a separate SAA AS2805 Interface process for each co-network to which it is linked. Co-networks A, B, and C are other BASE24 systems and would each require one SAA AS2805 Interface process. Co-network D is a non-BASE24 system and would require a proprietary interface that conforms to SAA AS2805 standards for message formats and processing.



An SAA AS2805 network can also be configured to allow for certain transactions to be exchanged directly between members of the SAA AS2805 network. An example of this is illustrated below. In this illustration, the configuration consists of a primary BASE24 system and three co-networks, two of which are other BASE24 systems. The primary BASE24 system can exchange transactions with each of the three co-networks; co-networks B and C, which are also BASE24 systems, are configured to exchange transactions directly with each other. Co-network A is a non-BASE24 system. In this type of arrangement, the primary BASE24 network would require three SAA AS2805 Interface processes, co-networks B and C would each require two SAA AS2805 Interface processes, and co-network A would require a proprietary interface that conforms to SAA AS2805 standards for message formats and processing.



Supported Transactions

The SAA AS2805 Interface provides support for both BASE24-atm and BASE24-pos transactions.

Different transaction sets are supported by the SAA AS2805 Interface depending on the products installed.

Supported BASE24-atm Transactions

If BASE24-atm is installed, the SAA AS2805 Interface supports the following transactions. These transactions are supported for checking, savings, and credit accounts. The SAA AS2805 Interface also supports transactions for which no account type has been specified.

- Balance inquiries
- Deposits
- Payments from/to
- Transfers
- Withdrawals

Supported BASE24-pos Transactions

If BASE24-pos is installed, the SAA AS2805 Interface supports the following transactions. These transactions are supported for checking, savings, and credit accounts. The SAA AS2805 Interface also supports transactions for which no account type has been specified:

- Balance inquiries
- Cash advances
- Merchandise returns
- Normal purchases
- Preauthorisations
- Preauthorisation completions
- Purchases with cash back

File Usage

The SAA AS2805 Interface process uses a number of BASE24 files. These files can be divided into control files and transaction files and are identified below.

Control Files

Control files contain configuration information and parameters that determine how the SAA AS2805 Interface process operates. The control files that contain configuration information for the SAA AS2805 Interface include:

- Acquirer Processing Code File (APCF)
- External Message File (EMF)
- Extended Interchange Configuration File (ICFE)
- Issuer Processing Code File (IPCF)
- Key File (KEYF) or Key 6 File (KEY6)
- Key T File (KEYT)
- Network Environment File (NEF)
- Logical Network Configuration File (LCONF)
- Token File (TKN)

For information on how these files are used to configure the SAA AS2805 Interface process, see section 2.

Transaction Files

Transaction files are used to store information about individual transactions processed through the SAA AS2805 Interface. The transaction files used by the SAA AS2805 Interface include:

- Interchange Log File (ILF)
- Store-and-Forward File (SAF)

For information on what the ILF contains when it is used by the SAA AS2805 Interface, refer to the ILF Data Definition Libraries (DDLs). For information on the SAF, refer to the SAF DDLs. The ILF and SAF DDLs can be accessed via DDLDOC. For information on DDLDOC, refer to the ***BASE24 Files and Record Structures Reference Manual***.

Supported Message Types

The SAA AS2805 Interface supports the following messages between BASE24 and the co-network. Notations appear for each message type to indicate its use or availability by BASE24 products.

Type	**Description**	**BASE24-atm**	**BASE24-pos**
0100	Pre-Authorisation Request	Y	Y
0110	Pre-Authorisation Request Response	Y	Y
0120	Pre-Authorisation Advice	Y	Y
0121	Pre-Authorisation Advice Repeat	Y	Y
0130	Pre-Authorisation Advice Response	Y	Y
0200	Financial Transaction Request	Y	Y
0210	Financial Transaction Request Response	Y	Y
0220	Financial Transaction Advice	Y	Y
0221	Financial Transaction Advice Repeat	Y	Y
0230	Financial Transaction Advice Response	Y	Y
0420	Acquirer Reversal Advice	Y	Y
0421	Acquirer Reversal Advice Repeat	Y	Y
0430	Reversal Advice Response	Y	Y
0520	Acquirer Reconciliation Advice	Y	Y
0521	Acquirer Reconciliation Advice Repeat	Y	Y
0530	Acquirer Reconciliation Advice Response	Y	Y
0800	Network/Key Management Request	Y	Y
0810	Network/Key Management Request Response	Y	Y
0820	Network Management Advice	Y	Y
0830	Network Management Advice Response	Y	Y

Interchange Reporting

The SAA AS2805 Interface produces two reports: the SETL01-SAA and the STAT09-SAA. The purpose of these reports is to facilitate settlement between BASE24 and the co-network. For information on these reports, refer to section 6.

Dynamic Key Management

The SAA AS2805 Interface uses session keys to protect transaction data moving between networks. Session keys are used to maintain the proper security to protect personal identification numbers (PINs) and verify Message Authentication Codes (MACs). Session keys should be changed regularly to ensure security. The monitoring and changing of these keys is known as key management and can be performed manually using text commands or automatically using dynamic key management.

Dynamic key management allows co-networks to automatically change session keys based on thresholds set for time, number of transactions and number of single or consecutive errors. One or more session keys can be changed when any one of the defined thresholds is surpassed

Session keys are generated from send and receive Key Exchange Keys (KEKs and KEKr), which in turn are derived from public or transport keys manually exchanged with the co-network.

Dynamic key management requires the use of hardware security modules to generate KEK and session keys and perform PIN encryption, and message authentication. It is not supported when these functions are performed in software.

KEK fields in the Key File (KEYF), or Key 6 File (KEY6) and the External Message File (EMF) must be set to specific values to allow for dynamic key management. See section 2 for a listing of KEYF, KEY6 and EMF settings.

For more information about dynamic key management see section 3 which discusses Security Management.

Dynamic KEK Management

The SAA AS2805 Interface also supports the random generation and exchange of Key Exchange Keys prior to the exchange of the session keys that are derived from them. The Key Exchange Keys are generated from public keys manually exchanged with the co-network.

Public key fields in the Key T File (KEYT) must be set to specific values to allow for dynamic KEK management. KEYT is only accessed when Hardware Security Module used is SCM. See section 2 for a listing of KEYT settings.

For more information about dynamic KEK management see section 3 which discusses Security Management.

Five-day Settlement

The SAA AS2805 Interface process can support 5-day settlement. An interface using 5-day settlement creates an ILF for Monday when it cuts over on Friday instead of creating one for Saturday. All transactions taking place after this cutover until the next cutover on Monday are logged to Monday's ILF. The totals calculations are also processed this way.

The cutover date can also be configured to allow for holiday dates. These dates are treated like weekends: when the interface cuts over on the day before the holiday, it creates the ILF for the day after the holiday. For example, when the interface cuts over on December 31, it creates an ILF for January 2. All transactions taking place on January 1 and January 2 are logged to the ILF for January 2.

Five-day settlement is configured on screen 2 of the Enhanced Interchange Configuration File (ICFE). See section 2 for more information about the ICFE screen and its field settings.

Note: Because the ILF for Monday will contain transactions for Saturday, Sunday, and Monday, the file extents for the ILF should be large enough to allow for an extra two days' volume.

Suppression of Discretionary Data

The SAA AS2805 Interface process has the ability to suppress discretionary data in the ILF log file for both issuer and acquirer transactions. Suppression of discretionary data is controlled by LCONF parameters and can be turned off for acquirer, issuer or acquirer and issuer transactions. The following paramters control the suppression of discretionary data for acquirer transactions:

SWI-ACQ-AVS-DATA-SUPPRS
SWI-ACQ-CVD2-DATA-SUPPRS
SWI-ACQ-DISC-DATA-SUPPRS

The following parameters control the suppression of discretionary data for issuer transactions:

SWI-ISS-AVS-DATA-SUPPRS
SWI-ISS-CVD2-DATA-SUPPRS
SWI-ISS-DISC-DATA-SUPPRS

For more information on setting the above parameters, refer to the LCONF section of this manual.

Format 2 File Support

The SAA AS2805 Interface process supports format 2 ILFS for D46 and higher Compaq NSK operating systems. Format 2 file support allows BASE24 customers the ability to format Enscribe disk file (ILF) partitions to accommodate format 2 files. Files with format 2 partitions can contain much more data than the existing format 1 files. Files, whether format 1 or format 2, can have up to 16 partitions. Format 1 files are limited to a 4 gigabytes (each partition can have up to 2 gigabytes). Format 2 files have a new block and label formats and can hold as much as 1024 gigabytes of data per partition and are currently limited by disk volume size. Large format 1 files, limited by the capacity of a single disk volume, have used partitioning as a means to increase file capacity.

Customers wishing to use format 2 ILFs should start at cutover. Create a new ILF format 2 template file and its altfile. To do so, create and obey file name FRMT2ILF with the following text:

- RESET
- SET TYPE E
- SET FORMAT 2
- SET EXT (4 PAGES, 32 PAGES)
- SET REC 4048
- SET BLOCK 4096
- SET ALTKEY ("PR", FILE 0, KEYOFF 0, KEYLEN 55)
- SET ALTFILE (0, $<VOL>.<SUBVOL>.xxYYMMDD)
- SET MAXEXTENTS <NUMBER OF MAXEXTENTS>
- CREATE $<VOL>.<SUBVOL>.xxYYMMDD

The format 2 ILF template file in the create command will need to be set to the value specified in the LCONF assign for the ILF for the interface process. The existing ILF template and altfile should be renamed, as the new files created should use the same locations.

Create the format 2 ILF and altfile with the following command
FUP/IN FRMT2ILF/

When the interface is cutover, it will create and begin using the new format 2 ILF and new format 2 altfile. The altfile can remain format 1 or change to format 2 along with the new format 2 ILF. Previous ILF files can be format 1 and current ILF files can be format 2.

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Section 2

Configuration and Control

This section describes the configuration and control of the SAA AS2805 Interface process and includes the following:

- Descriptions of files used by the SAA AS2805 Interface process
- Process configuration information
- Descriptions of institution and co-network-level controls
- A description of how stations are defined and used by the SAA AS2805 Interface process
- Descriptions of the timers used by the SAA AS2805 Interface process and the processing performed when each timer expires.

LCONF Assigns and Params

The SAA AS2805 Interface process uses a number of Logical Network Configuration File (LCONF) assigns and params in its processing. These assigns and params are identified below. For more information on configuring the SAA AS2805 Interface process via these assigns and params, refer to the ***BASE24 Logical Network Configuration File Manual.***

Assigns

The following assigns identify the names of files that contain SAA AS2805 Interface configuration and control information, or that the SAA AS2805 Interface uses during processing. The SAA AS2805 Interface process reads these assigns regardless of the products supported.

APCFEMT — Identifies the name of the Acquirer Processing Code File (APCF) extended memory table (APCFEMT), which contains APCF records.

EMF — Identifies the name of the External Message File (EMF).

ICF — Identifies the name of the Interchange Configuration File (ICF). This assign is required if the BASE24-REL param is not present or does not contain the value of 60.

ICFE — Identifies the name of the Enhanced Interchange Configuration File (ICFE).). This assign is required if the BASE24-REL param contains the value of 60.

IPCFEMT — Identifies the name of the Issuer Processing Code File (IPCF) extended memory table (IPCFEMT), which contains IPCF records.

ILF — Identifies the name of the Interchange Log File (ILF).

KEY6 — Identifies the name of the Key 6 File (KEY6).

KEYF — Identifies the name of the Key File (KEYF).

KEYT — Identifies the name of the Key T File (KEYT).

SAF — Identifies the name of the Store-and-Forward File (SAF).

LMCF — Identifies the name of the Log Message Configuration File (LMCF).

RPT — Identifies the name of the Report Perusal File (RPT). This assign is required if reports are to be run.

SW-RPT01-PROG-NAME

SW-RPT09-PROG-NAME — Identify the names of the interchange report programs. These entries are required if these reports are to be generated.

RPT-ERROR-FILE
SW-RPT-OBEY-FILE — Identify the name of the spooler location to receive report error messages and the name of the obey file used to start the interchange reports. These entries are required if reports are to be generated.

TKN — Identifies the name of the Token File (TKN).

Params

The SAA AS2805 Interface uses several params during processing. These params are as follows.

SW-ILF-FILE-RETENTION — The number of days an ILF is to remain on the system. After the specified number of days, the ILF is purged from the system by the SAA AS2805 Interface process. Valid values are 3–99. The default value is 7.

ATM-LN-CUTOVER — The time of day all BASE24-atm processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This parameter is required for BASE24-atm.

POS-LN-CUTOVER — The time of day all BASE24-pos processes in the entire logical network are cut over. All institutions must be cut over by this time. Valid values, in military time, are from 00:00 to 23:59. This parameter is required for BASE24-pos.

SW-RPT-STARTUP — The time the SAA AS2805 Interface process runs its report obey file. Valid values, in military time, are from 00:00 to 23:59.

SPAN-FLAG — Flag to indicate multiple networks. This flag is appended to the run command generated for the Interchange Reports. Valid values are “Y” and “N” (default)

DATE-DISPLAY-TYPE — A flag indicating the format of the month and day portion of the dates on reports. This parameter does not affect dates which include the year. Valid values are:

1 = DDMM

2 = MMDD (default)

DES-TRIPLE-SINGLE — A code indicating whether an Interchange Interface process translates PINs from encryption under a double length key using the Triple Data Encryption algorithm (T-DEA) to encryption under a single length key using the Data Encryption algorithm (DEA) and vice versa. Valid values are:

Y = Yes, translate PINs from T-DEA encryption to DEA encryption and vice versa
N = No, do not translate PINs from T-DEA encryption to DEA encryption and vice versa

If this param is set to a value of Y, the SAA AS2805 Interface process uses the Key 6 file (KEY6). Otherwise, it uses the Key File (KEYF).

ACCT-NUM-TYP — A flag indicating whether account numbers are stored in binary or decimal form. The binary form allows the account number to be up to 28 digits in length. The decimal form allows the account number to be up to 19 digits in length. Valid values are:

D = Decimal format (default)

B = Binary format

SAA-ILF-NOTIFY-INTERVAL — The number of writes made to the ILF between calculating the percentage of file full. Valid values are 10–32766. The default value is 50.

SAA-ILF-NOTIFY-PERCENTAGE — The percentage of file full at which the SAA AS2805 Interface process logs a message indicating the ILF is becoming full. The SAA AS2805 Interface process continues to log messages at the interval indicated in the ILF-NOTIFY-INTERVAL parameter until more file space is made available. Valid values are 10–99. The default value is 70.

SAA-SAF-NOTIFY-INTERVAL — The number of writes made to the SAF between calculating the percentage of file full. Valid values are 10–32766. The default value is 50.

SAA-SAF-NOTIFY-PERCENTAGE — The percentage of file full at which the SAA AS2805 Interface process logs a message indicating the SAF is becoming full. The SAA AS2805 Interface process continues to log messages at the interval indicated in the SAF-NOTIFY-INTERVAL parameter until more file space is made available. Valid values are 10–99. The default value is 70.

FRWD-INST-ID-CDE — The forwarding institution id code, it can be a maximum of 11 digits padded on the left with zeroes.

RCV-INST-ID-CDE — The receiving institution id code, it can be a maximum of 11 digits padded on the left with zeroes.

SAA-PROTOCOL-TYP — This is a protocol type indicator. Valid values are :

0 = no additional protocol processing needs to be performed by SAA (default)

1 = causes and ETX character to be added to the external message prior to sending it across the link.

SAA-DEFAULT-CRD-LN — The default card logical network. This is used to prime the 0220 internal message for report purposes as the co-network does not supply this field.

SAA-LOGON-ACTION — This parameter is used to determine what action to take when a logon message is received when the link is already considered to be up. Valid values are :

0 = respond to the logon but take no further action

1 = respond to the logon, log a message and insert a new logon timer.

2 = respond to the logon, send a logoff then, set a timer to logon in 5 seconds

SAA-CVV-FLAG — Card verification flag. This flag is used when priming the point of service entry mode. If this flag is set to “Y” and a VISA card is being processed the point of service entry mode is set to “901” otherwise it defaults to “021”. Valid values are :

“N” = CVV processing not supported (default)

“Y” = CVV processing supported

SAA-RACAL-TERM-MASTER — this parameter contains the RACAL Terminal Master Key which is used to generate the PIN key on a key exchange when using a RACAL box.
Valid values are 16 hexadecimal characters.

SAA-UNIQUE-STAN-FLAG — a flag to indicate if the STAN is to be unique to the interchange or primed by the ATM/POS sequence number. Valid values are :

“N” = non-unique STANs, use the ATM/POS sequence numbers to prime the external message.

“Y” = the interchange will generate a unique STAN for each external message. (default)

SAA-LOGON-REQUIRED — this parameter is used to determine if we require a logon from our co-network. Valid values are:

“N” = no logon required from our co-network. As soon as we receive a logon response the link is considered to be marked up and ready for requests.

“Y” = a logon is required from our co-network to establish the link. (default)

SAA-EBCDIC-CONVERSION — if this flag is set on, fields are converted from ascii to ebcdic for messages to our co-network; similarly, on receipt of messages from our co-network, fields are converted from ebcdic to ascii. Binary fields are exempt from this conversion. Valid values are :

“N” = do not convert the fields from ascii (default)

“Y” = convert the fields to ebcdic outbound and back to ascii for inbound messages.

SAA-REVERSE-BAL-INQ — flag to indicate if balance inquiries are allowed to be reversed. Valid values are :

“N” = do not send a reversal when the original request was a balance inquiry (default)

“Y” = allow reversals for balance inquiries.

SWI-ACQ-AVS-DATA-SUPPRS — flag to indicate if acquirer AVS data is to be suppressed before logging to the ILF. Valid values are :

“N” = do not suppress acquirer AVS data before logging to the ILF (default)

“Y” = suppress acquirer AVS data before logging to the ILF.

SWI-ACQ-CVD2-DATA-SUPPRS – flag to indicate if acquirer CVD2 data is to be suppressed before logging to the ILF. Valid values are:

“N” = do not suppress acquirer CVD2 data before logging to the ILF (default)

“Y” = suppress acquirer CVD2 data before logging to the ILF.

SWI-ACQ-DISCR-DATA-SUPPRS – flag to indicate if acquirer discretionary data is to be suppressed before logging to the ILF. Valid values are:
“N” = do not suppress acquirer discretionary data before logging to the ILF (default)
“Y” = suppress acquirer discretionary data before logging to the ILF.

SWI-ISS-AVS-DATA-SUPPRS – flag to indicate if issuer AVS data is to be suppressed before logging to the ILF. Valid values are:
“N” = do not suppress issuer AVS data before logging to the ILF (default)
“Y” = suppress issuer AVS data before logging to the ILF.

SWI-ISS-CVD2-DATA-SUPPRS – flag to indicate if issuer CVD2 data is to be suppressed before logging to the ILF. Valid values are:
“N” = do not suppress issuer CVD2 data before logging to the ILF (default)
“Y” = suppress issuer CVD2 data before logging to the ILF.

SWI-ISS-DISCR-DATA-SUPPRS – flag to indicate if issuer discretionary data is to be suppressed before logging to the ILF. Valid values are:
“N” = do not suppress issuer discretionary data before logging to the ILF (default)
“Y” = suppress issuer discretionary data before logging to the ILF.

Configuring a Co-network

Each co-network with which SAA AS2805 Interface communicates requires a record in the Enhanced Interchange Configuration File (ICFE). Co-networks are assigned stations and tied to the SAA AS2805 Interface processes with which they communicate using the ICFE.

A number of processing options is controlled at the co-network level, which enable institutions to set up co-networks with individual processing requirements. The majority of processing options available at the co-network level is stored in the ICFE. Other options are specified in the KEYF or the KEY6. The processing options controlled at the co-network level are described below.

Specifying Timer Lengths

The SAA AS2805 Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICFE, KEYF, or LCONF. Some timer intervals are specified once at the interface level, while others are specified at the interface and product level (e.g. you can specify one time limit for inbound BASE24-atm messages and a different time limit for inbound BASE24-pos messages).

The table below identifies each timer used by the SAA AS2805 Interface, along with the level at which it is defined and the file in which it is defined. Each of these timers are discussed later in this section.

Timer	**Interface Level**	**Interface/Product Level**
Network Management Message timer	ICFE	
Extended Network Management Message timer	ICFE	
Store-and-Forward Message timer	ICFE	
Wait-for-Traffic timer	ICFE	
Performance timer	ICFE	
Outbound Request timer		ICFE
Inbound Request timer		ICFE
Completion Acknowledgment timer		ICFE
PIN Key Exchange timer (*)		
MAC Key Exchange timer (*)		
Clear Old Key timer (*)		
Report Startup timer	LCONF	
Settlement Interval timer	ICF	

* Reserved for future use

Specifying Maximum Timeouts

Institutions must indicate the number of consecutive timeouts of non-network management and non-store-and-forward messages that can occur at a station before the station is marked down. For example, if 3 is specified, the SAA AS2805 Interface process marks the station down upon the third consecutive timeout. However, if only two consecutive timeouts have occurred, the station is not marked down.

Institutions control when a station is marked down because of timeouts via the MAXIMUM TIMEOUTS field on ICFE screen 3.

Initiating Network Management Messages

Institutions specify whether the SAA AS2805 Interface is allowed to initiate network management messages (that is, 0800 messages for logon, echo test, KEK change).

Institutions indicate whether the SAA AS2805 Interface can initiate network management messages using the NETWORK MANAGEMENT MESSAGE ENABLED field on ICFE screen 3.

Note: This option only controls whether the SAA AS2805 Interface can initiate the network management actions; the SAA AS2805 Interface always responds to the network management messages it receives.

Specifying Maximum Store-and-Forward Retries

Institutions set the maximum number of times the SAA AS2805 Interface process attempts to send a store-and-forward message to a co-network before sending that message to the log and deleting it from the SAF.

This option can be used to stop processing a message when, due to the number of timeouts on a store-and-forward message, it becomes apparent that the co-network cannot process the message. Institutions control this option via the MAX SAF RETRY field on ICFE screen 3.

Specifying Maximum Outbound Messages

The SAA AS2805 Interface allows institutions to specify the maximum number of request messages (that is, 0100 and 0200 messages) or advice messages (that is, 0120, 0220, 0420 and 0520 messages) that can be outstanding to an issuer co-network before the SAA AS2805 Interface stops sending additional messages. Outstanding messages are those messages that have not been received (for example, a 0210 for a 0200 or a 0230 for a 0220).

This option allows the SAA AS2805 Interface process to prevent processing when it becomes apparent that the co-network can not process a message within the time limits required because of the number of messages already outstanding to the co-network.

Once the number of outstanding messages reaches this limit, the SAA AS2805 Interface process marks any new request messages destined for the co-network as failed and returns them to their originating process. The SAA AS2805 Interface process places any new advice messages destined for the co-network in the SAF. At this point, the co-network does not receive additional request or advice messages until the messages in the queue are either responded to or have timed out.

Institutions control the number of outstanding request and advice messages to a co-network via the OUTBOUND field on ICFE screen 3.

Specifying Maximum Inbound Requests

The SAA AS2805 Interface also provides a parameter for limiting the number of outstanding 0100 and 0200 messages to a BASE24-atm or BASE24-pos Authorisation process. This parameter controls requests being sent from acquirer co-networks. Once the number of outstanding 0100 and 0200 messages reaches this limit, the SAA AS2805 Interface process declines any new 0100 and 0200 messages coming from the co-network and returns them as 0110 and 0210 messages.

The number of outstanding requests to an Authorisation process is controlled via the INBOUND field on ICFE screen 3.

Note: This feature only affects 0100 and 0200 messages. Other message types are not included in the count of outstanding messages, nor is their processing affected in any way.

Authorisation Process

The SAA AS2805 Interface processes handling requests from acquirer networks determine the Authorisation processes to which to route inbound transaction requests using the AUTH PROCESS field on ICFE screen 8 for BASE24-atm or ICFE screen 10 for BASE24-pos. Alternately the authorisation process can be determined by the transaction type of an inbound request, using the transaction allowed ACQUIRER TXN PROFILE field on on ICFE screen 8 for BASE24-atm or ICFE screen 10 for BASE24-pos. If the AUTHORISATION DESTINATION field on screen 2 of the APCF is non blank for the matching transaction type in the inbound request, then this field value is used as the authorisation process name.

Setting Up Co-network Stations

Stations serve as the links between BASE24 and a co-network. All messages sent to a co-network are actually sent to the stations defined for that co-network. Likewise, all messages received from a co-network are actually received from the stations defined to that co-network.

Up to 2 stations can be defined for each co-network. A station must first be defined as a co-network station in the Network Environment File (NEF). The station's symbolic name can then be placed in the ICFE to fully define the station.

The SAA AS2805 Interface process reads the ICFE at initialisation and builds a table of the stations defined for each co-network.

Station Types

Co-networks can define two types of stations for use by the SAA AS2805 Interface process: send/receive stations and receive-only/no output stations. These stations are described below.

Station types are defined via the STATION CONFIGURATION fields on ICF screen 13.

Send/Receive Stations

Send/receive stations allow for both incoming and outgoing messages. The SAA AS2805 Interface process can send messages to and receive messages from the co-network using this type of station.

Receive-Only/No Output Stations

Receive-only/no output stations only allow for messages coming in to BASE24. BASE24 never uses receive only/no output stations for sending messages to the co-network. The receive-only/no output station is used to avoid line connection when using a point-to-point or SNA protocol. In this case, communications are received on one station and sent from a different station.

Setting Up the ICFE Station Table

Stations are numbered STATION1 and STATION2 on screen 1 of the ICFE file maintenance screens.

When selecting a station to which to send a message, the SAA AS2805 Interface process goes through its station table from start to finish until it finds a station available to receive the message.

Selecting Stations for Sending Messages

To select the station to which to send a message, the SAA AS2805 Interface process begins with the first station in the ICFE station table for that co-network and evaluates each station based on the following criteria:

1. It must be a send/receive station and the station must be marked up. The SAA AS2805 Interface process never sends messages to receive only/no output stations.

2. Since the co-network is single-threaded, the station cannot be busy; that is it must not have a message pending.

The SAA AS2805 Interface process sends the message to the first station that meets the above criteria. To send the next message, the SAA AS2805 Interface process performs its station selection in the same manner, starting with the next station in the table. For example, if the last message was sent to the second station in the table, the SAA AS2805 Interface would begin its station selection with the first station.

Configuring Timers

The SAA AS2805 Interface process uses internal timers to control co-network message traffic. The length of each of these timers is specified in the ICFE.

Network Management Message Timer

A Network Management Message timer is set each time an 0800 message is sent to a co-network station. The Network Management Message timer limits the amount of time that the SAA AS2805 Interface process waits for a response after transmitting an 0800 message. The number of seconds to use for this timer is taken from the NETWORK MANAGEMENT field on ICFE screen 3.

The actions taken when a Network Management Message timer expires depend on the type of message for which the timer was set. If the message was a logon, logoff, or echo test message, the SAA AS2805 Interface performs as follows:

1. Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the SAA AS2805 Interface process marks the co-network down.
2. Sets an Extended Network Management Message timer for the station. If the message was a 0820 dynamic key management message, the SAA AS2805 Interface performs as follows:
   a. Adds one to the count of consecutive timeouts accumulated for the station.
   b. Checks the number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICFE screen 1.
      1) If the current number of consecutive timeouts is less than the maximum allowed, processing continues with step c.
      2) If the current number of consecutive timeouts is equal to the maximum allowed, the SAA AS2805 Interface process marks the station down and verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the SAA AS2805 Interface process marks the co-network down.
   c. Sets an Extended Network Management Message timer for the station.
   d. If the message that timed out was a new key or a change key message, a message is sent to the network logging facility indicating that the message expired and that the message will automatically be retried.

Extended Network Management Message Timer

An Extended Network Management Message timer is set each time a co-network station is marked down and each time a new key or change key message times out. The Extended Network Management Message timer specifies the following intervals:

- The amount of time that the SAA AS2805 Interface waits after a station is marked down before attempting to reestablish communications with the station.
- The amount of time the SAA AS2805 Interface waits after a new key or change key message times out before attempting to resend the message.

The number of seconds to use for this timer is taken from the EXTENDED NETWORK field on ICFE screen 3.

If an Extended Network Management Message timer that was set when a station was marked down expires before any messages are received from the station, the SAA AS2805 Interface process:

1. Sends an echo test message to the station.
2. Sets a Network Management Message timer for the echo test message.

When an Extended Network Management Message timer that was set after a new key or change key message time out expires, the SAA AS2805 Interface determines whether there is a station available to which to send the message. For more information on how the SAA AS2805 Interface selects a station for sending a message, see the topic “Setting up Co-Network Stations” earlier in this section. Depending on whether a station is available, the SAA AS2805 Interface performs as follows:

- If a station is available, the SAA AS2805 Interface sends the new key message or change key message to that station. The SAA AS2805 Interface sets a Network Management Message timer to wait for the response to the new key or change key message.
- If no stations are available, the SAA AS2805 Interface, sets an Extended Network Management Message timer. When this timer expires, the SAA AS2805 Interface process will attempt to send the message again.

Store-and-Forward Message Timer

A Store-and-Forward Message timer is set each time a store-and-forward message is sent to a co-network. The Store-and-Forward Message timer limits the amount of time that the SAA AS2805 Interface waits for an acknowledgment after transmitting a store-and-forward message. The number of seconds to use for this timer is taken from the STORE AND FORWARD field on ICFE screen 8 for BASE24-atm and ICFE screen 10 for BASE24-pos.

If this timer expires before an appropriate acknowledgment is received in response, the SAA AS2805 Interface process:

1. Deletes the message from internal memory.
2. Adds one to the SAF retries counter. The SAF retries count indicates the number of times the SAF record has been sent to the co-network.
3. Checks the SAF retries count against the maximum allowed by the MAX SAF RETRY field on ICFE screen 3. If the SAF retries count is equal to the maximum allowed, the SAA AS2805 Interface process:
    a. Deletes the record from the SAF.
    b. Resets the SAF retry counter.

Wait-for-Traffic Timer

A Wait-for-Traffic timer is set each time a station is marked up and each time a previously set Wait-for-Traffic timer expires. The Wait-for-Traffic timer specifies the amount of time that the SAA AS2805 Interface waits for traffic on the line before sending an echo test message. The number of seconds to use for this timer is taken from the WAIT FOR TRAFFIC field on ICFE screen 3.

The SAA AS2805 Interface process sets Wait-for-Traffic timers for its send/receive stations only. Wait-for-Traffic timers are not set for receive-only/no output stations.

When this timer expires, the SAA AS2805 Interface process determines whether there has been any message traffic to or from the station since the timer was set. Depending on whether there has been traffic on the line, the SAA AS2805 Interface process performs as follows:

1. If there has been message traffic, the SAA AS2805 Interface process simply sets another Wait-for-Traffic timer.
2. If there has not been message traffic, the SAA AS2805 Interface process:
    a. Sends an echo test message to the station.
    b. Sets a Network Management Message timer for the echo test message.

Performance Timer

A performance timer is set for each co-network at the start of that co-network’s new performance statistics accumulation period. The Performance timer specifies the interval the SAA AS2805 Interface uses for gathering performance statistics. The number of minutes to use for this timer is taken from the PERFORMANCE PERIOD field on ICFE screen 3.

For each co-network, the SAA AS2805 Interface process tracks the number of requests sent to the co-network, the number and percentage of requests that time out, and the average response time. These statistics are tracked on an interval basis and are available for the most current four intervals (interval zero through interval three with interval zero being the most current). The Performance timer triggers the end of one interval and the beginning of the next.

When a Performance timer expires, the SAA AS2805 Interface process:

1. Moves the performance statistics from interval two to interval three.
2. Moves the performance statistics from interval one to interval two.
3. Moves the performance statistics from interval zero to interval one.
4. Zeros out the performance statistics in interval zero to begin accumulating statistics again.
5. Sets a Performance timer for the co-network.

Note: These statistics are accessed via a Performance command entered from the Network Control Supervisor (NCS) screen. For information concerning the entry of the Performance command and the display of the statistics, please refer to the description of the Performance command in the ***BASE24 Text Command Reference Manual.***

Outbound Request Timer

An Outbound Request timer is set each time a 0100, 0200 message is sent to a co-network. The Outbound Request timer limits the amount of time that the SAA AS2805 Interface waits for a response from the co-network after transmitting a request message. The number of seconds to use for this timer is taken from the OUTBOUND field on the product-specific ICF screen.

If this timer expires before a response is received to the message, the SAA AS2805 Interface process performs as follows.

1. Performs product-specific processing.
   a. If the message is a BASE24-atm message, the SAA AS2805 Interface process performs as follows:
      1) Retrieves the original message from memory.
         The SAA AS2805 Interface process retains a copy of the internal message for each message it sends to the co-network until it is notified that the message was received.
      2) Sets the STM to indicate the destination was not available. This identifies the 0200 message as a failure.
      3) Sets the STM to cause the Authorisation process to generate a 0220 message if it subsequently stands in and authorises the transaction.
      4) Sends the STM to the process where the message originated.

b. If the message is a BASE24-pos message, the SAA AS2805 Interface performs as follows:

   1) Retrieves the 0200 message from memory (all 0100 messages are handled as 0200 messages internally).
      The SAA AS2805 Interface process retains a copy of the PSTM for each 0100 and 0200 message it sends to the co-network until it is notified that the message was received.

   2) Sets the PSTM to indicate the destination was not available. This identifies the 0200 message as failed.

   3) Sets the PSTM to cause the Authorisation module to generate a 0200 message if it subsequently stands in and authorises the transaction.

   4) Sends the PSTM to the process where the message originated.

2. Adds one to the count of consecutive timeouts accumulated for the station.

3. Adds one to the number of request timeouts for the co-network.
   The SAA AS2805 Interface process tracks this information as part of its co-network performance statistics.

4. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICFE screen 3.

   a. If the current number of consecutive timeouts is equal to the maximum allowed, no further action is taken.

   b. If the current number of consecutive timeouts is equal to the maximum allowed, the SAA AS2805 Interface process:

      1) Marks the station down.

      2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the SAA AS2805 Interface process marks the co-network down.

      3) Sets a Network Management Message timer for the station.

Inbound Request Timer

An Inbound Request timer is set each time a 0100 or 0200 message is sent to a BASE24-atm or BASE24-pos Authorisation process. The Inbound Request timer limits the amount of time that the SAA AS2805 Interface waits for a response from a BASE24 Authorisation process after transmitting a request message. The number of seconds to use for this timer is taken from the INBOUND field on the product-specific ICFE screen.

If an Inbound Request timer expires before a response message is received, the SAA AS2805 Interface process:

1. Retrieves the 0200 message from memory.
   The SAA AS2805 Interface process retains a copy of the external message for each 0200 message it sends to the Authorisation process until a response is received (BASE24-pos handles all 0100 messages as 0200 messages internally).
2. Formats a 0110 or 0210 external message to send to the co-network using the information from the request. The SAA AS2805 Interface process:
   a. Sets the Responder Code in the BASE24 External Message Header to 6, indicating that the SAA AS2805 Interface process is responding to the request.
   b. Sets the Response Code (P-39) field in the message to 05 (denied).
3. Sends the external message to the co-network.
4. Add one to the number of request timeouts for the SAA AS2805 Interface.

Completion Acknowledgment Timer

A Completion Acknowledgment timer is set each time a 0120, 0220, 0420, or 0520 message is sent to a co-network. The Completion Acknowledgment timer limits the amount of time that the SAA AS2805 Interface waits for an acknowledgment from the co-network after transmitting an advice message. The number of seconds to use for this timer is taken from the COMPLETION ACK field on the product-specific ICFE screen.

If a Completion Acknowledgment timer expires before an appropriate acknowledgment is received, the SAA AS2805 Interface process performs as follows:

1. Retrieves a copy of the external message from memory.
   The SAA AS2805 Interface process retains a copy of the external message for each message it sends to the co-network until it is notified that the message was received.
2. Changes the message type from 0120, 0220, 0420, or 0520 to 0121, 0221, 0421, or 0521 respectively then places the external message in the SAF.
3. Adds one to the count of consecutive timeouts accumulated for the station.
4. Adds one to the number of request timeouts for the co-network.
5. Checks the current number of consecutive timeouts accumulated for the station against the maximum allowed by the MAXIMUM TIMEOUTS field on ICF screen 3.
   a. If the current number of consecutive timeouts is less than the maximum allowed, the SAA AS2805 Interface process takes no further action.
   b. If the current number of consecutive timeouts is equal to the maximum allowed, the SAA AS2805 Interface process:
      1) Marks down the station.

2) Verifies that the co-network still has a station available for sending messages. If no send/receive stations are left up for the co-network, the SAA AS2805 Interface process marks the co-network down.

3) Sets a Network Management Message timer for the station.

Report Startup Timer

The SAA AS2805 Interface uses a report startup timer to start the SETL01 and STAT09 reports. The SAA AS2805 Interface starts the timer at initialisation and at interchange cutover. The time for this timer is taken from the LCONF parameter SW-RPT-STARTUP.

When the report startup timer expires, the obey file containing the commands to run the report programs is generated and executed.

ATM Settlement Timer

The ATM settlement timer specifies the time at which the 0520 cutover message will be sent to the co-network for ATM traffic. This timer is inserted at initialisation or warmboot and is reset after expiration.

It is set from the ATM SETTLE TIME on screen 12 of the ICFE.
Refer to the section 6 Cutover and Settlement for full details.

POS Settlement Timer

The POS settlement timer specifies the time at which the 0520 cutover message will be sent to the co-network for POS traffic. This timer is inserted at initialisation or warmboot and is reset after expiration.

It is set from the ATM SETTLE TIME on screen 12 of the ICFE.
Refer to the section 6 Cutover and Settlement for full details.

Switch Settlement Timer

The switch settle timer specifies the time at which we cutover the co-network when no 0520 message is expected from them. This timer is inserted at initialisation or warmboot and is reset after expiration.

It is set from the SETTLEMENT HOUR/MINUTE on screen 2 of the ICFE.
Refer to the section 6 Cutover and Settlement for full details.

Configuring External Messages

The SAA AS2805 Interface processes the EMF file to create and interpret external messages.

The message structures are created and modified in the EMF. The SAA AS2805 Interface allows incoming and outgoing messages to be configured individually by an institution, depending on the information the co-network chooses to send and receive.

Data elements can be defined in the EMF as mandatory, conditional, or not required.

- If a data element is defined as mandatory it is always sent in outgoing external messages and must be present in incoming external messages.
- If a data element is defined as conditional, it is sent in outgoing messages under certain conditions. If the element contains data and the data is valid, the SAA AS2805 Interface sends the element in its outgoing external messages. If the element contains blanks or zeros, it is not sent. Likewise, the data element may or may not be present in incoming messages.
- If a data element is defined as not required, the field in the EMF used to define a data element as mandatory or conditional is left blank. Data elements defined in this manner are not sent in outgoing external messages and are not expected in incoming external messages.

Message Type

The message type identifies the external message type being defined by the record. Any external message type allowed by the SAA AS2805 Interface is valid in this field, except for repeat messages. A repeat message contains the same information as the original message. The value entered in this field can only be used with certain values entered in the PROD # field and IN-OUT-IND field.

Changing the External Message Structures

A separate EMF record can be created for each message type. To create/modify the values for any message type, the following steps must be performed. These steps assume that you are logged on to the BASE24 CRT access system.

1. Display EMF screen 1 by entering EMF in the FILE DESTINATION field at the bottom of the CRT access screen. From a menu screen press the F1 key. From a file screen, press the F16 key.

2. Read the EMF record being changed by completing the field information shown below. The default values can be displayed by pressing the F7 key.
   - Set the INTERF TYP to SAA
   - Set the PROCESS NAM field to the symbolic name of the SAA AS2805 Interface process
   - Set the PROD # to 00 for BASE, 01 for ATM, 02 for POS
   - Set MSG TYP to a valid message type
   - Set IN-OUT-IND to O – outbound, I – inbound, B – for both in and out
3. Review the values for the message type being displayed. If the value for a specific data element needs changing, enter an M (mandatory), a C (conditional), or a blank (not required) in the field directly to the right of the appropriate data element number.
4. Repeat steps 2 through 7 for any remaining message types to be added to the EMF.
5. Reinitialise the SAA AS2805 Interface process in order to read the new values.

Configuring Tokens to be Sent in the External Message

A *token* is a collection of related data required to perform a specific function. As transactions are processed and additional information is required, BASE24 processes add tokens to the internal message. The Token File (TKN) is used to configure the following token processing:

- The tokens that get logged to the transaction log files. The transaction log files affected are the BASE24-atm Transaction Log File (TLF), POS Transaction Log File (PTLF), and the Interchange Log Files (ILFs).
- The tokens that get extracted from the transaction log files by the Super Extract process. Extracting allows the token information to be processed by the co-network.
- The tokens that get sent to the co-network with each transaction in the AS2805 based external message.

For more information on configuring the tokens that get logged, and extracted in the external message, please refer to the ***BASE24 Tokens Manual***.

Configuring Key Management

The SAA AS2805 Interface process reads the Key File (KEYF) for the information it needs to handle PIN encryption, PIN translation, message authentication, and dynamic key management. The KEYF contains one record for each SAA AS2805 Interface. The ***BASE24 Transaction Security Manual*** describes in detail the configuration of PIN encryption, PIN translation, message authentication, and dynamic key management using Atalla and Racal Security Modules. The ***BASE24 ERACOM Security Control Module (SCM) Reference Manual*** and ***ERACOM Security Module (ESM) Reference Manual*** describe the configuration of these functions for ERACOM ProtectHost White Security Module options.

The key management options available for the SAA AS2805 Interface are described briefly in the paragraphs that follow and further expanded in the section Security Management.

Co-network Encryption Type

The institution can specify the type of PIN encryption used for messages going to and coming from the co-network. There are three options:

- Clear PIN processing. All PINs handled by the co-network are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by the co-network are encrypted under the secure keys of a security module. The security modules that can be used include the Racal-Guardata Host Security Module (formerly VISA Security Module), ERACOM ProtectHost White Mark II (ESM) Security Module and ERACOM ProtectHost White APCA (SCM) Security Module.
- Encrypted PINs—software DES PIN processing. All PINs handled by the co-network are encrypted via software, using the DES algorithm. No security module is in use.

This option is controlled via the ENCRYPT TYPE field on KEYF or KEY6 screen 1.

BASE24 Encryption Type

BASE24 and the co-network are not required to handle PIN encryption in the same manner. Each has its own PIN encryption options that are controlled separately. BASE24 can handle PIN encryption in two different ways:

- Clear PIN processing. All PINs handled by BASE24 are in the clear.
- Encrypted PINs—security module accessed for PIN processing. All PINs handled by BASE24 are encrypted under the secure keys of a security module.

This option is controlled via the BASE24 ENCRYPT TYPE field on KEYF or KEY6 screen 1.

PIN Block Type

The institution can specify the type of PIN block used during encryptions. PINs are not simply fed into the encryption routine as is. Rather, they are presented to the encryption routines in what is called a PIN block. A PIN block is a 16-digit series of hexadecimal characters, of which the PIN is a part.

The SAA AS2805 Interface supports two types of PIN blocks—ANSI PIN blocks and PIN/PAD PIN blocks. The interchange can specify one of these two PIN block types or specify clear PINS, indicating PINs are not encrypted.

This option is controlled via the PIN BLOCK FORMAT field on KEYF or KEY6 screen 1.

PAN Characters Used in PIN Block

If the ANSI PIN block is selected, the institution can specify which PAN characters to use in formatting the PIN block. There are three options:

- Use the 12 right-most digits of the PAN, excluding the check digit
- Use the 12 right-most digits of the PAN, including the check digit
- Use the 12 left-most digits of the PAN

This option is controlled via the ANSI PAN FORMAT field on KEYF or KEY6 screen 1.

PAD Character Used in PIN Block

If the PIN/PAD PIN block is selected, the institution must specify the PAD character to use in formatting the PIN block. The PAD character is a single hexadecimal character used to fill out the remaining positions of the PIN block.

The PAD character is specified via the PIN PAD CHARACTER field on KEYF or KEY6 screen 1.

Configuring Message Authentication

Message authentication options are configured using parameters set up in two files: the External Message File (EMF) and the Key File (KEYF) or Key 6 File(KEY6). The paragraphs that follow discuss the configuration options for using message authentication with an SAA AS2805 Interface process.

MAC Encryption Type

The message authentication encryption type indicates whether authentication is supported, and if so where the encryption is done. BASE24 and the co-network are required to handle the encryption of the message authentication code (MAC) in the same manner. Currently, BASE24 only supports message authentication using a security device.

This option is controlled using the MAC ENCRYPT TYPE field on KEYF or KEY6 screen 1.

MACing Option

While BASE24 supports a range of message authentication options on the KEYF and EMF files, the SAA AS2805 Interface process only allows for full Macing to be switched on or off on an individual message basis. For EMF records configured for only primary bit map fields setting field 64 (P-64) to M (for mandatory) switches message authentication on, while for EMF records configured for both primary and secondary bit map fields setting field 128 (S-128) to M (for mandatory) switches message authentication on.

Configuring Processing Options

The following options are used to determine how the SAA AS2805 Interface process is configured for transactions allowed, stand-in authorization and automatic logons.

Transactions Allowed

The SAA AS2805 Interface process ensures that inbound transactions from the co-network are allowed using the Acquirer Processing Code File (APCF) prior to forwarding the transaction to the authorization process. Similarly, the interface ensures that outbound transactions are allowed to be sent to the co-network using the Issuer Processing Code File (IPCF) prior to forwarding the transaction to theco-network.

The SAA AS2805 Interface process can be set up to specify the transactions that can be sent to or from the co-network for authorization. Transactions allowed are configured using the transaction profiles in the Enhanced Interchange Configuration File (ICFE). These profiles correspond to records in the Acquirer Processing Code File (APCF) and the Issuer Processing Code File (IPCF).

Outbound Transactions

Institutions define the transactions that can be sent to a particular co-network by specifying an issuer transaction profile value in the ISSUER TXN PROFILE field on screen 8 (ATM) and screen 10 (POS) of the Enhanced Interchange Configuration File (ICFE). The value of the issuer transaction profile is used to read the Issuer Processing Code File (IPCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an issuer transaction profile is not specified, the Interchange Interface process uses a value of all asterisks (wildcard characters) to read the IPCF extended memory table.

Issuer Transaction Profile Processing. Once the issuer transaction profile value used to read the IPCF extended memory table has been determined, the SAA AS2805 Interface process attempts to read the IPCF extended memory table using the issuer transaction profile value, the message category code from the transaction, and the AS2805 processing code for the transaction. The SAA AS2805 Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding AS2805 processing code before reading the IPCF extended memory table.

Using the issuer transaction profile, message category, and AS2805 processing code, the SAA AS2805 Interface process steps through the records in the IPCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	**Issuer Transaction Profile**	**Message Category**	**ISO Processing Code**
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	****************	Exact match	Exact match
1	****************	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed. If the ON-US TRANSACTION ALLOWED field on IPCF screen 2 is set to a nonzero value, the transaction is allowed. If the transaction is allowed and the COMPLETION REQUIRED TO HOST field on IPCF screen 2 is set to a value of Y, the Router/Authorization module will generate a completion message indicating whether the transaction is approved or declined and send it to the host after the transaction has been authorized.

Inbound Transactions

The ACQUIRER TXN PROFILE field on ICFE screens 8 and 10 defines the transactions allowed from the co-network. The value of the acquirer transaction profile is used to read the Acquirer Processing Code File (APCF) extended memory table, which defines all the transactions allowed for the profile and the accounts on which the transactions can be performed. If an acquirer transaction profile is not specified, the SAA AS2805 Interface process uses a value of all asterisks (wildcard characters) to read the APCF extended memory table. If BASE24 receives a transaction from an interchange that has not been defined as allowed, the SAA AS2805 Interface process denies the transaction.

Acquirer Transaction Profile Processing. Once the acquirer transaction profile value used to read the APCF extended memory table has been determined, the Interchange Interface process attempts to read the APCF extended memory table using the acquirer transaction profile value, the message category code from the transaction, and the AS2805 processing code for the transaction. The SAA AS2805 Interface process calls a utility to map the internal BASE24 transaction code for the transaction to the corresponding AS2805 processing code before reading the APCF extended memory table.

Using the acquirer transaction profile, message category, and AS2805 processing code, the SAA AS2805 Interface process steps through the records in the APCF extended memory table looking for a match. If an exact match is not found, certain other alternatives are allowed. Below is the hierarchy used for selection.

Preference	Acquirer Transaction Profile	Message Category	ISO Processing Code
1	Exact match	Exact match	Exact match
2	Exact match	*	Exact match
3	****************	Exact match	Exact match
1	****************	*	Exact match

Once a match is found, the record is read to determine whether the transaction is allowed.

The APCF also allows you to specify an authorization destination for a transaction other than the BASE24-atm Authorization process or BASE24-pos Device Handler/Router/Authorization process specified in the AUTH PROCESS field on ICFE screen 8 or 10, respectively. This mechanism allows you to route transactions received from an interchange to an application process. When a transaction is routed in this way, you can also specify whether the transaction response returned from the authorization destination is logged to the transaction log file.

Stand-in Authorisation

The SAA AS2805 Interface can be set up to specify whether the acquirer can stand in to authorise transactions if the link to the co-network is down. This option is controlled via the STAND-IN AUTH field on ICFE screen 12.

Automatic Logon

The SAA AS2805 Interface can be set up to automatically log on to the co-network at startup. If this option is selected, no operator intervention is required to log on to the co-network. It is done automatically as part of the initialisation of the SAA AS2805 Interface process. This option is controlled via the AUTO SIGNON START field on ICFE screen 3.

ICFE Screens

This section contains a listing of ICFE fields used by the SAA AS2805 Interface. The ICFE is used by the institution to define institution and terminal interchange sharing holidays, transaction handling in offline and online situations, and settlement handling. ICFE screens 1 through 11 are shared by multiple Interchange Interfaces. ICFE screens 12 and 13 are interchange-specific, which means there is an ICFE screen 12 and 13 used exclusively for the SAA AS2805 Interface.

The following tables indicate the ICFE fields used by the SAA AS2805 Interface and the expected value. Values in quotation marks (“ ”) are literal values that must be placed in the corresponding fields. The SAA AS2805 Interface does not use all the fields on all the ICFE screens. These fields are noted with “not used” in the Values column.

For documentation on ICFE screens 1 through 11 and their corresponding field descriptions, refer to the ***BASE24 Base Files Maintenance Manual***. ICFE screens 12 and 13 and descriptions of their fields are provided later in this section.

Note: The information contained in the following tables does not indicate whether a field is required, but rather if a field can be used if you so choose. The field may or may not be required.

ICFE Screen 1

ICFE screen 1 contains fields establishing interchange identifiers, station names, default values for certain fields, and report title names to be printed on the SETL01-SAA and STAT09-SAA reports. The fields on ICFE screen 1 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 1	
Field	**Value**
INTERCHANGE FIID	Interchange FIID
PROCESS	Process name as defined in the NEFS
SWITCH TYPE	“SAA ”
INTERCHANGE LOGICAL NET	Logical network identifier
REPORTING NAME	SAA AS2805 Interface name as it appears on reports
INSTITUTION ID	Not used
SWITCH ID	Not used

ICFE SCREEN 1	
Field	**Value**
STATION 1	First Station name as it appears in the NEF
STATION 2	Second Station name as it appears in the NEF
SIC CODE	SIC code of this interchange
CURRENCY CODE	Currency code of this interchange.
DEFAULT TERM NUM	Not used
DEFAULT ACQUIRER ID NUM	Acquirer Id Num
CUSTOMER BALANCE DISPLAY	Not Used

ICFE Screen 2

ICFE screen 2 contains interchange settlement information and pertinent holiday dates. The fields on ICFE screen 2 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 2	
Field	**Value**
SETTLEMENT HOUR	00–23
SETTLEMENT MINUTE	00–59
SETTLEMENT DAYS	0 for five-day settlement, 1 for seven-day settlement
REPORT PRIORITY	0–255
REPORT CPU	CPU number
SWITCH POSTING DATE	YYMMDD
HOLIDAY DATES	YYMMDD (used with five-day settlement only)

ICFE Screen 3

ICFE screen 3 is split up into two sections. The first section defines transaction message timer limits used by the SAA AS2805 Interface. The second section defines the processing options used by the SAA AS2805 Interface. The fields on ICFE screen 3 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 3	
Field	**Value**
NETWORK MANAGEMENT	0–9999 (seconds)
ACQUIRER	Y or N
EXTENDED NETWORK	0–9999 (seconds)
ISSUER	Y or N
WAIT FOR TRAFFIC	0–9999 (seconds)
PROCESSING MODE	0 if no totals are to be kept, 1
PERFORMANCE PERIOD	0–9999 (minutes in performance monitoring period)
AUTO SIGNON START	Y or N
MAXIMUM TIMEOUTS	0–9999 (timeouts allowed)
MAX SAF RETRY	0–9999 (SAF retries allowed)
OUTBOUND	0–9999 (outbound transactions allowed)
ACK TO SWITCH	Not Used
INBOUND	0–9999 (inbound transactions allowed)
ACK FROM SWITCH	Not Used
NETWORK MANAGEMENT MESSAGE ENABLED	Not Used
TYPE OF INTERCHANGE REPORTS	Not used
ILF EXTRACT NUMBER	Not Used

ICFE Screen 8

ICFE screen 8 defines ATM transaction message timer limits, ATM Authorisation process name, and ATM sharing groups. The fields on ICFE screen 8 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 8	
Field	**Value**
AUTH PROCESS	Authorisation process name
DEFAULT ROUTING GROUP	Terminal routing group number
ACQUIRER TXN PROFILE	Acquirer transaction profile for BASE24-atm processing
ISSUER TXN PROFILE	Issuer transaction profile for BASE24-atm processing
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	0-9999 (seconds)
COMPLETION ACK	0-9999 (seconds)
SHARING GROUPS	Sharing group codes

ICFE Screen 10

ICFE screen 10 defines POS transaction message timer limits, POS Authorisation process name, POS default retailer ID, and POS timeout action. The fields on ICFE screen 10 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 10	
Field	**Value**
AUTH PROCESS	Authorisation process name
REFERRAL PHONE NUMBER	Telephone number
RETAILER ID DEFAULT	Not used
TIMEOUT ACTION	Not Used
SETTLE ENTITY	0,1
ACQUIRER TXN PROFILE	Acquirer transaction profile for BASE24-pos processing
ISSUER TXN PROFILE	Issuer transaction profile for BASE24-pos processing
STORE AND FORWARD	0–9999 (seconds)
OUTBOUND	0–9999 (seconds)
INBOUND	0–9999 (seconds)
COMPLETION	0-9999 (seconds)
COMPLETION ACK	0-9999 (seconds)

ICFE Screen 11

ICFE screen 11 defines POS preauthorisation default time and amount, approval code length, and services allowed to the co-network. The fields on ICFE screen 11 used by the SAA AS2805 Interface and their expected values are indicated in the following table.

ICFE SCREEN 11	
Field	**Value**
DEFAULT PRE-AUTH AMOUNT	Default pre-authorisation amount
APPROVAL CODE LENGTH	2–6
PRE-AUTH HOLD INCREMENT	Not Used
PRE-AUTH HOLD TIME	00–99
ALLOWED SERVICES	Service code

ICFE Screen 12

ICFE screen 12 is an interchange-specific screen used exclusively by the SAA AS2805 Interface. It defines options used at logon to determine processing requirements for the SAA AS2805 Interface. The SAA AS2805 Interface can use all fields on ICFE screen 12. ICFE screen 12 is shown below followed by descriptions of its fields.

```
BASE24-BASE  ICFE FILE MAINTENANCE    llll        98/03/17  13:49  12 OF 13
INTERCHANGE FIID: SAA                    PROCESS: P1A^SAA

                         ICF KEY PARAMETERS

 ATM SETTLE TIME: 18:00  POS SETTLE TIME: 18:00  STAND IN ALLOWED: Y (Y/N)
  CUTOVER METHOD: 0 (SEND/RECEIVE 0520)  RECON FLAG: 0 (BOTH ATM AND POS)
 ATM DEFAULT ACC: 11 CHEQUE              POS DEFAULT ACC: 00 DEFAULT

                     SESSION KEY ROLLOVER THRESHOLDS
          KEY EXCHANGE BAD PINS:   5       BAD MACS:    5
   NUM TRANS FOR TOTALS REWRITE: 100       NUM TRANS KEY EXCHANGE:  100
                SAF TIME UPDATE: N (Y/N)   CALC MAC ON REPEAT: Y (Y/N)

        KEY EXCHANGE INFORMATION                             INCLUDE
 TYPE OF KEY EXCHANGE: 1 (STATIC)                    DATA KEY      : N (Y/N)
 TAPE FILE FOR KEYS:                                 KVC PROCESSING: N (Y/N)
 TAPE DATE:        (YYMMDD)                          PROOF ENDPOINT: N (Y/N)


 *********************************** BASE24 ********************************
  NEW PAGE:        FILE DESTINATION:       NEW LOGICAL NETWORK ID:
                              F12-HELP
```

ICFE KEY PARAMETERS

The following fields contain information relevant to the functionality for the SAA AS2805 Interface.

ATM SETTLE TIME - The Time at which the 0520 cutover message is sent to the co-network for ATM traffic. This field is used in conjunction with the fields "Cutover Method", and "Recon Flag". Valid values are 00:00 through 23:59.

Field Length: 4 numeric
Required Field: Yes
Default Value: 00:00
Data Name: ICFSAA.ATM-SETL-TIM

POS SETTLE TIME - The Time at which the 0520 cutover message is sent to the co-network for POS traffic. This field is used in conjunction with the fields "Cutover Method", and "Recon Flag". Valid values are 00:00 through 23:59.

Field Length: 4 numeric
Required Field: Yes
Default Value: 00:00
Data Name: ICFSAA.POS-SETL-TIM

STAND-IN ALLOWED - A flag that indicates whether BASE24, when acting as an acquirer, is allowed to perform stand-in authorisation for the co-partner if the link to the co-network is unavailable. Valid values are:

Y - Stand-in is allowed
N - Stand-in is not allowed

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: N
Data Name: ICFSAA.STAND-IN-AUTH

CUTOVER METHOD - A flag the indicates whether or not 0520 reconciliation messages will be exchanged between the two partners before it is considered that cutover is complete for the day, and the ILF can be closed for the day. Valid values are;

0 - A cutover message is required to be sent to the co-network, and one must be received from the co-network.
1 - A cutover message must be sent to the co-network, but a cutover message is not required from the co-network.
2 - A cutover message will not be sent to the co-network, but a cutover message is expected to be received from the co-network.
3 - No cutover messages will be exchanged between the partners.

Field Length: 1 alpha-numeric

Required Field: Yes
Default Value: 0
Data Name: ICFSAA.CUTOVER-METHOD

RECON FLAG - A flag indicating the type of 0520 cutover messages to send. Valid values are;

0 - Separate ATM and POS cutover messages
1 - Send ATM cutover messages
2 - Send POS cutover messages
3 - Send combined ATM and POS totals in a single cutover message

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.RECONCILIATION-FLG

ATM DEFAULT ACCT - The default account type to set the STM account type to if an account type is not received in the external message from the co-network. Valid values are;

“00” - Default
“01” - Savings
“11” - Cheque
“31” - Credit

Field Length: 2 alpha-numeric
Required Field: Yes
Default Value: “00”
Data Name: ICFSAA.DEFAULT-ACCT-ATM

POS DEFAULT ACCT - The default account type to set the PSTM account type to if an account type is not received in the external message from the co-network. Valid values are;

“00” - Default
“01” - Savings
“11” - Cheque
“31” - Credit

Field Length: 2 alpha-numeric
Required Field: Yes
Default Value: “00”
Data Name: ICFSAA.DEFAULT-ACCT-POS

SESSION KEY ROLLOVER THRESHOLDS

KEY EXCHANGE BAD PINS - The number of consecutive bad PINs to occur to force new session keys to be exchanged automatically. Valid values in the range 0 thru 999.

Field Length: 3 numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.TRANS-COUNT-OPTIONS.NUM-BAD-KEYS-PIN

BAD MACS - The number of consecutive bad MACs to occur to force new session keys to be exchanged automatically. Valid values in the range 0 thru 999.

Field Length: 3 numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.TRANS-COUNT-OPTIONS.NUM-BAD-KEYS-MAC

NUM TRANS FOR TOTALS REWRITE - The number of transactions that need to occur before the first record of the ILF is updated. Valid values in the range of 0 thru 999.

Field Length: 3 numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.TRANS-COUNT-OPTIONS.NUM-TRANS-REWRITE

NUM TRANS KEY EXCHANGE - The number of transactions that need to occur before a new set of session keys are exchanged automatically. Valid values in the range of 0 thru 999.

Field Length: 3 numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.TRANS-COUNT-OPTIONS.NUM-TRANS-KEY-EXCHANGE.

SAF TIME UPDATE - A flag to indicate whether or not to update the transaction date and time on external messages that have come from the SAF. Valid values are;

N - Do not update the transaction time.
Y - Update the transaction time.

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: N

Data Name: ICFSAA.SAF-TIM-UPDT.

CALC MAC ON REPEAT - A flag to indicate whether the MAC should be recalculated when sending repeat messages. Valid values are;

N - Do not re-calculate the MAC.
Y - Re-calculate the MAC..

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: Y
Data Name: ICFSAA.MAC-REPEAT-CALC.

KEY EXCHANGE INFORMATION

TYPE OF KEY EXCHANGE - Indicates what type of session key management is to take place. Valid values are;

“0” - No key management scheme is in use.
“1” - Static keys will be used
“2” - Tape keys will be used
“3” - Dynamic key exchange will be used
“4” - Dynamic KEK exchange will be used

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: 0
Data Name: ICFSAA.KEY-EXCHNG-TYP.

DATA KEY - A flag to indicate whether or not session key exchange will include the exchange of data session keys. Valid values are;

N - Data keys will not be exchanged, only PIN and MAC keys.
Y - Data keys will also be exchanged.

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: N
Data Name: ICFSAA.INCL-DATA-KEY.

TAPE FILE FOR KEYS - The external data file name of the tape key file if the “Key Exchange Type” indicates tape key exchange (“2”).

Field Length: 24 alpha-numeric
Required Field: No
Default Value: None
Data Name: ICFSAA.TKEY-FILE-NAME.

KVC PROCESSING - A flag to indicate whether or not KVCs will be included and checked on key exchange messages. Valid values are;

N - Do not check KVCs.
Y - Check KVC values.

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: N
Data Name: ICFSAA.INCL-KVC.

TAPE DATE - The date the tape key file was created. This date is compared to the date in the physical tape key file (TKEY) to make sure the correct file is being used. It must be a valid YYMMDD date.

Field Length: 6 numeric
Required Field: No
Default Value: None
Data Name: ICFSAA.TAPE-DATE.

PROOF OF ENDPOINTS - A flag to indicate whether or proof of endpoints will be performed on logon request/response messages. Valid values are;

N - Do not perform proof of endpoints.
Y - Perform proof of endpoints.

Field Length: 1 alpha-numeric
Required Field: Yes
Default Value: N
Data Name: ICFSAA.PROOF-ENDPOINT.

ICFE Screen 13

ICFE screen 13 is an interchange-specific screen used exclusively by the SAA AS2805 Interface. It defines the foreign currency support options and the station configuration. The SAA AS2805 Interface can use all fields on ICFE screen 13. ICFE screen 13 is shown below followed by descriptions of its fields.

```
 BASE24-BASE  ICFE FILE MAINTENANCE    llll        98/03/17  13:52  13 OF 13
 INTERCHANGE FIID: SAA                         PROCESS: P1A^SAA




           FOREIGN CURRENCY SUPPORT                 STATION CONFIGURATION
       POS  1    NATIONAL               SIM^SAA            0 SEND/RECEIVE
       ATM  1    NATIONAL




  *********************************** BASE24 ***************************
   NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                              F12-HELP
```

FOREIGN CURRENCY SUPPORT – Two flags, one for POS and one for ATM, that indicates what type of support this interchange has for transactions from other countries. The transaction currency code in the request from the co-network is compared with the currency code in screen 1 of the ICF to determine whether a transaction is local, or international. Valid values are;

“0” - Transactions not allowed at all
“1” - National only. Only local transactions allowed
“2” – Either national or international transactions allowed
“3” – International only. Only foreign transactions allowed.

Field Length:	1 alpha-numeric each
Required Field:	Yes
Default Value:	“1”
Data Name:	ICFSAA.POS-NATL-INTL, ICFSAA.ATM-NATL-INTL

STATION CONFIGURATION – Two flags, one for each station defined on screen 1 of the ICF, that indicates what type of communications control this interchange has. It indicates which of the two stations will be used to send messages out on, and which will be used to receive messages from. Note that regardless of the station configuration, each station will be polled (or echo tested) separately. Echo test requests and responses will be sent and received on each station. Valid values are;

"0" – Send/Receive. This station can be used to send and receive financial transactions.
"1" – Receive only. This station will be used to receive messages from the co-network, but it will not be used to send transactions to the co-network

Field Length:	1 alpha-numeric each
Required Field:	Yes
Default Value:	"0"
Data Name:	ICFSAA.STA.STA1-TYP, ICFSAA.STA.STA2-TYP

KEYF and KEY6 Screens

This section contains KEYF screens and descriptions of KEYF screen fields used by the SAA AS2805 Interface. The KEY6 contains the same fields as the KEYF, except that session key fields on KEY6 screens allow for double-length keys (i.e., 32 hexadecimal characters instead of 16 hexadecimal characters) to be used in triple DES encryption. Thus PIN session key fields take up all of KEY6 screen 2, MAC session key fields take up all of KEY6 screen 3, and the dynamic key management fields (not used by SAA AS2805 Interface) are on KEY6 screen 4 instead of screen 3.

For further documentation on the KEYF screens and their corresponding field descriptions, refer to the ***BASE24 Files Maintenance Manual***. For information on BASE24 security, refer to the ***BASE24 Transaction Security Manual*** for Atalla and Racal Security Modules, ***BASE24 ERACOM Security Control Module (SCM) Reference Manual*** for SCM and ***ERACOM Security Module (ESM) Reference Manual*** for ESM.

KEYF Screen 1

KEYF screen 1 defines the number of PIN and MAC keys currently being used by the interface, the length of the Key Exchange Keys (KEKs) used by the interface, and the PIN and MAC exchange keys used by the interface. In addition, screen 1 contains security information used to set up the PIN and MAC parameters associated with the SAA AS2805 Interface process. KEYF screen 1 is shown below followed by descriptions of its fields used by the SAA AS2805 Interface.

Note: SCM related fields such as the KVC will now occupy a different location in DDLFKEYF for this release (R6.0 V10). Unused USER-FLDS in DDLFKEYF have been used to accommodate a number of SCM related fields that were retired in previous releases of KEYF.

```
BASE24-BASE  KEY FILE                Y2K1        98/03/17  13:54  01 OF 04
         DPC/FIID: SAA           INTERFACE PROCESS: P1A^SAA

    ENCRYPT TYPE: 1 (SECURE DEVICE)    BASE24 ENCRYPT TYPE: 1 (SECURE DEVICE)
PIN BLOCK FORMAT: 1 (ANSI)                 ANSI PAN FORMAT: 0 (12 RIGHT/NO CHK)
MAC ENCRYPT TYPE: 1 (SECURE DEVICE)      PIN PAD CHARACTER: F
   MAC DATA TYPE: 0 (ASCII)                 NUMBER OF KEYS: 2 (SEPARATE)
      KEY LENGTH: 1 (SINGLE)             FULL MESSAGE MAC: N (SELECTED FIELDS)

                                INTERMEDIATE KEYS
          CLEAR: 0000000000000000                    CHECK DIGITS: 0000
      ENCRYPTED: 0000000000000000

                                  EXCHANGE KEYS
        PIN KEY: 0000000000000000  0000000000000000   CHECK DIGITS: 0000
        MAC KEY: 0000000000000000  0000000000000000   CHECK DIGITS: 0000
        KEKs: 00 DDEDC7101C239494  6A3D61C2B7DAFCC0   KVC: FB0975
        KEKr: 00 44496B0B20DA9614  C371FCDEC8FEB971   KVC: F39C09

 RECORD LAST CHANGED: 98/02/09  16:39   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 **********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                         F12-HELP
RECORD RETRIEVED FROM  \ACIP.$SYSTEM.Y2K1DATA.KEYF                       0000
```

DPC NUMBER OR FIID - Identifier for the financial institution associated with this record.

If this record is an institution record, the value entered in this field must match a value in the FIID field in the IDF. The Authorisation process accesses this file after obtaining the FIID from the IDF. Authorisation then obtains the KEY and PIN field values for the institution.

If this record is an interchange interface record, the value entered in this field must match a value in the FIID field in the ICF. Interchange interface processes access this file after obtaining the FIID from the ICF. The process then obtains the KEY and PIN processing values for the interchange.

If this record is a host interface record, the value entered in this field must match a value in the DPC NUMBER field in the HCF. Host interface processes access this file after obtaining the DPC NUMBER from the HCF. The process then obtains the KEY and PIN processing values for the Data Processing Center.

Field Length:	1-4 alphanumeric characters
Required Field:	Yes
Default Value:	None
Data Name:	KEYF.PRIKEY-ICF.FIID

INTERFACE PROCESS - The name of the interface process associated with the FIID for interchange interface records or DPC NUMBER for host interface processes.

Field Length:	16 alphanumeric characters
Required Field:	Yes
Default Value:	None
Data Name:	KEYF.PRIKEY-ICF.SWI-PRO

ENCRYPT TYPE - Specifies the type of encryption between the DPC or interchange and the interface. Valid values are:

0 = No PIN encryption
1 = Secure device PIN encryption
2 = Software DES PIN encryption

Field Length:	1 numeric character
Required Field:	Yes
Default value:	0 (no encryption)
Data Name:	KEYF.INTERFACE.ENCRYPT-TYP

BASE24 ENCRYPT TYPE - Specifies the type of PIN encryption between the interface process and BASE24. The Standard Internal Message (STM) indicates the type of encryption between the interface process and BASE24. Valid values are:

0 = No encryption, clear PINs
1 = Secure device PIN encryption

Field Length: 1 numeric character
Required Field: Yes
Default value: 0 (no encryption)
Data Name: KEYF.INTERFACE.B24-ENCRYPT-TYP

PIN BLOCK FORMAT - Specifies what PIN block format will be utilised by the DPC or interchange. Valid values are:

0 = Clear, PINs are not encrypted
1 = ANSI PIN block
3 = PIN/PAD and PIN block

Field Length: 1 numeric character
Required Field: Yes
Default value: 0 (no encryption)
Data Name: KEYF.INTERFACE.PIN-BLK

ANSI PAN FORMAT - Specifies the pan digits utilised in the formation of an ANSI PIN block. Valid values are:

0 = 12 right-most digits of the pan, excluding the check digit
1 = 12 right-most digits of the pan, including the check digit
2 = 12 left-most digits of the pan

Field Length: 1 numeric character
Required Field: Yes
Default value: 0
Data Name: KEYF.INTERFACE.ANSI-PAN

MAC ENCRYPT TYPE - Specifies the type of MAC encryption between the interface process and BASE24. Valid values are:

0 = No MAC processing
1 = Security Module MAC processing
2 = Software MAC Processing
This field must contain a 1 if using dynamic MAC Key management.

Field Length: 1 numeric character
Required Field: Yes
Default value: 0
Data Name: KEYF.INTERFACE.MAC-TYP

PIN PAD CHARACTER - Specifies the PIN block PAD character for PIN/PAD PIN blocks. Valid values are A through F.

Field Length: 1 alphanumeric character
Required Field: Yes
Default value: F

Data Name: KEYF.INTERFACE.PINPAD-CHAR

MAC DATA TYPE - Specifies the character set in which messages are transmitted between the DPC or Interchange and the interface.
THIS FIELD IS NOT USED BY THIS SWITCH.
Valid values are

0 – ASCII
1 - EBCDIC

Field Length: 1 numeric character
Required Field: Yes
Default value: 0
Data Name: KEYF.INTERFACE.MAC-DATA-TYP

NUMBER OF KEYS - Specifies whether the inbound and outbound keys are combined or separate.
THIS FIELD IS NOT USED BY THIS SWITCH.

Valid values are:
1 – Combined
2 - Separate

Field Length: 1 numeric character
Required Field: Yes
Default value: 1
Data Name: KEYF.INTERFACE.NUM-KEYS

KEY LENGTH - Specifies whether single- or double-length Key Exchange Keys (KEK) are being used with this interface.
THIS FIELD IS NOT USED BY THIS SWITCH.

Valid values are:
1 - Single-length KEKs
2 - Double-length KEKs

Field Length: 1 numeric character
Required Field: Yes
Default value: 1
Data Name: KEYF.INTERFACE.KEY-LGTH

FULL MESSAGE MAC - Specifies whether selected data elements or the entire message will be used to construct the MAC.
THIS FIELD IS NOT USED BY THIS SWITCH.

Valid values are:
N - Selected fields authenticated.
Y - Full message authentication.

Field Length: 1 alphanumeric character
Required Field: Yes

Default value: N
Data Name: KEYF.INTERFACE.FULL-MSG-MAC

INTERMEDIATE KEYS

INTERMEDIATE KEYS - If an encrypted PIN must be transmitted in the clear, a secure device is utilised to translate the PIN to be encrypted under the ENCRYPTED field and then the CLEAR key is used in software to decrypt the PIN.

CLEAR - The clear version of the Intermediate Key. PINs that are received by BASE24 encrypted under another key, are decrypted, and re-encrypted under this key.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.INTERM.KEY-CLEAR

CHECK DIGITS - These correspond to the value in the Encrypted Intermediate Key field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.INTERM-KEY-CHK-VALUES

ENCRYPTED - The encrypted version of the Intermediate Key, used together with the encrypted PIN and the key the PIN is currently encrypted under, to decrypt the PIN.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.KEY-ENCRYPT

EXCHANGE KEYS

EXCHANGE KEYS - These are master keys for the interchange. When PIN keys and/or MAC keys are automatically exchanged between the interchange interface and the interchange they are encrypted under these keys. These keys must be manually exchanged between the two entities.

PIN KEY - The encrypted version of the key used to exchange PIN keys. The first set is used for single-length keys and both sets are used for double-length keys.

Field Length: 16 hexadecimal characters (for both sets)
Required Field: Yes

Default value: 0000000000000000 (for both sets)
Data Name: KEYF.INTERFACE.EXCHNG-KEY
KEYF.INTERFACE.EXCHNG-KEY-EXTND

CHECK DIGITS - The check digit corresponding to the value in the PIN Key field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.EXCHNG-KEY-CHK-VALUES

MAC KEY - The encrypted version of the key used to exchange MAC keys. The first set is used for single-length keys and both sets are used for double-length keys.

Field Length: 16 hexadecimal characters (for both sets)
Required Field: Yes
Default value: 0000000000000000 (for both sets)
Data Name: KEYF.INTERFACE.MAC-EXCHNG-KEY
KEYF.INTERFACE.MAC-EXCHNG-KEY-EXTND

CHECK DIGITS - The check digit corresponding to the value in the MAC Key field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.MAC-EXCHNG-KEY-CHK-VALUE

KEKs - The encrypted version of the sending key encrypting key used by the SCM security module. If an ESM is being used for dynamic key exchange this field on KEY6 screen contains the index of the KIS in the right-most 2 digits.

Field Length: 34 hexadecimal characters (for both sets)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 0000000000000000
Data Name: KEYF.INTERFACE.E-KMV7-KEKS

KVC - The key verification code of the sending key encrypting key used by the SCM security module.

Field Length: 6 hexadecimal characters
Required Field: No
Default value: 000000
Data Name: KEYF.INTERFACE.OUTBOUND.PIN.KEKS-KVC

KEKr - The encrypted version of the receiving key encrypting key used by the SCM security module. If an ESM is being used for dynamic key exchange this field on KEY6 screen contains the index of the KIR in the right-most 2 digits

Field Length: 34 hexadecimal characters (for both sets)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 0000000000000000
Data Name: KEYF.INTERFACE.E-EKMV8-KEKR

KVC - The key verification code of the receiving key encrypting key used by the SCM security module.

Field Length: 6 hexadecimal characters
Required Field: No
Default value: 000000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEKR-KVC

KEYF Screen 2

KEYF screen 2 contains inbound and outbound working information for PINs and MACs. KEYF screen 2 is shown below followed by descriptions of its fields used by the SAA AS2805 Interface.

```
BASE24-BASE  KEY FILE                 Y2K1        98/03/17  13:54  02 OF 04
         DPC/FIID: SAA           INTERFACE PROCESS: P1A^SAA

                               PIN KEY INFORMATION
OUTBOUND KEYS:
   PIN KEY1: 0000000000000001    CHECK DIGITS1: FFFF    CURRENT INDEX: 1
   PIN KEY2: 0000000000000000    CHECK DIGITS2: FFFF      KEY COUNTER: 000000
INBOUND KEYS:
   PIN KEY1: 0DCF7332DB3A261E    CHECK DIGITS1: FFFF    CURRENT INDEX: 1
   PIN KEY2: 0000000000000000    CHECK DIGITS2: FFFF      KEY COUNTER: 000000

                               MAC KEY INFORMATION
OUTBOUND KEYS:
   MAC KEY1: 461D73DEB95345F5    CHECK DIGITS1: FFFF    CURRENT INDEX: 1
   MAC KEY2: 0000000000000000    CHECK DIGITS2: FFFF      KEY COUNTER: 000000
INBOUND KEYS:
   MAC KEY1: E60F0FD5753CC2FC    CHECK DIGITS1: FFFF    CURRENT INDEX: 1
   MAC KEY2: 0000000000000000    CHECK DIGITS2: FFFF      KEY COUNTER: 000000

 RECORD LAST CHANGED: 98/02/09  16:39   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                        F12-HELP
```

PIN KEY INFORMATION

OUTBOUND KEYS

The PIN Key fields which follow contain the PIN encryption key used to encrypt PINs sent from BASE24 to the Host DPC or the Interchange Interface.

If an Atalla Security Module is being used for PIN encryption then the PIN Key fields contain working keys encrypted under the Atalla Master File Key.

If an ESM is being used for PIN encryption then the PIN Key fields contain the index of the KIS in the right-most 2 digits.

PIN KEY1 - Used for the encryption of PINs in transactions sent to the DPC or interchange. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length:	16 hexadecimal characters
Required Field:	Yes
Default value:	0000000000000000
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.KEY1

CHECK DIGITS1 - The check digits corresponding to the value in the PIN KEY1 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length:	4 hexadecimal characters
Required Field:	Yes
Default value:	0000
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.KEY-CHK-VALUE1

CURRENT INDEX - Indicates which PIN encryption key is currently being used. Valid values are 1 or 2.

Field Length:	1 numeric character
Required Field:	Yes
Default value:	1
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.CURR-INDX

PIN KEY2 - Used for the encryption of PINs in transactions sent to the DPC or interchange. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length:	16 hexadecimal characters
Required Field:	Yes
Default value:	0000000000000000
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.KEY2

CHECK DIGITS2 - The check digits corresponding to the value in the PIN KEY2 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length:	4 hexadecimal characters
Required Field:	Yes
Default value:	0000
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.KEY-CHK-VALUE2

KEY COUNTER - Indicates the number of times the Outbound PIN Key has been changed by Dynamic Key management processing.
THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length:	6 numeric characters
Required Field:	Yes
Default value:	000000
Data Name:	KEYF.INTERFACE.OUTBOUND.PIN.KEY-CNTR

INBOUND KEYS

The PIN Key fields which follow contain the PIN encryption key used to encrypt PINs sent from the Host DPC or the Interchange interface to BASE24.

If an Atalla Security Module is being used for PIN encryption then the PIN Key fields contain working keys encrypted under the Atalla Master File Key.

If an ESM is being used for PIN encryption then the PIN Key fields contain the index of the KIR in the right-most 2 digits.

PIN KEY1 - Used for the encryption of PINs in transactions sent to BASE24. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length:	16 hexadecimal characters
Required Field:	Yes
Default value:	0000000000000000
Data Name:	KEYF.INTERFACE.INBOUND.PIN.KEY1

CHECK DIGITS1 - The check digits corresponding to the value in the PIN KEY1 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length:	4 hexadecimal characters
Required Field:	Yes
Default value:	0000
Data Name:	KEYF.INTERFACE.INBOUND.PIN.KEY-CHK-VALUE1

CURRENT INDEX - Indicates which PIN encryption key is currently being used. Valid values are 1 or 2.

Field Length:	1 numeric character
Required Field:	Yes
Default value:	1
Data Name:	KEYF.INTERFACE.INBOUND.PIN.CURR-INDX

PIN KEY2 - Used for the encryption of PINs in transactions sent to BASE24. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length:	16 hexadecimal characters
Required Field:	Yes
Default value:	0000000000000000
Data Name:	KEYF.INTERFACE.INBOUND.PIN.KEY2

CHECK DIGITS2 - The check digits corresponding to the value in the PIN KEY2 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length:	4 hexadecimal characters
Required Field:	Yes
Default value:	0000
Data Name:	KEYF.INTERFACE.INBOUND.PIN.KEY-CHK-VALUE2

KEY COUNTER - Indicates the number of times the Inbound PIN Key has been changed by Dynamic Key management processing.
THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 6 numeric characters
Required Field: Yes
Default value: 000000
Data Name: KEYF.INTERFACE.INBOUND.PIN.KEY-CNTR

MAC KEY INFORMATION

OUTBOUND KEYS

The MAC Key fields which follow contain the MAC generation key used to generate MACs sent from BASE24 to the Host DPC or the Interchange Interface.

If an ESM is being used for MAC generation and verification then the MAC Key fields contain a working key encrypted under a Domain Master Key variant.

Note: The MAC keys should be security module encrypted.

MAC KEY1 - Used for the encryption of MACs in transactions sent to the DPC or interchange. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEY1

CHECK DIGITS1 - The check digits corresponding to the value in the MAC KEY1 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEY-CHK-VALUE1

CURRENT INDEX - Indicates which MAC encryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character
Required Field: Yes
Default value: 1
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.CURR-INDX

MAC KEY2 - Used for the encryption of MACs in transactions sent to the DPC or interchange. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEY2

CHECK DIGITS2 - The check digits corresponding to the value in the MAC KEY2 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEY-CHK-VALUE2

KEY COUNTER - Indicates the number of times the Inbound MAC Key has been changed by Dynamic Key management processing.
THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 6 numeric characters
Required Field: Yes
Default value: 000000
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.KEY-CNTR

INBOUND KEYS

The MAC Key fields which follow contain the MAC encryption key used to encrypt MACs sent from the Host DPC or the Interchange interface to BASE24.

MAC KEY1 - Used for the encryption of MACs in transactions sent to BASE24. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.INBOUND.MAC.KEY1

CHECK DIGITS1 - The check digits corresponding to the value in the MAC KEY1 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.INBOUND.MAC.KEY-CHK-VALUE1

CURRENT INDEX - Indicates which MAC encryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character
Required Field: Yes
Default value: 1

Data Name: KEYF.INTERFACE.INBOUND.MAC.CURR-INDX

MAC KEY2 - Used for the encryption of MACs in transactions sent to BASE24. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters
Required Field: Yes
Default value: 0000000000000000
Data Name: KEYF.INTERFACE.INBOUND.MAC.KEY2

CHECK DIGITS2 - The check digits corresponding to the value in the MAC KEY2 field. THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 4 hexadecimal characters
Required Field: Yes
Default value: 0000
Data Name: KEYF.INTERFACE.INBOUND.MAC.KEY-CHK-VALUE2

KEY COUNTER - Indicates the number of times the Inbound MAC Key has been changed by Dynamic Key management processing.
THIS FIELD IS NOT USED BY THIS SWITCH.

Field Length: 6 numeric characters
Required Field: Yes
Default value: 000000
Data Name: KEYF.INTERFACE.INBOUND.MAC.KEY-CNTR

KEYF Screen 3

KEYF Screen 3 consists of Dynamic Key management parameters.

The SAA AS2805 interface does not currently use this screen for dynamic key management. Please refer to screen 12 of the ICF for configuration parameters.

For further details for KEYF screen 3 refer to the BASE24 BASE File Maintenance Manual.

KEYF Screen 4

KEYF screen 4 contains inbound and outbound SCM Static Key information for PINs and MACs. KEYF screen 4 is shown below followed by descriptions of its fields used by the SAA AS2805 Interface.

Note: SCM related fields such as the KM-ID will now occupy a different location in DDLFKEYF for this release (R6.0 V10). Unused USER-FLDS in DDLFKEYF have been used to accommodate a number of SCM related fields that were retired in previous releases of KEYF.

```
BASE24-BASE  KEY FILE                Y2K1        98/03/17  13:55  04 OF 04
        DPC/FIID: SAA          INTERFACE PROCESS: P1A^SAA
```

```
                               SCM STATIC KEY SUPPORT

OUTBOUND KEYS:

   PIN KEY1 00 BD780E34464D9026  PIN KEY2 00 BD780E3446D49026  CURRENT INDEX: 1
   MAC KEY1 00 37AC5C1CC12CC9D8  MAC KEY2 00 37AC5C1CC12CC9D8  CURRENT INDEX: 1


INBOUND KEYS:

   PIN KEY1 00 BEC88507D4C5BEE9  PIN KEY2 00 BEC88507D4C5BEE9  CURRENT INDEX: 1
   MAC KEY1 00 D9F4F07F5849A706  MAC KEY2 00 D9F4F07F5849A706  CURRENT INDEX: 1




 RECORD LAST CHANGED: 98/02/09  16:39   BY USER: 0255 , 00000255   CHANGE
 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                         F12-HELP
```

SCM STATIC KEY SUPPORT

OUTBOUND KEYS

The fields which follow contain the keys to support SCM Static Key Management which are used to generate MACs and PINs sent from the interchange process to the co-network.

PIN KEY1 - Used for the encryption of PINs in transactions sent to the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.OUTBOUND.PIN.SKEY1 (Key)
KEYF.INTERFACE.OUTBOUND.PIN.SCM-OPIN.KM-ID-S1 (KM Id)

PIN KEY2 - Used for the encryption of PINs in transactions sent to the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.OUTBOUND.PIN.SKEY2 (Key)
KEYF.INTERFACE.OUTBOUND.PIN.SCM-OPIN.KM-ID-S2 (KM Id)

CURRENT INDEX - Indicates which PIN encryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character
Required Field: Yes

Default value: 1
Data Name: KEYF.INTERFACE.OUTBOUND.PIN.CURR-INDX

MAC KEY1 - Used for the encryption of MACs in transactions sent to the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.OUTBOUND.MAC.SKEY1 (Key)
KEYF.INTERFACE.OUTBOUND.MAC.SCM-OMAC.KM-ID-S1 (KM Id)

MAC KEY2 - Used for the encryption of MACs in transactions sent to the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.OUTBOUND.MAC.SKEY2 (Key)
KEYF.INTERFACE.OUTBOUND.MAC.SCM-OMAC.KM-ID-S2 (KM Id)

CURRENT INDEX - Indicates which MAC encryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character
Required Field: Yes
Default value: 1
Data Name: KEYF.INTERFACE.OUTBOUND.MAC.CURR-INDX

INBOUND KEYS

The fields which follow contain the keys to support SCM Static Key Management which are used to verify MACs and PINs received by the interchange from the co-network.

PIN KEY1 - Used for the decryption of PINs in transactions received from the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.INBOUND.PIN.SKEY1 (Key)

KEYF.INTERFACE.INBOUND.PIN.SCM-IPIN.KM-ID-S1 (KM Id)

PIN KEY2 - Used for the decryption of PINs in transactions received from the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.INBOUND.PIN.SKEY2 (Key)
KEYF.INTERFACE.INBOUND.PIN.SCM-IPIN.KM-ID-S2 (KM-Id)

CURRENT INDEX - Indicates which PIN decryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character
Required Field: Yes
Default value: 1
Data Name: KEYF.INTERFACE.INBOUND.PIN.CURR-INDX

MAC KEY1 - Used for the decryption of MACs in transactions received from the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 1.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.INBOUND.MAC.SKEY1 (Key)
KEYF.INTERFACE.INBOUND.MAC.SCM-IMAC.KM-ID-S1 (KM-Id)

MAC KEY2 - Used for the decryption of MACs in transactions received from the co-network. The interface process will use this key when the CURRENT INDEX field contains a value of 2.

Field Length: 16 hexadecimal characters (Key)
2 hexadecimal characters (KM Id)
Required Field: Yes for SCM or ESM sites
Default value: 00 0000000000000000 (KM Id + Key)
Data Name: KEYF.INTERFACE.SCM.INBOUND.MAC.SKEY2 (Key)
KEYF.INTERFACE.INBOUND.MAC.SCM-IMAC.KM-ID-S2 (KM-Id)

CURRENT INDEX - Indicates which MAC decryption key is currently being used. Valid values are 1 or 2.

Field Length: 1 numeric character

Required Field:	Yes
Default value:	1
Data Name:	KEYF.INTERFACE.INBOUND.MAC.CURR-INDX

KEYT Screen

This section contains the KEYT screen and descriptions of KEYT screen fields used by the SAA AS2805 Interface.

KEYT Screen 1

The KEYT screen 1is used to store public keys that are used for SCM Dynamic KEK Exchange.

```
BASE24-BASE  KEYT FILE                 ____  ____  __/__/__  __:__  01 OF 02
            FIID: ____                  INTERFACE PROCESS: ________________

 PIN BLOCK FORMAT: _ (____)               ANSI PAN FORMAT:   (_______________)
                                          PIN PAD CHARACTER: _

 PUBLIC KEY:
       ________________  ________________  ________________  ________________
       ________________  ________________  ________________  ________________
       ________________  ________________  ________________  ________________
       ________________  ________________  ________________  ________________
 SECRET KEY INDEX: __

                        KEY ENCRYPTING KEY INFORMATION
 KEY LENGTH: _   (______)      KM ID: __
      KEKs: ________________  ________________          CHECK DIGITS: ______

 KEY LENGTH: _   (______)      KM ID: __
      KEKr: ________________  ________________          CHECK DIGITS: ______

 *********************************** BASE24 ***********************************
  NEW PAGE:        FILE DESTINATION:        NEW LOGICAL NETWORK ID:
                            F12-HELP
```

DPC NUMBER OR FIID - Identifier for the financial institution associated with this record.

If this record is an institution record, the value entered in this field must match a value in the FIID field in the IDF.   The Authorisation process accesses this file after obtaining the FIID from the IDF.   Authorisation then obtains the KEY and PIN field values for the institution.

If this record is an interchange interface record, the value entered in this field must match a value in the FIID field in the ICF.   Interchange interface processes access this file after obtaining the FIID from the ICF.   The process then obtains the KEY and PIN processing values for the interchange.

If this record is a host interface record, the value entered in this field must match a value in the DPC NUMBER field in the HCF. Host interface processes access this file after obtaining the DPC NUMBER from the HCF.   The process then obtains the KEY and PIN processing values for the Data Processing Center.

Field Length: 1-4 alphanumeric characters
Required Field: Yes

Default Value: None
Data Name: KEYT.PRIKEY-ICF.FIID

INTERFACE PROCESS - The name of the interface process associated with the FIID for interchange interface records or DPC NUMBER for host interface processes.

Field Length: 16 alphanumeric characters
Required Field: Yes
Default Value: None
Data Name: KEYT.PRIKEY-ICF.SWI-PRO

PUBLIC KEY - Specifies the Interchange Partners Public key generated which is used to calculate the dynamic kek at interchange dynamic kek exchange.

Field Length: 256 hex character
Required Field: Yes
Default value: N/A
Data Name: KEYT.INTERFACE.KPR

SECRET KEY INDEX - Specifies the Interchanges Secret key index used to generate it's Public key which was sent to the Interchange partner, used to calculate the dynamic kek at interchange dynamic kek exchange.

Field Length: 2 numeric character
Required Field: Yes
Default value: 00 (index 00)
Data Name: KEYT.INTERFACE.KSS-INDEX

The remaining fields on this screen are redundant and not used by any BASE24 Interchange processing. SCM Dynamic Key exchange uses equivalent double length key fields on Screen 1 of the KEY6 file.

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Section 3

Network Management and Operations

Network management messages can be divided into two categories: those used to manage the operational status of the communications lines between BASE24 and the co-network, and those used to manage PIN and message authentication keys between BASE24 and the co-network.

The SAA AS2805 Interface supports four types of network management messages:

- Logon messages
- Echo test messages
- Logoff messages
- Dynamic key management messages

Network management messages are sent as 0800/0820 messages and require 0810/0830 messages in response.

This section describes the message flows that occur when network management messages are sent.

Monitoring Co-networks and Stations

The SAA AS2805 Interface process keeps track of the status of all co-networks and stations with which it communicates via its station table. This station table is built at initialisation and is updated continuously during processing.

Stations are marked up and down in the table depending on whether the SAA AS2805 Interface process can communicate with them. For instance, if a station fails to respond appropriately to a message, the SAA AS2805 Interface process checks the maximum retries allowed for the co-network's stations. If the maximum retries allowed is reached, the SAA AS2805 Interface process marks the station down and takes the necessary network management actions to reestablish the link with the station. Once the link is re-established, the SAA AS2805 Interface process marks the station up.

The SAA AS2805 Interface process also tracks the status of the co-networks. A co-network is marked as up if the SAA AS2805 Interface process can send messages to and receive messages from at least one of the co-network's stations. If all of the stations to a co-network are marked down, the co-network itself is marked down. Additionally, the SAA AS2805 Interface process tracks such things as whether the co-network is logged off, whether store-and-forward processing is in progress for the co-network.

The SAA AS2805 Interface process keeps track of station and co-network information at all times and logs a message when the status of the co-network changes (for example, from up to down or at the start of store-and-forward processing).

Note: The SAA AS2805 Interface process only tracks the logical ability of BASE24 to communicate with each station. If a station responds to a message, the SAA AS2805 Interface process considers it to be up. If a station fails to respond to a message the number of times specified in the MAXIMUM TIMEOUTS field on ICF screen 3, the SAA AS2805 Interface process considers the station to be down.

Also, echo test messages do not substitute for logon messages. The SAA AS2805 Interface process does not consider the co-network to be logged on and available for processing transactions until it receives a logon message.

Logon Messages

Logon messages are network management messages used to manage communications access between BASE24 and a co-network. They are sent as 0800 messages and require 0810 messages in response.

Logon messages are distinguished by a value of 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Logon Messages From BASE24

The SAA AS2805 Interface process initiates logon messages at initialisation and upon receipt of a logon command from the Network Control Supervisor (NCS) screen.

At both initialisation and/or on receipt of a logon command, the SAA AS2805 Interface process sends a logon message to the first station in its station table set-up as a send/receive type.

Logon Messages From a Co-network

As part of a logon sequence or after a period of being logged off, co-networks can send logon messages to log on to BASE24.

If the SAA AS2805 Interface process receives a logon message from the co-network after it has been logged off, the SAA AS2805 Interface process marks the co-network as logged on and begins sending echo test messages to its stations. The SAA AS2805 Interface process then returns a logon response to the co-network.

If the SAA AS2805 Interface process receives a logon message when the co-network is not logged off, it determines what action to take depending on the LCONF parameter SAA-LOGON-ACTION :

0 = send a logon response to the co-network

1 = send a logon response to the co-network, log a message and insert a new logon timer.

2 = send a logon response to the co-network, send a logoff then, set a timer to logon in 5 seconds

Note: In its responses, the SAA AS2805 Interface process always sets the Response Code (P-39) field in the BASE24 External Message to 00 to indicate an approval.

Echo Test Messages

Echo test messages are network management messages used to establish whether a station is available for message processing. They are sent as 0800 messages and require 0810 messages in response.

Echo tests messages are distinguished by a value of 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Echo Test Messages From BASE24

The SAA AS2805 Interface process initiates echo tests under the following circumstances:

1. Upon expiration of a Wait-for-Traffic timer. This processing is described in section 2.
2. When an echo test has timed out but the MAXIMUM TIMEOUTS as determined by the value on screen 2 of the ICF has not yet been equalled.

Echo Test Messages From a Co-network

Co-networks can send echo test messages at any time. The SAA AS2805 Interface process always returns an echo test response.

Note: In its responses, the SAA AS2805 Interface process sets the Response Code (P-39) field in the BASE24 External Message to 00 to indicate an approval unless the link has been marked down in which case it sets the response code to 12.

Logoff Messages

Logoff messages are network management messages used to stop communications between BASE24 and a co-network. They are sent as 0820 messages and require 0830 messages in response.

Logoff messages are distinguished by a value of 002 in the Network Management Information Code (S-70) field of the BASE24 External Message. They can be sent by either BASE24 or a co-network. In either situation, once a co-network is logged off, BASE24 does not send echo test messages or non-network management messages to any of the co-network’s stations. Any non-network management messages that must be forwarded to the co-network are placed in the SAF instead.

BASE24 only attempts to reopen the link to a co-network that is logged off if a logon command is received from the Network Control Supervisor (NCS) screen. The co-network, on the other hand, can reopen the link at any time by sending a logon message.

Note: When a co-network is logged off, any transactions in progress are allowed to complete.

Logoff Messages From BASE24

BASE24 sends logoff messages to a co-network as a result of one of the following:

- On receipt of an XPNET stop message
- On receipt of a logon message when the link is already marked up and the LCONF parameter SAA-LOGON-ACTION is set to 2
- On receipt of an echo test which has a non-approved response code
- On receipt of a logon message where proof of endpoints do not validate
- On receipt of a Key Exchange Response message which has a non-approved response code
- On receipt of a Key Exchange Response message where the KVCs can not be validated
- On receipt of an NCS logoff command
- When a Key Exchange times out
- When the number of bad MACs equal the value specified in BAD MACS on screen 12 of the ICF

BASE24 performs no action if it receives a logoff response from the co-network.

Logoff Messages From a Co-network

When a co-network must log off because of a scheduled down or unavailable period, it can transmit a logoff message from any of its stations.

Upon receiving a logoff message, the SAA AS2805 Interface process marks the co-network down and logged off and returns a logoff response to the co-network.

Note: In its responses, the SAA AS2805 Interface process always sets the Response Code (P-39) field in the BASE24 External Message to 00 to indicate an approval.

Interactive Network Management Message Flows

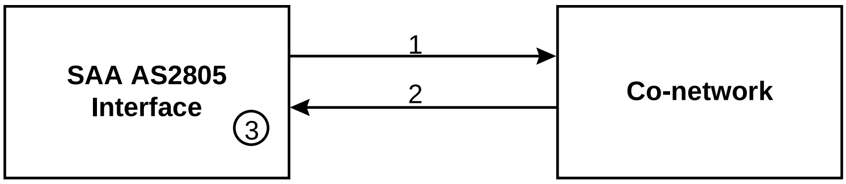
The remainder of this section contains examples of network management and dynamic key management message flows and describes the processing performed by BASE24 for each.

Logon Message Flow

The following pages provide examples of logon sequences and describe the processing performed by the SAA AS2805 Interface process.

BASE24-Initiated Logon

In this example, the SAA AS2805 Interface process logs on to the co-network.



1. The SAA AS2805 Interface process sends a logon message to the co-network on the first send/receive station in its station table.
    a. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.
    b. If the interchange is configured for Dynamic Key Exchange with Proof of Endpoints enabled the security box is called to generate a proof of endpoint and this is stored in Private Data field 48. See Security Management for further details.
    c. In addition, the SAA AS2805 Interface sets an Extended Network Management Message timer.
2. The co-network returns a logon response. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.
3. The SAA AS2805 Interface process receives a logon response and checks the Response Code (P-39) field in the message.
    a. If the Response Code field contains zeros, indicating an approval, the SAA AS2805 Interface process marks the co-network as up and logged on.

        1) If the interchange is configured for Dynamic Key Exchange with Proof of Endpoints enabled it validates the proof of endpoint data

returned in field 48. If this is invalid it sends a logoff message and inserts another logon timer to restart the logon sequence. See the section on Security Management for further details.

2) If the co-network is logged on and/or is configured not to send a logon it marks the station up.

3) If the interchange is configured for Dynamic KEKs, then insert a KEK Change timer to expire immediately.

4) If the interchange is configured for Dynamic Keys, then insert a Key Change timer to expire immediately.

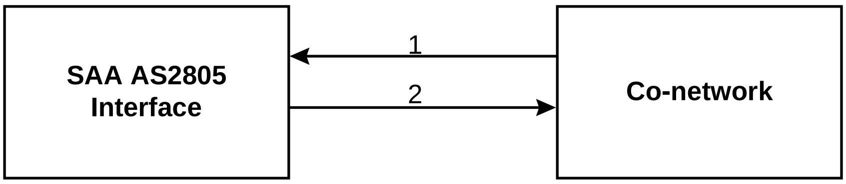
5) The SAA AS2805 Interface also sets a Wait-for-Traffic timer on all send/receive stations.

6) If the interchange is not configured for Dynamic Keys or Dynamic KEKs then the logon sequence is considered complete so mark the link up and proceed to recovery processing.

b. If the Response Code field does not contain zeros, the SAA AS2805 Interface process logs a message, and inserts a new logon timer to restart the logon sequence.

Co-network-Initiated Logon

In this example, the co-network logs on to BASE24.



1. The co-network sends a logon message to reestablish communications. The logon message is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.
2. The SAA AS2805 Interface process receives the message and performs the following steps:
   a. If the interchange is configured for Dynamic Keys with Proof of Endpoint enabled it validates Private Data Field 48. See the Security Management section for further details. If the Proof of Endpoint is found to be invalid it logs a message and sets the Response Code field in the external message to 12.
   b. Sends a logon response to the co-network. The logon response is identified by the value 001 in the Network Management Information Code (S-70) field of the BASE24 External Message.
   c. Marks the co-network as logged on.
   d. If the interchange has not sent a logon , it initiates a logon.

   e. If the interchange is logged on, it marks the station up.

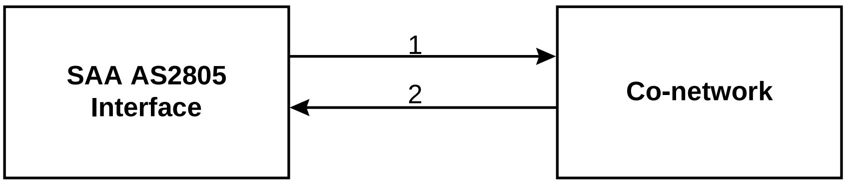
3. If the interchange is configured for Dynamic KEKs, it inserts a KEKs Change timer to expire immediately.
4. If the interchange is configured for Dynamic Keys, it inserts a Key Change timer to expire immediately.
5. If the interchange is not configured for Dynamic Keys or Dynamic KEKs then the logon sequence is considered complete so mark the link up and proceed to recovery processing.
6. Sets a Wait-for-Traffic timer for each send/receive station defined for the co-network.

Logoff Message Flow

The following pages provide examples of logoff sequences and describe the processing performed by the SAA AS2805 Interface process.

BASE24-Initiated Logoff

In this example, the SAA AS2805 Interface process logs off from the co-network.



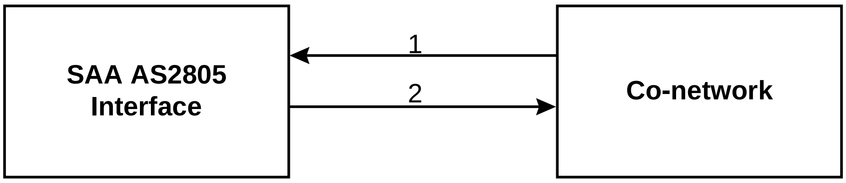
1. The SAA AS2805 Interface process receives a logoff command from the Network Control Supervisor (NCS) screen.:
   a. The SAA AS2805 interface clears the following timers running for the co-network and its stations:
      1) Network Management Message Timer
      2) Wait for traffic timer
      3) Logon timer
      4) Key Change timers

      All other timers are left running to ensure that transactions in progress complete.
   b. The SAA AS2805 Interface sends a logoff message to the co-network. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.
   c. The SAA AS2805 Interface process marks the co-network and BASE24 as logged off and the link set to down.

2. The co-network returns a logoff response (from one or each station). The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Co-network-Initiated Logoff

In this example, the co-network logs off from BASE24:

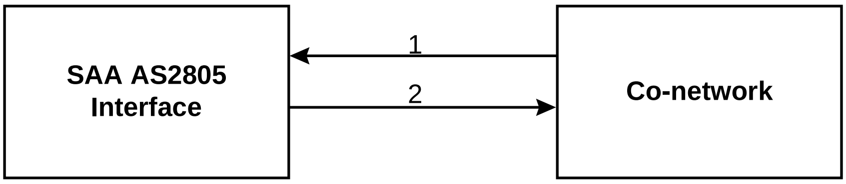


1. The co-network sends a logoff message to BASE24. The logoff message is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.
2. The SAA AS2805 Interface process receives the logoff message and performs the following steps:
   a. Marks the co-network as logged off.
   b. Clears the following timers running for the co-network and its stations:
      1) Network Management Message timers
      2) Wait-for-Traffic timers
      3) Logon timer
      4) Key Change timers

      All other timers are left running to ensure that current transactions in progress complete.
3. Sends a logoff response to the co-network. The logoff response is identified by the value 002 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Echo Test Message Flow

The following example describes how the SAA AS2805 Interface process handles echo test messages from the co-network. In this case, the co-network is logged on, but BASE24 considers the station to be down.



1. The co-network sends an echo test message from Station A. The echo test message is identified by the value 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.

2. The SAA AS2805 Interface process receives the message and returns an echo test response. The echo test response is identified by the value 301 in the Network Management Information Code (S-70) field of the BASE24 External Message.

Security Management

The SAA AS2805 interface supports a number of security management schemes which it uses to encrypt and decrypt PIN blocks, and perform Message Authentication (MAC'ing). The Security management schemes supported are: Static keys, Tape keys, Dynamic keys and Dynamic KEKs. Furthermore Static and Dynamic keys and KEKs can be used to support triple DES, as well as single DES, encryption.

The "SECURE-DEV-TYP" param in the LCONF determines the security device type used and contains one of the following values:

ASM	Atalla
ESM	Eracom Type 1 (Mark II)
RSM	Racal
SCM	Eracom Type 2 (APCA/SCM)
TSS	Transaction Security Services
N/A	No security box

The "DES-TRIPLE-SINGLE" param in the LCONF determines whether double length PIN keys sent and received by the co-network will be translated into and from single length PINs within BASE24.

The "SUBTYPE" field associated with the "SECURE-DEVn" assign in the LCONF determines whether double length keys and triple DES functions are to be utilised in calls to the SCM or ESM. Values of S2 and E2 respectively in subtype indicate these functions are to be used.

Static Keys

This scheme is configured by setting the "Key Exchange Type" on screen 12 of the ICF to "1" (Static Keys).

The Static Keys scheme involves deriving a pair of initial PIN and MAC keys externally to BASE24 which have been encrypted under the relevant master key of the security box. These keys are loaded into screen 2 of the KEYF (screen 4 for SCM) and are used by the SAA AS2805 interface until they are again manually derived and re-entered. A current index is also set on screen 2 of the KEYF (screen 4 for SCM) to indicate which of the pair of keys will be used.

On initialisation, the SAA AS2805 interface loads these keys and index into an internal table.

There is no session key exchange performed online between the two partners.

On financial messages, the index of the current key set being used (1 or 2) is sent in field 53 (Security Data) of the external message. This index is changed and the other set of keys used under the following conditions;

- Number of consecutive bad pins as defined on screen 12 of the ICFE

- Number of consecutive bad macs as defined on screen 12 of the ICFE
- Number of financial transactions as defined on screen 12 of the ICFE
-

Tape Keys

This scheme is configured by setting the “Key Exchange Type” on screen 12 of the ICF to “2” (Tape Keys) and the “Tape Key File Name” to the file that contains the 256 pairs of keys. The date contained in the first record of the file must also be primed in the “TAPE DATE” field for verification purposes.

The Tape Keys scheme involves creating a tape of 256 sets of PIN and MAC keys externally to BASE24 which have been encrypted under the relevant master key of the security box. This tape is exchanged between the two partners where the keys are encrypted under a transport key.

On initialisation, the SAA AS2805 interface loads these 256 sets of keys into an internal table and generates a random index between 1 and 256 which it also stores in an internal table.

There is no session key exchange performed online between the two partners at logon.

On financial messages, the index of the current key set being used (1 thru 256) is sent in field 53 (Security Data) of the external message. This index can be changed by either partner independently of each other under the following conditions;

- Number of consecutive bad pins as defined on screen 12 of the ICFE
- Number of consecutive bad macs as defined on screen 12 of the ICFE
- Number of financial transactions as defined on screen 12 of the ICFE

Dynamic Keys

This scheme is configured by setting the “Key Exchange Type” on screen 12 of the ICF to “3” (Dynamic Keys).

ESM:

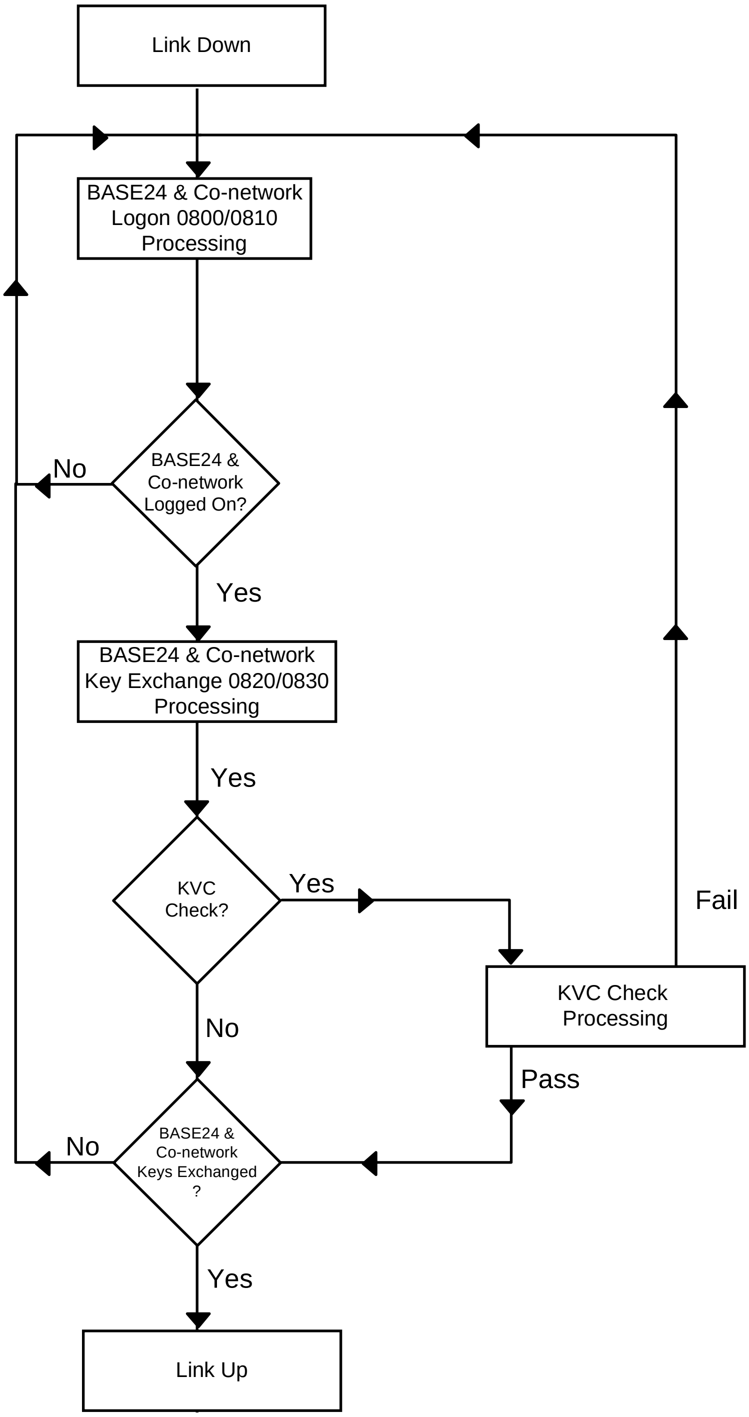
The ESM Dynamic Keys scheme refers to the process of exchanging PIN and MAC working keys encrypted under a KIS (Interchange Send Key) online between the two partners whilst the link is established. These keys are then periodically changed online. An initial set of KISn (Interchange key for sending messages) and KIRn (Interchange key for receiving messages) must first be loaded into the security box and the associated institution index entered in the inbound and outbound pin and mac fields on screen 2 of the KEYF for single DES or KEKs and KEKr fields on screen 1 of the KEY6 for triple DES. More information regarding the setting up of these keys can be obtained by referring to the ***BASE24 ERACOM Security Module (ESM) Reference Manual***.

The SAA AS2805 interface supports KVC checking during its logon sequence.

On initialisation, the SAA AS2805 interface loads the institution index for KISn and KIRn into an internal table. It also generates a random number between 1 and 256 which it then adds to the “Number of Trans Key Exchange” field on screen 12 of the ICFE to determine the number of transactions that need to occur before a key exchange is performed.

Logon

The logon sequence involves both partners exchanging 0800/0810 logon messages followed by 0820/0830 key exchange messages.



Key Exchange

A key exchange is performed under the following conditions:

- After a successful logon
- Number of consecutive bad PINs as defined on screen 12 of the ICFE
- Number of consecutive bad MACs as defined on screen 12 of the ICFE
- Number of financial transactions as calculated during initialisation

Either partner can initiate a key exchange. The SAA AS2805 interface formats an 0820 Key Exchange Advice by placing a call to the security box for each of the keys to be sent (PIN, MAC and optionally DATA) to perform the KEY-

GENERATE function. This function is performed in order to generate random outbound PIN, MAC and DATA keys respectively. It returns one set of these keys encrypted under KISn and another set encrypted under the KM of the security box. The KISn set of keys is sent to the co-partner in field 48 of the key exchange advice message. The KM set of keys is stored in an internal table for later use to perform MAC generation and PIN encryption.

When the SAA AS2805 interface receives a key exchange advice it places a call to the security box for each of the keys received (PIN, MAC and optionally DATA) to perform the KEY-TRANSLATE function. This function receives as input the inbound MAC, PIN and optionally DATA keys from field 48 of the key exchange advice and decrypts it under KIRn and then re-encrypts them under the KM of the security box. The KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

If the DATA key is not to be included in the Key Exchange Advice to or from the co-network, it can be turned off by setting the field “DATA KEY” to “N” on screen 12 of the ICFE. While the SAA AS2805 interface does not use the DATA key for any purpose, it may be required to support the addition of this field in the external message from partners.

If DES-TRIPLE-SINGLE set to “Y” then function II-KEY-GEN (EE0402) is first performed to generate a single length transport key (for inbound double to single length PIN block translation) and then to generate double length session keys, which are exchanged in field 48 of the key exchange advice, function II-KEY-RCV (EE0403) performed to translate the exchanged keys and the KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

KVC Processing

The MAC, PIN and DATA keys can be validated by setting “KVC Processing” on screen 12 of the ICFE to “Y”. When the SAA AS2805 interface formats an 0820 key exchange advice it will place a call to the security box to perform the KEY-GENERATE function. This function not only generates a random working key encrypted under KISn and a KM version but also additionally produces a key verification code of the working key. This code is stored in an internal table for later validation on the key exchange response message.

When the SAA AS2805 interface receives an 0820 key exchange advice a call is placed to the security box to perform the KEY-TRANSLATE function. This function receives as input the MAC, PIN and optionally DATA keys received in field 48 of the key exchange advice as mentioned in the above section on key exchange. In addition to deriving a KM version of the working key received it also generates a key verification code. The SAA AS2805 interface then returns these key verification code to the co-partner in the key exchange acknowledgment.

When the SAA AS2805 interface receives an 0830 key exchange acknowledgment, it compares the KVC values returned in field 48 with those previously stored in the internal table. If they are not the same the key exchange sequence is aborted and a logoff is sent to the co-network. A timer is then set to restart the logon sequence.

Financial Messages

When formatting financial messages the SAA AS2805 interface encrypts the PIN block with the stored outbound single or double length PIN session key previously obtained in the key exchange. The data is MAC'ed using the outbound single or double length MAC session key also previously obtained.

When receiving financial messages the SAA AS2805 interface sends a single DES PIN block as it was received from the co-network to AUTH/RTAU along with the stored inbound single length PIN session key previously obtained in the key exchange. For financial messages containing a triple DES PIN block encypted under a double length session key, the PIN block is translated to a single DES PIN block using a special single length transport key generated at logon and sent to AUTH/RTAU along with the transport key. The MAC is verified using the inbound single or double length MAC session key also previously obtained.

SCM:

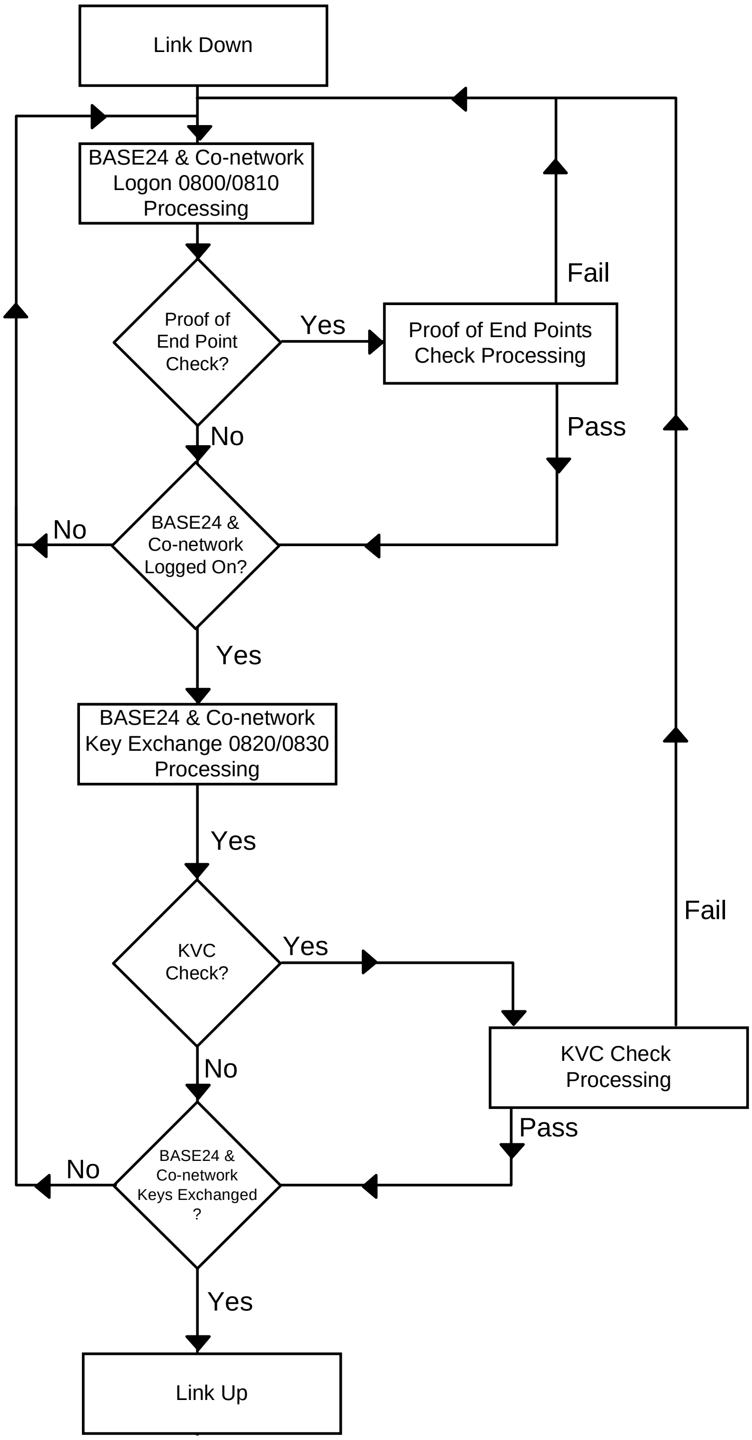
The SCM Dynamic Keys scheme refers to the process of exchanging PIN and MAC session keys encrypted under a KEK online between the two partners whilst the link is established. These keys are then periodically changed online. An initial pair of KEKs (KEK send key for sending messages) and KEKr (KEK receive key for receiving messages) keys must first be set up on screen 1 of the KEYF. This can be done using the ESMCOM utility located on the BA60ESM subvolume. More information regarding the setting up of these keys can be obtained by referring to the ***BASE24 ERACOM Security Control Module (SCM) Reference Manual***.

The SAA AS2805 interface supports proof of end points and/or KVC checking during its logon sequence.

On initialisation, the SAA AS2805 interface loads the KEKs and KEKr into an internal table. It also generates a random number between 1 and 256 which it then adds to the "Number of Trans Key Exchange" field on screen 12 of the ICFE to determine the number of transactions that need to occur before a key exchange is performed.

Logon/Proof of Endpoints

The logon sequence involves both partners exchanging 0800/0810 logon messages followed by 0820/0830 key exchange messages.



During logon, the KEKs and KEKr can be validated between the two partners online by setting the "Proof of Endpoint" flag on screen 12 of the ICFE to "Y". This will cause the SAA AS2805 interface to perform a call to the security box to perform the NODEPROOF function while formatting a logon request. This function generates two variants of a single random length key. Variant 7 is sent to the co-partner in field 48 of the external message. Variant 8 is stored in an internal table and is compared against the value returned in field 48 on the logon acknowledgment. If the values are not the same the logon cycle will be restarted.

Similarly, when the co-network sends a logon to the SAA AS2805 interface a call is made to the security box to perform a NODERESP function. This function decrypts the random key sent in the co-networks logon under variant 7 of the KEKr and re-encrypts under variant 8. The result is then returned in field 48 of the logon acknowledgment.

Once both partners have successfully exchanged logons the SAA AS2805 interface will initiate a Key Exchange.

If proof of endpoints is not required the “Proof of Endpoint” flag on screen 12 of the ICF must be set to “N”.

Key Exchange

A key exchange is performed under the following conditions:

- After a successful logon
- Number of consecutive bad PINs as defined on screen 12 of the ICFE
- Number of consecutive bad MACs as defined on screen 12 of the ICFE
- Number of financial transactions as calculated during initialisation

Either partner can initiate a key exchange. The SAA AS2805 interface formats an 0820 Key Exchange Advice by placing a call to the security box to perform the NODEKEYGEN function. This function generates random outbound PIN, MAC and DATA keys. It returns one set of these keys encrypted under KEKs and another set encrypted under the KM of the security box. The KEKs set of keys is sent to the co-partner in field 48 of the key exchange advice message. The KM set of keys is stored in an internal table for later use to perform MAC generation and PIN encryption.

When the SAA AS2805 interface receives a key exchange advice it places a call to the security box to perform the RTMK function. This function receives as input the inbound MAC, PIN and DATA keys from field 48 of the key exchange advice and decrypts it under KEKr and then re-encrypts them under the KM of the security box. The KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

If the DATA key is not to be included in the Key Exchange Advice to or from the co-network it can be turned off by setting the field “DATA KEY” to “N” on screen 12 of the ICF. While the SAA AS2805 interface does not use the DATA key for any purpose, it may be required to support the addition of this field in the external message from partners.

If DES-TRIPLE-SINGLE set to “Y” then function NODEKEYGEN is first performed to generate a single length transport key (for inbound double to single length PIN block translation) and then to generate double length session keys, which are exchanged in field 48 of the key exchange advice, function RTMK performed to translate the exchanged keys and the KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

KVC Processing

The MAC, PIN and DATA keys can be validated by setting "KVC Processing" on screen 12 of the ICF to "Y". When the SAA AS2805 interface formats an 0820 key exchange advice it will place a call to the security box to perform the KVCREQSIN function. This function receives as input the MAC, PIN and DATA keys encrypted under KEKs and returns a key verification code. This code is stored in an internal table for later validation on the key exchange response message.

When the SAA AS2805 interface receives an 0820 key exchange advice a call is placed to the security box to perform the KVCREQSIN function. This function receives as input the MAC, PIN and DATA keys received in field 48 of the key exchange advice which were previously decrypted using KEKr and re-encrypted using the KM of the security box. It outputs a set of key verification codes which the SAA AS2805 interface then returns to the co-partner in the key exchange acknowledgment.

When the SAA AS2805 interface receives an 0830 key exchange acknowledgment it compares the KVC values returned in field 48 with those previously stored in the internal table. If they are not the same the key exchange sequence is aborted and a logoff is sent to the co-network. A timer is then started to restart the logon sequence.

Financial Messages

When formatting financial messages the SAA AS2805 interface encrypts the PIN block with the stored PIN outbound single or double length session key previously obtained in the key exchange. The data is MAC'ed using the MAC outbound single or double length session key also previously obtained.

When receiving financial messages the SAA AS2805 interface sends a single DES PIN block as it was received from the co-network to AUTH/RTAU along with the stored PIN inbound session key previously obtained in the key exchange. For financial messages containing a triple DES PIN block encypted under a double length session key, the PIN block is translated to a single DES PIN block using a special single length transport key generated at logon and sent to AUTH/RTAU along with the transport key. The MAC is verified using the MAC inbound session key also previously obtained.

Dynamic KEKs

This scheme is configured by setting the "Key Exchange Type" on screen 12 of the ICFE to "4" (Dynamic KEKs). The "SECURE-DEV-TYP" parameter in the LCONF must be set to "SCM".

The Dynamic KEKs scheme refers to the process of generating and exchanging KEKs prior to exchanging PIN and MAC session keys encrypted under this generated KEK online between the two partners whilst the link is established. Both the KEK and the session keys are then periodically changed online. A

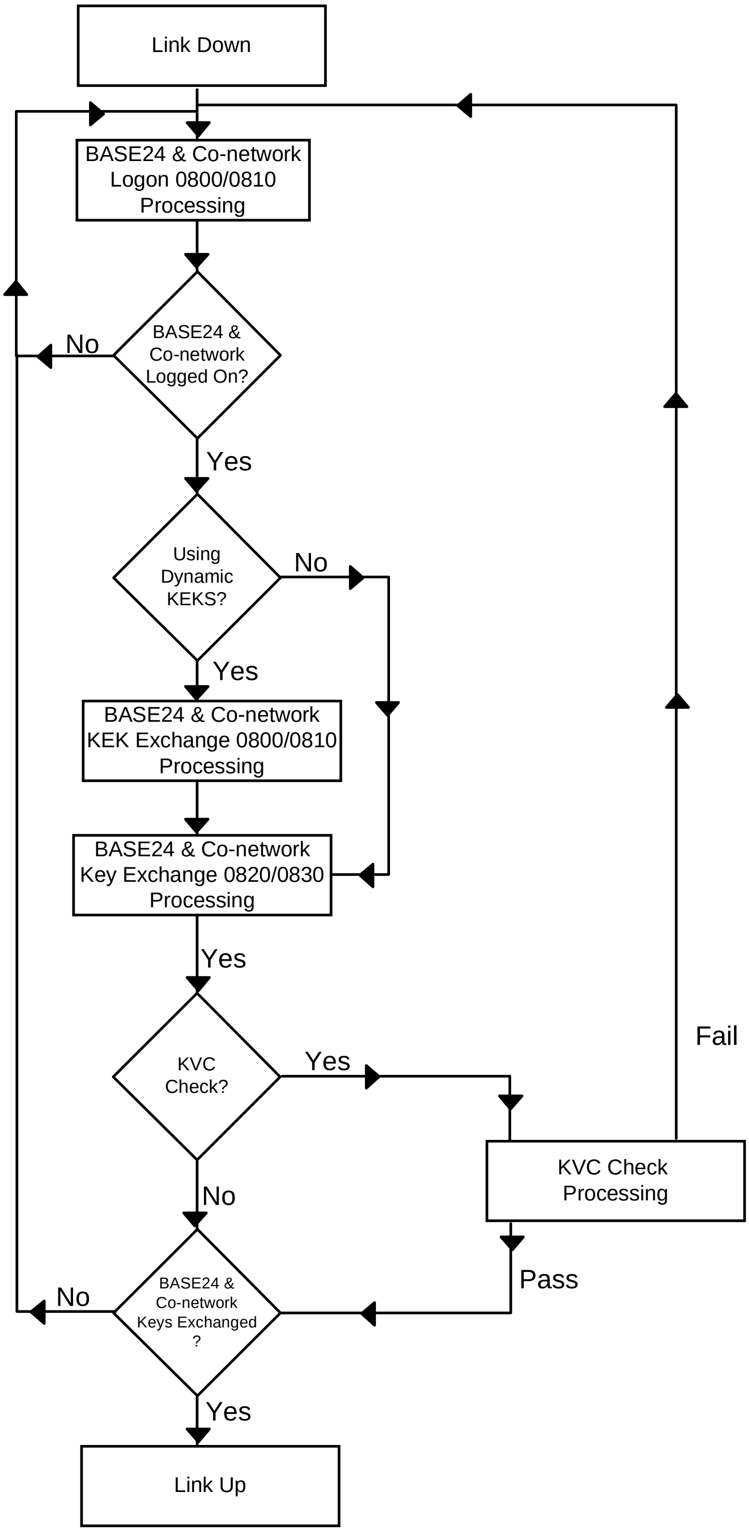
Public Key must first be exchanged by both partners and injected in to the KEYT. This can be done using the ESMCOM utility located on the BA60ESM subvolume. More information regarding the setting up of a Public Key can be obtained by referring to the ***BASE24 ERACOM Security Control Module (SCM) Reference Manual***.

The SAA AS2805 interface does not support proof of end points when it is configured for Dynamic KEKs. However, it does support KVC checking on key exchange messages.

On initialisation, the SAA AS2805 interface loads the Public Key into an internal table. It also generates a random number between 1 and 256 which it then adds to the “Number of Trans Key Exchange” field on screen 12 of the ICFE to determine the number of transactions that need to occur before a key exchange is performed.

Logon

The logon sequence involves both partners exchanging 0800/0810 logon messages followed by 0800/0810 KEK Exchange messages and lastly 0820/0830 key exchange messages.



Once both partners have successfully exchanged logons the SAA AS2805 interface will initiate a KEK Exchange.

KEK Exchange

A KEK Exchange can be initiated for the following reasons:

- Successful exchange of logons between both partners
- As a result of an operator command (KEKCHANGE)

The SAA AS2805 interface will format a KEK exchange by placing a call to the security box to perform the NODEKEKGEN function. This function receives as input the Public Key previously stored in an internal table and generates a random KEK Send key and KVC encrypted under the KM of the security box and the public key. The KEK send key and KVC are stored in an internal table and passed to the co-network in field 48 of the KEK Exchange Advice message. The KEKs is stored in an internal table for later use in key exchange processing.

On receipt of a KEK exchange advice message the SAA AS2805 interface places a call to the security box to perform the NODEKEKTRANS function. This function receives as input the Public Key previously stored in an internal table and the KEKs and KVC extracted from field 48 of the KEK exchange advice. The KEK is decrypted using the public key and re-encrypted using the KM of the security box to derive the session KEKr. The KEKr is stored in an internal table for later use in key exchange processing.

On successful KEK Exchange from both partners as a result of a logon a Key Exchange is then initiated. For operator initiated KEK exchanges the new KEK will not take effect until the next key exchange is required. See the following paragraph on Key Exchange for the conditions.

Key Exchange

A key exchange is performed under the following conditions:

- After a successful logon/KEK exchange
- Number of consecutive bad PINs as defined on screen 12 of the ICFE
- Number of consecutive bad MACs as defined on screen 12 of the ICFE
- Number of financial transactions as calculated during initialisation

Either partner can initiate a key exchange. The SAA AS2805 interface formats an 0820 Key Exchange Advice by placing a call to the security box to perform the NODEKEYGEN function. This function generates random outbound PIN, MAC and DATA keys. It returns one set of these keys encrypted under KEKs and another set encrypted under the KM of the security box. The KEKs set of keys is sent to the co-partner in field 48 of the key exchange advice message. The KM set of keys is stored in an internal table for later use to perform MAC generation and PIN encryption.

When the SAA AS2805 interface receives a key exchange advice it places a call to the security box to perform the RTMK function. This function receives as input the inbound MAC, PIN and DATA keys from field 48 of the key exchange advice, decrypts it under KEKr and then re-encrypts them under the KM of the security box. The KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

If the DATA key is not to be included in the Key Exchange Advice to or from the co-network it can be turned off by setting the field “DATA KEY” to “N” on screen 12 of the ICF. While the SAA AS2805 interface does not use the DATA key for any purpose it may be required to support the addition of this field in the external message from partners.

If DES-TRIPLE-SINGLE set to “Y” then function NODEKEYGEN is first performed to generate a single length transport key (for inbound double to single length PIN block translation) and then to generate double length session keys, which are exchanged in field 48 of the key exchange advice, function RTMK performed to translate the exchanged keys and the KM set of keys is stored in an internal table for later use to perform MAC verification and PIN decryption.

KVC Processing

The MAC, PIN and DATA keys can be validated by setting “KVC Processing” on screen 12 of the ICF to “Y”. When the SAA AS2805 interface formats an 0820 key exchange advice it will place a call to the security box to perform the KVCREQSIN function. This function receives as input the MAC, PIN and DATA keys encrypted under KEKs and returns a key verification code. This code is stored in an internal table for later validation on the key exchange response message.

When the SAA AS2805 interface receives an 0820 key exchange advice a call is placed to the security box to perform the KVCREQSIN function. This function receives as input the MAC, PIN and DATA keys received in field 48 of the key exchange advice which were previously decrypted using KEKr, and re-encrypted using the KM of the security box. It outputs a set of key verification codes which the SAA AS2805 interface then returns to the co-partner in the key exchange acknowledgment.

When the SAA AS2805 interface receives an 0830 key exchange acknowledgment it compares the KVC values returned in field 48 with those previously stored in the internal table. If they are not the same the key exchange sequence is aborted and a logoff is sent to the co-network. A timer is then started to restart the logon sequence.

Financial Messages

When formatting financial messages the SAA AS2805 interface encrypts the PIN block with the stored outbound single or double length PIN session key previously obtained in the key exchange. The data is MAC’ed using the outbound single or double length MAC session key also previously obtained.

When receiving financial messages the SAA AS2805 interface sends a single DES PIN block as it was received from the co-network to AUTH/RTAU along with the stored inbound PIN session key previously obtained in the key exchange.

For financial messages containing a triple DES PIN block encypted under a double length session key, the PIN block is translated to a single DES PIN block using a special single length transport key generated at logon and sent to AUTH/RTAU along with the transport key. The MAC is verified using the inbound MAC session key also previously obtained.

Section 4

Store-and-Forward Processing

Store-and-forward processing allows for storing transactions that BASE24 has processed when a co-network is down or unavailable.

The SAA AS2805 Interface process keeps track of each station and co-network. If a message bound for a co-network can not be transmitted, the SAA AS2805 Interface process places the message in the Store-and-Forward File (SAF) for later transmission. When communications with the co-network resume, the messages in the SAF are sent to the co-network.

This section describes store-and-forward processing. It includes information on what messages can be placed in the SAF, when messages are placed in the SAF and when messages from the SAF are sent to the co-network. It also contains messages flows that illustrate store-and-forward processing.

The Store-and-Forward File (SAF)

The Store-and-Forward File (SAF) temporarily stores messages bound for a co-network until those messages can be sent to the intended co-network. The SAF is a relative file and the messages are processed as first-in, first-out, as described below.

The store-and-forward cycle used by the SAA AS2805 Interface is single-threaded, so the oldest SAF records are sent to the co-network first. Records arrive at the co-network in the same order in which they were added to the SAF.

Messages in the SAF are stored in the format that they are to be sent to the co-network.

Messages Placed in the SAF

The SAA AS2805 Interface process places the following message types in the SAF:

0120 — Pre-Authorisation Advice
0220 — Financial Transaction Advice
0221 — Financial Transaction Advice Repeat
0420 — Reversal Advice
0520 — Acquirer Reconciliation Advice

0120, 0220, 0420 and 0520 messages are placed in the SAF if the intended co-network is marked down at the time the message is to be sent. If the SAA AS2805 Interface process receives one of these message to be sent to a co-network, but discovers the co-network is marked down, the SAA AS2805 Interface process places the message in the SAF without attempting to send it to the co-network.

If the SAA AS2805 Interface process sends a 0120, 0220 0420 or 0520 message to a co-network, but the message times out, the SAA AS2805 Interface process changes the 0120 message to a 0121, 0220 to a 0221, 0420 to a 0421 or 0520 to a 0521 message and places it in the SAF.

Messages Not Placed in the SAF

The SAA AS2805 Interface process never places the following message types in the SAF:

0100 — Authorisation Request
0110 — Authorisation Request Response
0130 — Authorisation Advice Response
0200 — Financial Transaction Request
0210 — Financial Transaction Request Response
0230 — Financial Transaction Advice Response
0430 — Reversal Advice Response
0530 — Acquirer Reconciliation Advice Response

0100, 0110, 0200, 0210 messages are interactive messages and their timing is crucial. If they are not processed within a given period of time, the transactions they represent are declined. Therefore, these messages are not placed in the SAF.

0130, 0230, 0430, 0530 messages are acknowledgments or responses to other messages. Here again, timing is crucial. If a co-network does not receive a required acknowledgment or response within a given time, the co-network assumes that BASE24 never received the message and might then send it again as a duplicate.

Deleting Messages from the SAF

Records are deleted from the SAF under the following circumstances:

- When the SAA AS2805 Interface process receives a response message for the SAF record that was forwarded to it.
- When the SAA AS2805 Interface process determines that the SAF record can not be processed by the co-network.

  If the message is returned as failed from the co-network or the same SAF message has timed out the number of times specified in the MAX SAF RETRY field on ICF screen 3, the SAA AS2805 Interface process assumes the co-network can not process the message. In this case, the SAA AS2805 Interface process writes the message to its log and deletes it from the SAF.

Forwarding SAF Messages to a Co-network

The SAA AS2805 Interface process checks for the presence of store-and-forward messages at the following times:

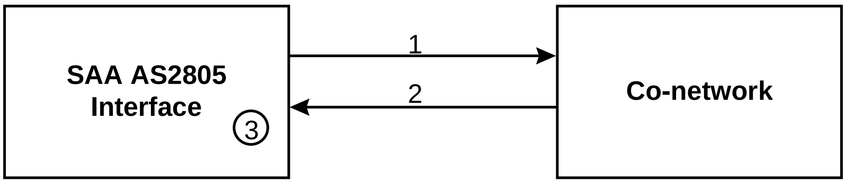
- Each time we receive a completion or message be it from XPNET or from the co-network.
- Every time a timer expires.

Store-and-Forward Message Processing

The general processing flow for SAF messages is the same as for normal messages. To the co-network, the messages appear to be happening as they are sent—they are indistinguishable from normal messages. Internally, the SAA AS2805 Interface uses different timers that identify the messages as SAF records. If the specified timer expires, these messages are left on the SAF to be resent during store-and-forward processing.

Normal Store-and-Forward Messages

The following diagram illustrates the message exchanges expected by BASE24 when store-and-forward messages are sent to a co-network.



1. The SAA AS2805 Interface process sends a store-and-forward message to the co-network. To do this, it performs the following steps:
   a. Reads a record from the SAF.
   b. The SAA AS2805 Interface process selects the oldest record in the file for the co-network. The record is a 0120, 0220, 0420, 0520 message that is already in external message format.
   c. Sets the message type in the external message to 0121 for 0120, 0221 for 0220, 0421 for 0420 and 0521 for 0520.
   d. Sets a flag to indicate that store-and-forward processing is in progress.
   e. Locates an available send/receive station and sends the message.
   f. Sets a Store-and-Forward Message timer for the message.
2. The co-network processes the 0120, 0121, 0220, 0221, 0420, 0421, 0520 or 0521 message and returns a 0130, 0230, 0430 or 0530 message.
3. Upon receipt of a 0130, 0230, 0430 or 0530 message the SAA AS2805 Interface process performs the following steps:
   a. Locates and deletes the corresponding record in the SAF.
   b. Resets the SAF retry counter.
   c. Resets a flag to indicate that store-and-forward processing is no longer in progress.
   d. If the response code is not set to approved or no action, log a message then update the ILF record with a sub-state indicating that an invalid acknowledgment has been received.

Section 5

Transaction Handling

This section describes how the SAA AS2805 Interface process handles various types of transactions. It describes the basic flow of transaction processing and provides transaction flow diagrams illustrating how messages to and from the co-network are handled.

Note: Transaction flow diagrams for cutover messages are provided in section 6.

Introduction

The following transaction flows are described in this section. In most cases, these flows are described twice—once when the message flow is initiated by BASE24 and once when the message flow is initiated by the co-network.

- Requests
- Reversals
- Timeouts
- Force Posts
- Logon
- Logoff
- Echo Test

Logging to the ILF

The SAA AS2805 Interface logs the following types of messages to the Interchange Log File (ILF):

- Requests containing error conditions
- Responses
- Reversals
- Force posts
- Reconciliation totals

Whenever a reversal occurs, the SAA AS2805 Interface updates the original ILF record if present, otherwise the reversal is added to the ILF. If MACs are used, the SAA AS2805 Interface attempts to verify the MAC.

Tracing Transactions

In order to trace transactions, the SAA AS2805 Interface uses the STAN (P-11) from the external message. The STAN is assigned by a message initiator to uniquely identify a transaction.

If SAA is configured to use unique STANs, every message between the two partners will have a different and unique STAN per business day. When BASE24 initiates a transaction, or a reversal to a transaction, the STAN is determined by the SAA AS2805 interface, which keeps in an internal table of the last STAN used.

Reversals will contain the STAN of the original transaction it is reversing in the ORIG DATA ELEMENTS (S-90) from the external message.

If SAA is configured to not use unique STANs, reversals will have the same STAN as the original request. When BASE24 initiates a transaction, or a reversal to a transaction, the terminal sequence number is used as the STAN. This number remains unchanged for all messages throughout the life of the transaction.

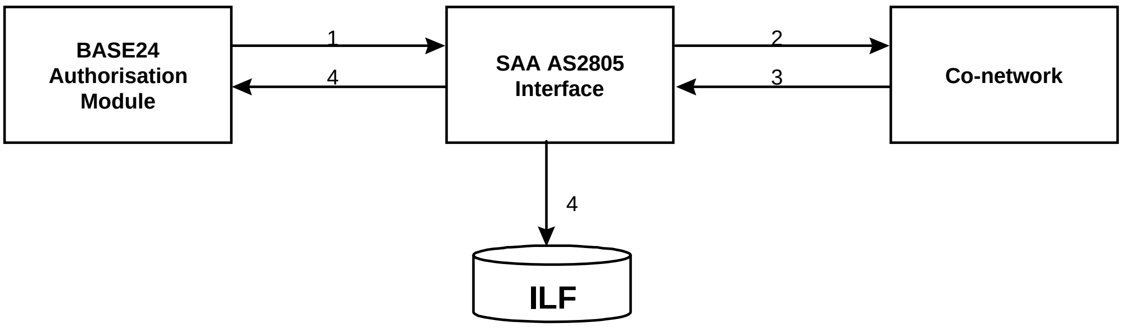
Message Flows to the Co-network

The following pages describe the message flows for the following transactions outbound from BASE24 to the co-network.

- Requests
- Reversals
- Force Posts
- Timeouts

Request Message to the Co-network

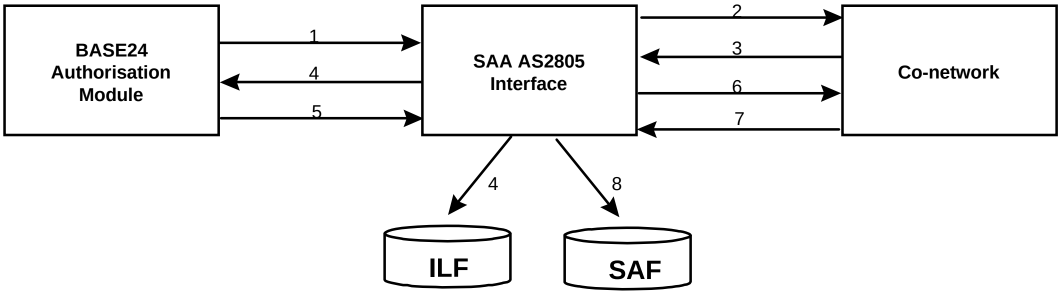
The diagram below illustrates the message flow when a transaction is initiated by the BASE24 network and must be sent to the co-network for authorisation. The steps in the diagram are described below.



1. A BASE24 Authorisation process sends the request to the SAA AS2805 Interface in the 0200 internal message format.
2. The SAA AS2805 Interface reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorisation and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authoriser. After the transaction has been approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the SAA AS2805 Interface.
4. The SAA AS2805 Interface deletes the outbound request timer, formats a 0210 internal response message, sends the response to the BASE24 Authorisation process to be delivered to the ATM or POS device and logs a copy of the transaction to the Interchange Log File (ILF).

Reversal to the Co-network

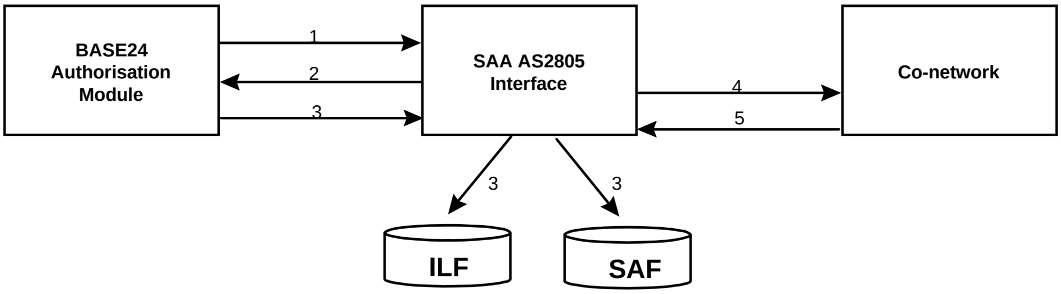
When an ATM or POS device connected to the local BASE24 network fails to complete a transaction as authorised, a reversal message must be sent to the co-network. The diagram below illustrates message routing when a reversal is sent to the co-network. The steps in the diagram are described below.



1. A BASE24 Authorisation process sends the request to the SAA AS2805 Interface in the 0200 internal message format.
2. The SAA AS2805 Interface reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorisation and starts an outbound request timer to wait for a response from the co-network.
3. The co-network routes the transaction to the appropriate authoriser. After the transaction is approved, the co-network formats the response into the 0110 or 0210 external message format and sends it to the SAA AS2805 Interface.
4. The SAA AS2805 Interface deletes the outbound request timer, formats a 0210 internal response message, returns the response to the BASE24 Authorisation process to be delivered to the ATM or POS device and logs a copy of the transaction to the Interchange Log File (ILF).
5. The BASE24 Authorisation process returns a 0420 message to the SAA AS2805 Interface, indicating that the transaction failed to complete as authorised. The SAA AS2805 Interface locates the ILF record for the original transaction and updates the ILF record with the reversal information.
6. The SAA AS2805 Interface formats a 0420 external to be sent to the co-network.
7. The co-network responds with an 0430 reversal acknowledgment.
8. If the acknowledgment fails to be received in time, the reversal is logged to the SAF and will be retried during SAF processing.

Force Post Messages to the Co-network

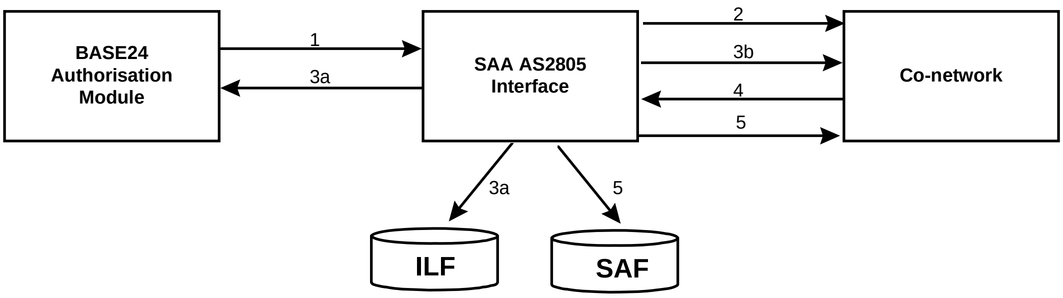
When the local BASE24 network performs stand-in authorisation, it sends the co-network a force post message for each transaction it authorises. The diagram below illustrates message routing when a force post message is sent to the co-network. The steps in the diagram are described below.



1. A BASE24 Authorisation process sends the request to the SAA AS2805 Interface in the 0200 internal message format.
2. The SAA AS2805 Interface determines that the co-network is not available and sends the message back to the BASE24 Authorisation process to be authorised.
3. The BASE24 Authorisation process sends the SAA AS2805 Interface a force post message in the 0220 internal message format. The SAA AS2805 Interface reformats the message into the 0220 external message. The message is added to the Interchange Log File(ILF).
4. the link is up, the SAA AS2805 Interface sends the force post transaction to the co-network and an outbound timer is set to wait for an acknowledgment from the co-network. If the link is not up, the message is put on the SAF and will be sent to the co-network as part of store and forward processing.
5. the message was sent to the co-network, the co-network sends a 0230 message to the SAA AS2805 Interface.
6. an acknowledgment is not received within the time limit, the message is either resent when the link is re-established (if it was already on the SAF), or it is put on the SAF and will be sent to the co-network as part of store and forward processing.

Timeouts of Requests to the Co-network

Each time the SAA AS2805 Interface sends a 0100 or a 0200 financial transaction request to the co-network for authorisation, the SAA AS2805 Interface starts an outbound request timer. If the outbound request timer expires before a response is received from the co-network, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. BASE24 Authorisation process sends the request to the SAA AS2805 Interface in the 0200 internal message format.
2. SAA AS2805 Interface reformats the message into the 0100 or 0200 external message format, sends the external message to the co-network for authorisation and starts an outbound request timer to wait for the response from the co-network.
3. If the timer expires before a response is received from the co-network, the SAA AS2805 Interface checks to determine whether stand-in authorisation is allowed.
   a. stand-in authorisation is allowed, the SAA AS2805 Interface returns the transaction to a BASE24 Authorisation process as a 0200 message with the RTE-STAT field set to indicate this condition.

      If stand-in authorisation is not allowed, the SAA AS2805 Interface formats a 0210 internal response message denying the transaction, returns the response to the BASE24 Authorisation process and logs a copy of the transaction to the Interchange Log File (ILF).
   b. SAA AS2805 Interface formats a 0420 external reversal message and sends this to the co-network if the link is up or adds it to the SAF if the link is down to send to the co-network during SAF Processing.
4. the co-network sends a 0110 or 0210 late response to the SAA AS2805 Interface, the SAA AS2805 Interface drops the late response as it has already sent a response back to the acquirer and reversal to the co-network.
5. an acknowledgment is not received within the time limit, the message is either resent when the link is re-established (if it was already on the SAF) or it is put on the SAF and will be sent to the co-network as part of store and forward processing.

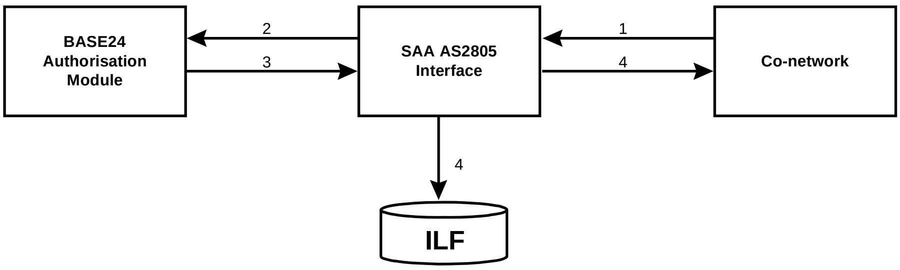
Message Flows from the Co-network

The following pages describe the message flows for the following transactions inbound to BASE24 from the co-network.

- Requests
- Reversals
- Force Posts
- Timeouts

Request Message from the Co-network

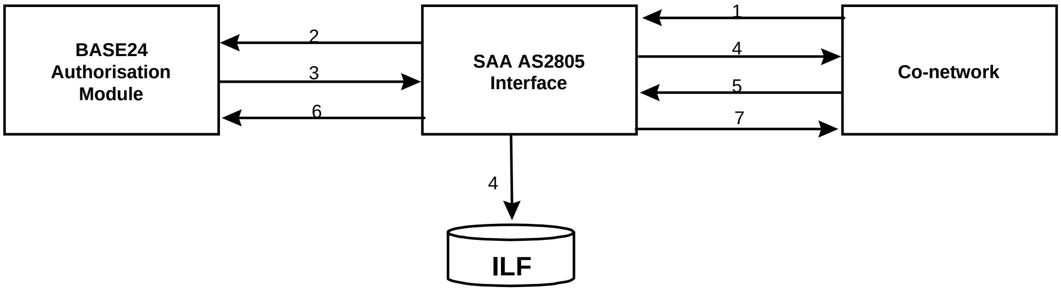
The diagram below illustrates message routing when a transaction is sent from the co-network to BASE24 for authorisation. The steps in the diagram are described below.



1. The co-network sends the request to the SAA AS2805 Interface in the 0100 or 0200 external message format.
2. The SAA AS2805 Interface reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorisation process for authorisation and starts an inbound request timer to wait for the response.
3. The BASE24 Authorisation process sends the response to the SAA AS2805 Interface in the 0210 internal message format.
4. The SAA AS2805 Interface deletes the inbound request timer, formats a 0110 or 0210 external response message, returns the response to the co-network to be completed and logs a copy of the transaction to the Interchange Log File (ILF).

Reversal from the Co-network

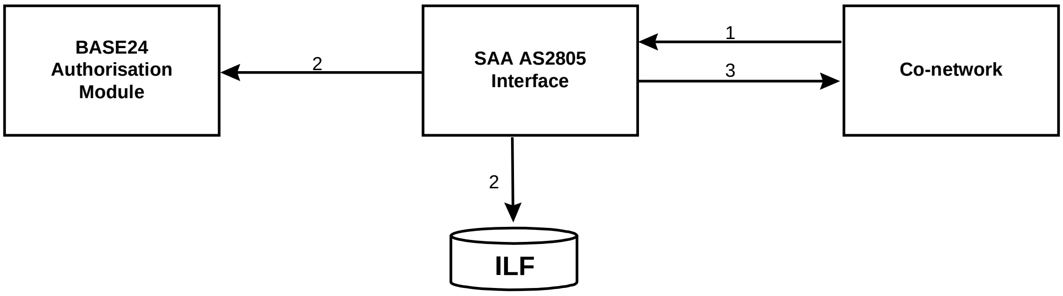
When an ATM or POS device connected to the co-network fails to complete a transaction as authorised, a reversal message must be sent to BASE24. The diagram below illustrates message routing when a reversal is sent to BASE24. The steps in the diagram are described below.



1. The co-network sends the request to the SAA AS2805 Interface in the 0100 or 0200 external message format.
2. The SAA AS2805 Interface reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorisation process for authorisation and starts an inbound request timer to wait for a response from the BASE24 Authorisation process that received the internal message.
3. The BASE24 Authorisation process sends the transaction response to the SAA AS2805 Interface in the 0210 internal message format.
4. The SAA AS2805 Interface deletes the inbound request timer, formats a 0110 or 0210 external response message, returns the response to the co-network to be delivered to the ATM or POS device and logs a copy of the transaction to the Interchange Log File (ILF).
5. The co-network returns a 0420 message to the SAA AS2805 Interface, indicating that the transaction failed to complete as authorised.
6. The SAA AS2805 Interface formats a 0420 internal message and sends the internal message to the BASE24 Authorisation process. It also updates the ILF record of the original transaction if the original transaction was approved.
7. The SAA AS2805 interface sends an 0430 reversal acknowledgment to the co-network.

Force Post Messages from the Co-network

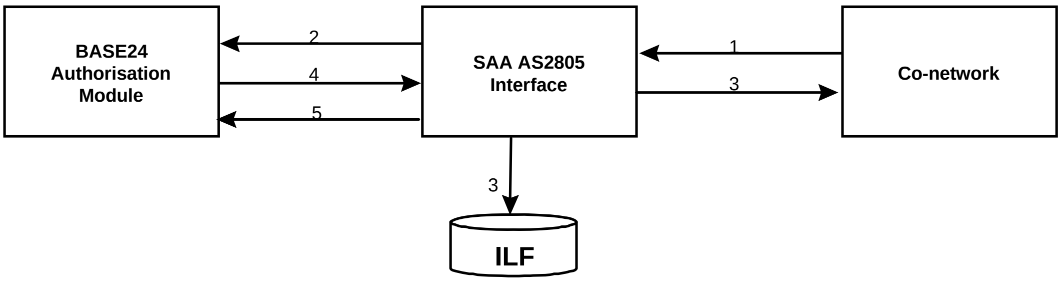
The diagram below illustrates message routing when a force post message is sent to BASE24. The steps in the diagram are described below.



1. The co-network sends the SAA AS2805 Interface a force post message in the 0220 external message format.
2. The SAA AS2805 Interface reformats the message into the 0220 internal message format, sends the internal message to a BASE24 Authorisation process and logs a copy of the transaction to the Interchange Log File (ILF).
3. The SAA AS2805 Interface sends the co-network a 0230 completion acknowledgment.

Timeouts of Request Messages from the Co-network

Each time the SAA AS2805 Interface sends a financial transaction to a BASE24 Authorisation process for authorisation, the SAA AS2805 Interface starts an inbound request timer. If the inbound request timer expires before a response is received from BASE24, the transaction is timed out. The diagram below illustrates message routing when a financial transaction times out. The steps in the diagram are described below.



1. The co-network sends the request to the SAA AS2805 Interface in the 0100 or 0200 external message format.
2. The SAA AS2805 Interface reformats the message into the 0200 internal message format, sends the internal message to a BASE24 Authorisation process for authorisation and starts an inbound request timer to wait for a response from the Authorisation process.
3. When the inbound request timer expires, the SAA AS2805 Interface formats a 0110 or 0210 external response message denying the transaction and sends the 0110 or 0210 message to the co-network. It also logs this denied response to the ILF.
4. The BASE24 Authorisation process sends a late 0210 response to the SAA AS2805 Interface. If the transaction was denied by the BASE24 Authorisation process, the SAA AS2805 Interface drops the late response.
5. If the transaction was approved, the SAA AS2805 Interface formats a 0420 internal reversal message and sends the reversal to a BASE24 Authorisation process.

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Section 6

CUTOVER AND SETTLEMENT

Overview

On any one day, an SAA AS2805 interface can be processing transactions that will be settled with the cardholder and the merchant today, or tomorrow. The date the transaction is settled between the parties is determined by the settlement date in external request messages from the co-network or the posting date in the STM/PSTM request messages from BASE24. BASE24 or the co-network can at any time intermingle today's transactions with tomorrow's transactions. This is due to the fact that some merchants may close their current business day before BASE24 or the co-network closes their business day. Therefore, it always keeps two ILF files open at all times.   The acquirer of a transaction (i.e. the owner of the terminal) always determines which date a transaction will be settled on.

Settlement is the process of informing the other party that they, as an acquirer, will no longer be sending any new request messages for the current date. It also informs the other partner of the dollar value of all transactions it believed it acquired and sent for authorisation for that day. Settlement messages have a message type of 0520 (refer to AS2805 standards for message layout). On receipt of the settlement message, the other partner must compare the dollar value of the transactions it believes it approved for that day, with that received in the settlement message and respond with either an agreement acknowledgment or a disagreement acknowledgment. If they disagree, manual procedures must be performed to reconcile this value.

Cutover is the term used when referring to the time of day that both the co-network and BASE24 have stopped processing for the current date. Both partners must have previously sent a settlement message indicating their completion of processing for the current day. At this stage, the ILF for the current date is closed. The ILF for the next day then becomes the current date ILF and a new ILF is created for the following day.

Configuration Options

The SAA AS2805 interface can be configured to perform cutover in a number of ways using the following parameters from the ICF;

ATM SETTLEMENT TIME

This parameter is set on screen 12 of the ICF. It indicates the time at which BASE24-atm has finished processing transactions for the current date and therefore the time at which the SAA AS2805 interface can start its settlement and cutover strategy. This time MUST be set to a time later than the ATM logical network cutover time.

POS SETTLEMENT TIME

This parameter is set on screen 12 of the ICF. It indicates the time at which BASE24-pos has finished processing transactions for the current date and therefore the time at which the SAA AS2805 interface can start its settlement and cutover strategy. This time MUST be set to a time later than the POS logical network cutover time.

SWITCH SETTLEMENT TIME

This parameter is set on screen 2 of the ICF. It indicates the time at which the co-network should have completed all transactions for the current day and therefore the time at which the SAA AS2805 interface can start its settlement and cutover strategy. This field is only used when the Cutover Method on screen 12 of the ICF is set to not receive settlement messages from the co-network and the Issuer flag on screen 3 of the ICF is set to indicate that BASE24 is acting as an issuer. This time MUST be set to a time later than the latest time at which the co-network can send acquired transactions to BASE24 for the current day.

CUTOVER METHOD

This flag indicates the method involved to determine cutover and settlement. It can have the following values;

“0” – Send and Receive 0520 settlement messages.

If the interface is configured to support ATM traffic, an ATM settlement message will be sent from BASE24 to the co-network at the ATM SETTLE TIME as specified on screen 12 of the ICF. Similarly for the POS settlement message if the network is configured to perform POS transactions. The co-network must also send its 0520 settlement message at a time determined by them. Once both sides have sent and received settlement messages, the interface will perform cutover and close the current ILF.

“1” – Send 0520 messages only.

If the interface is configured to support ATM traffic, an ATM settlement message will be sent from BASE24 to the co-network at the ATM SETTLE TIME as specified on screen 12 of the ICF. Similarly for the POS settlement message if the network is configured to perform POS transactions.   The co-network is not required to send an 0520 settlement message. The interface will assume that no further transactions for today will be received from the co-network at the time specified by the SETL TIME on screen 2 of the ICF. However, if the co-network does send a settlement message it will be logged to the ILF and ignored. Once BASE24 has sent its settlement message and it is passed the time specified in the SETL TIME of the ICF, the interface will perform cutover and close the current ILF.

“2” – Receive 0520 messages only.

BASE24 is not required to send a settlement message to the co-network. If the interface is configured to support ATM traffic, an ATM settlement message will be constructed by the interface at the ATM SETTLE TIME as specified on screen 12 of the ICF. However, it will not send this message to the co-network, but will merely log the message to the ILF. Similarly for the POS settlement message if the network is configured to perform POS transactions.   The co-network must still send its 0520 settlement message at a time determined by them. Once both of these things occur, the interface will perform cutover and close the current ILF.

“3” – No exchange of 0520 messages is required.

BASE24 is not required to send a settlement message to the co-network. If the interface is configured to support ATM traffic, an ATM settlement message will be constructed by the interface at the ATM SETTLE TIME as specified on screen 12 of the ICF. However, it will not send this message to the co-network, but will merely log the message to the ILF. Similarly for the POS settlement message if the network is configured to perform POS transactions. The co-network is not required to send an 0520 settlement message. The interface will assume that no further transactions for today will be received from the co-network at the time specified by the SETL TIME on screen 2 of the ICF. However, if the co-network does send a settlement message, it will be logged to the ILF and ignored. Once both of these activities have occurred, the interface will perform cutover and close the current ILF.

RECONCILIATION FLAG

This flag is used to determine whether this interchange processes ATM transactions and/or POS transactions and whether there will be separate settlement messages for each or combined totals of both will be included in the settlement message. It can have the following values;

“0” – Separate ATM and POS settlement messages.

The interface will send and expect to receive separate ATM and POS settlement messages before it considers that cutover can occur.

“1” – Only ATM settlement messages.

The interface will send and expect to receive only ATM settlement messages before it considers that cutover can occur.

“2” – Only POS settlement messages.

The interface will send and expect to receive only POS settlement messages before it considers that cutover can occur.

“3” – Combined ATM and POS settlement messages.

The interface will send and expect to receive combined ATM and POS totals in the one settlement message. It will use the later of the ATM settlement time and the POS settlement time on screen 12 of the ICF to determine at what time to send its settlement message to the co-network.

ACQUIRER

This parameter is set on screen 3 of the ICF. The value in this field indicates whether BASE24 is acquiring transactions on behalf of the co-network. When set to “Y”, the above fields are used to determine the method for cutover and settlement in relation to sending settlement messages to the co-network. When it is set to “N”, no settlement messages will be generated by the SAA AS2805 interface.

ISSUER

This parameter is set on screen 3 of the ICF. The value in this field indicates whether BASE24 is the issuer for transactions which the co-network has acquired. When set to “Y”, the above fields are used to determine the method for cutover and settlement in relation to receiving settlement messages from the co-network. When it is set to “N”, no settlement messages will be expected from the co-network.

PROCESSING MODE

This parameter is set on screen 3 of the ICF. This flag indicates whether or not totals are being kept for this link. Valid values are;

“0” – Totals are not being kept. Settlement messages will still be sent to and received from the co-network, but they will not have totals.

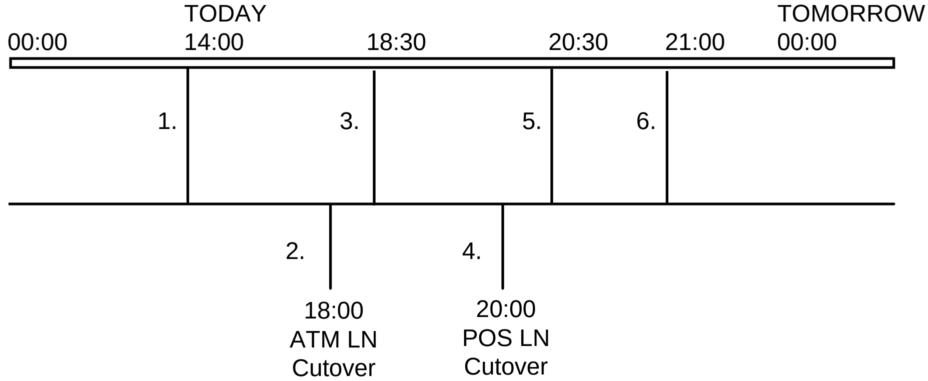
“1” – Totals are being kept.

Configuration Examples

The following are some examples of how to configure an interface to perform settlement and cutover with different options and the result of this configuration.

1. An interchange acts as an acquirer and an issuer for POS and ATM transactions. It will send and receive settlement messages separately for ATM and POS totals. We will assume that ATM logical network cutover is at 18:00, POS logical network cutover is at 20:00 and that the co-network will send its ATM settlement message at 18:30 and POS settlement message at 20:30.

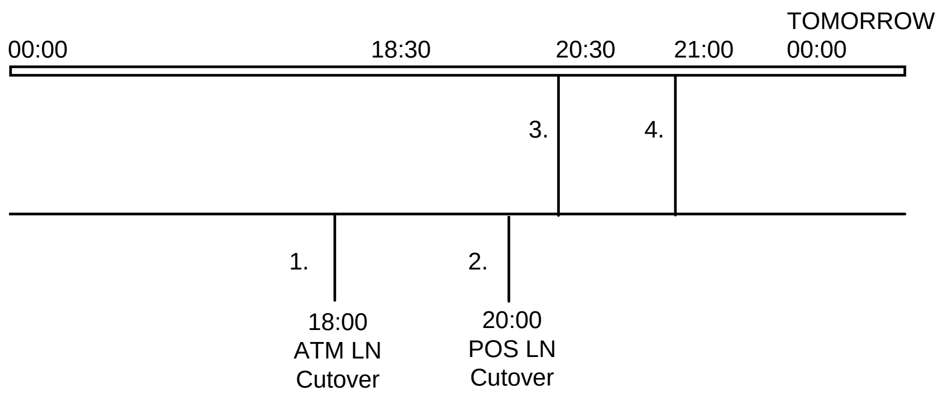
PARAMETER	***VALUE***
ATM Settle Time	18:30
POS Settle Time	20:30
Switch Settle Time	Not applicable – co-network sends settlement
Cutover Method	“0” – send and receive
Reconciliation Mode	“0” – ATM and POS separately
Acquirer	“Y” – we are an acquirer
Issuer	“Y” – we are an issuer
Processing Mode	“1” – we are keeping totals



1) Co-network sends their ATM acquired settlement message after which no more ATM transactions will be received from them for this processing date.
2) ATM logical network cutover occurs, after which no more ATM transactions will be received by the SAA AS2805 interface from BASE24 for today's processing date.
3) The SAA internal ATM settle timer goes off. SAA sends its ATM acquired settlement message.
4) POS logical network cutover occurs, after which no more POS transactions will be received by the SAA AS2805 interface from BASE24 for today's processing date.
5) The SAA internal POS settle timer goes off. SAA sends its POS acquired settlement message.
6) Co-network sends their POS acquired settlement message after which no more POS transactions will be received from them. The SAA AS2805 interface, according to its configuration, determines that all settlement has occurred for this processing date and the ILF for today can be closed, tomorrows ILF can be rolled into today's ILF and a new ILF can be created for tomorrow. Cutover is then considered complete.

2. An interchange acts as an acquirer and an issuer for POS and ATM transactions. It will only send settlement messages, it will not receive them. It will send combined ATM and POS totals in its settlement message. We will assume that ATM logical network cutover is at 18:00 and POS logical network cutover is at 20:00 and that the co-network will not send any further transactions with the current processing date after 21:00.

PARAMETER	***VALUE***
ATM Settle Time	18:30
POS Settle Time	20:30
Switch Settle Time	21:00
Cutover Method	“1” – send only
Reconciliation Mode	“3” – Combined totals
Acquirer	“Y” – we are an acquirer
Issuer	“Y” – we are an issuer
Processing Mode	“1” – we are keeping totals

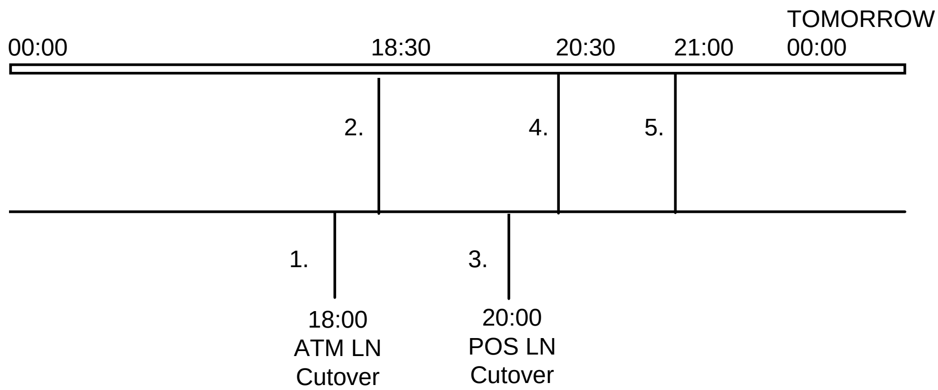


1) ATM logical network cutover occurs, after which no more ATM transactions will be received by the SAA AS2805 interface from BASE24 for today’s processing date.
2) POS logical network cutover occurs, after which no more POS transactions will be received by the SAA AS2805 interface from BASE24 for today’s processing date.
3) The SAA internal POS settle timer goes off. SAA uses the later of the ATM and POS settle times to determine when to send the settlement message as it is a combined total. SAA sends its combined ATM and POS acquired settlement message.

4) The SAA internal switch settle timer goes off which informs SAA that there will be no more transactions with the current processing days date received from the co-network. SAA creates a settlement message which reflects what it believes to be the co-network's acquired total and logs this to the ILF. The SAA AS2805 interface, according to its configuration, determines that all settlement has occurred for this processing date and the ILF for today can be closed, tomorrows ILF can be rolled into today's ILF and a new ILF can be created for tomorrow. Cutover is then considered complete.

3. An interchange acts as an acquirer and an issuer for POS and ATM transactions. It will not expect to send or receive settlement messages. It will keep separate ATM and POS totals internally. We will assume that ATM logical network cutover is at 18:00, POS logical network cutover is at 20:00 and that the co-network will not send and further transactions with the current processing date after 21:00.

PARAMETER	***VALUE***
ATM Settle Time	18:30
POS Settle Time	20:30
Switch Settle Time	21:00
Cutover Method	“3” – No send or receive
Reconciliation Mode	“0” – Separate ATM and POS totals
Acquirer	“Y” – we are an acquirer
Issuer	“Y” – we are an issuer
Processing Mode	“1” – we are keeping totals



1) ATM logical network cutover occurs, after which no more ATM transactions will be received by the SAA AS2805 interface from BASE24 for today’s processing date.
2) The SAA internal ATM settle timer goes off. The SAA AS2805 interface creates and logs the settlement message for ATM to the ILF. However, it does not send it to the co-network.
3) POS logical network cutover occurs, after which no more POS transactions will be received by the SAA AS2805 interface from BASE24 for today’s processing date.
4) The SAA internal POS settle timer goes off. The SAA AS2805 interface creates and logs the settlement message for POS to the ILF. However, it does not send it to the co-network.

5) The SAA internal switch settle timer goes off which informs SAA that there will be no more transactions with the current processing days date received from the co-network. SAA creates a settlement message which reflects what it believes to be the co-network's acquired total and logs this to the ILF. The SAA AS2805 interface, according to its configuration, determines that all settlement has occurred for this processing date and the ILF for today can be closed, tomorrows ILF can be rolled into today's ILF and a new ILF can be created for tomorrow. Cutover is then considered complete.

SAA AS2805 Totals

The SAA AS2805 Interface maintains a number of running totals that reflect the approved transactions processed between BASE24 and the co-network since the last business day cutover. The totals maintained by the SAA AS2805 Interface include:

- Debit amount — the total amount of all debit transactions
- Debit count — the total number of debit transactions
- Credit amount — the total amount of all credit transactions
- Credit count — the total number of credit transactions
- Reversal debit amount — the total amount of all reversal debit transactions
- Reversal debit count — the total number of reversal debit transactions
- Reversal credit amount — the total amount of all reversal credit transactions
- Reversal credit count — the total number of reversal credit transactions
- Authorisation count — the total number of authorisation transactions
- Inquiries count — the total number of inquiry transactions
- Transfers count — the total number of transfer transactions
- Transfer reversal count — the total number of transfer reversal transactions
- Net settlement amount — the net settlement amount for the cutover date just completed

These totals are kept for the current processing date and the next processing date. The totals kept by the SAA AS2805 Interface are sent to the co-network in the 0520 and 0530 reconciliation messages and are used by the interchange report programs.

As each transaction is processed by the SAA AS2805 Interface, the appropriate totals are updated based on the type of transaction and the posting date for the transaction. The following tables summarise how each transaction impacts the totals listed above. For the purposes of impacting totals, force post transactions are treated the same as normal transactions and adjustments are treated the same as reversals.

BASE24-atm Transactions

Transaction	RECONCILIATION TOTALS												
	Amounts					Counts							
	Cash Amounts	Debits	Credits	Debit Reversals	Credit Reversals	Debits	Credits	Debit Reversals	Credit Reversals	Inquiries	Transfers	Transfer Reversals	Authorisations
Withdrawal	O	O				+1							
Withdrawal Reversal	CA			O				+1					
Deposit			O				+1						
Deposit Reversal					O				+1				
Payment From/To											+1		
Payment From/To Reversal												+1	
Transfer											+1		
Transfer Reversal												+1	
Balance Inquiry										+1			

Legend:

O = Original Transaction Amount
CA = Actual completion amount of the transaction
+1 = Adds one to specified transaction count

BASE24-pos Transactions

Transaction	RECONCILIATION TOTALS												
	Amounts					Counts							
	Cash Amounts	Debits	Credits	Debit Reversals	Credit Reversals	Debits	Credits	Debit Reversals	Credit Reversals	Inquiries	Transfers	Transfer Reversals	Authorisations
Normal Purchase		O				+1							
Normal Purchase Reversal				O				+1					
Preauthorisation													+1
Preauthorisation Completion			O			O							
Preauthorisation Completion Reversal				O				+1					
Mail/Phone Orders		O				+1							
Mail Phone Orders Reversal				O				+1					
Merchandise Return			O				+1						
Merchandise Return Reversal									+1				
Cash Advance	CA	0				+1							
Cash Advance Reversal	CA			0				+1					
Purchase W/Cash Back	CA	0				+1							
Purchase W/Cash Back Reversal	CA			0				+1					
Balance Inquiry										+1			

Legend:

O = Original Transaction Amount
CA = Cash Out Amount
-CA = Subtract Cash Out amount
+1 = Adds one to specified transaction count

Interchange Totals in the ILF

The SAA AS2805 Interface process accumulates its reconciliation totals in memory during the business day. These accumulated totals are stored in the first record of the Interchange Log File (ILF) for a business day. The SAA AS2805 Interface process updates the first record when it has processed the number of transactions specified in the NUM TRANS FOR TOTALS REWRITE field on ICF screen 12.

Each time the totals in the ILF are updated, the keys (i.e. addresses) of the last ILF records included in the accumulated totals are recorded in the first record with the totals figures. During warmboot or process initialisation, the SAA AS2805 Interface process locates the address of the last accumulated transaction for the ILFs for its current and next posting days. It then reads from that record to the end of the file, updating the totals in memory for the current and next local BASE24 business days. In this way, the SAA AS2805 Interface process being restarted can rebuild its memory-resident totals without having to read any ILF in its entirety.

Logging Reversals to the ILF

When the SAA AS2805 interface logs reversals, the original transaction is updated (if present) with the reversal on the ILF. Because the transaction that is being updated in place may have already been included in the last ILF first record update, the reversal could be missed in restart processing after a process failure. For this reason, a reversal causes an immediate update of the ILF first record with the address of the last transaction in the ILF.

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Appendix A

SAA AS2805 Interface Process Startup and Control

This appendix provides information on the configuration of the SAA AS2805 Interface process and network control.

Startup

The SAA AS2805 Interface process should be assigned a startup logic of AUTOMATIC in the Network Environment Source File (NEFS). AUTOMATIC means that the SAA AS2805 Interface process will be started automatically when the resource node is initiated. This ensures that the SAA AS2805 Interface process is ready when it receives its first transaction.

Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The SAA AS2805 Interface process routes all of its log messages with a standard routing code of zero.

The SAA Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator’s primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

Network Control Commands

This section provides a list of the BASE24 commands that can be received by the SAA AS2805 Interface process. Many of these commands can be entered by an operator via the Network Control Supervisor (NCS) screen or Network Control Program User Interface (NCPCom) and serve as a tool that operators can use to perform network management functions such as changing processing parameters, warmbooting processes and monitoring processes.

For detailed descriptions and the syntax of all network control commands, refer to the BASE24 Text Command Reference Manual.

- Debug — Places the SAA AS2805 Interface process in the DEBUG state. The debug message places the SAA AS2805 Interface into Inspect and brings up an Inspect prompt on the home terminal (the terminal on which the network was brought up). Inspect is a Tandem programming tool. This command should only be used to interactively debug a running process when in a test environment. Executing this command in a production environment can result in a significant impact on system performance.
- KEKChange — When an interchange is configured for Dynamic KEKs, an operator can initiate a KEKChange. This results in the interchange generating new KEKs and sending them to the co-network in an 0800 KEK Change Advice message. The new KEK will not take effect until the next key change occurs.
- Logoff — Initiates a logoff sequence. It causes the SAA AS2805 Interface to send a logoff message to the co-network
- Logon — Initiates a logon sequence. It causes the SAA AS2805 Interface to send a logon message to the co-network.
- Performance — Requests that the performance statistics kept in memory by the SAA AS2805 Interface be sent to the log. The SAA AS2805 Interface tracks performance statistics for each station assigned to it. The statistics are displayed for the SAA AS2805 Interface process as a whole. These statistics are gathered on an interval basis—the interval length being established in the ICF—and statistics are available for the last five intervals.
- Status — Requests a summary of the status of the SAA AS2805 Interface process. The requested information is sent to the current logging facility.
- Status Detail — Requests a detailed status of the SAA AS2805 Interface process. The requested information is sent to the current logging facility.
- Trace Off — Stops the trace function within the SAA AS2805 Interface. For a description of the trace function, please refer to the TRACE ON message description.

- Trace On — Starts a trace function within the SAA AS2805 Interface. This trace function generates log messages at periodic intervals while the trace function is enabled. The log messages display the name of the procedure from which it was written or other helpful processing information. The trace function should only be enabled in a test environment. Enabling the trace function in a production environment can result in a significant impact on system performance.
- Warmboot — Warmboots the SAA AS2805 Interface. Warmboot processing is similar to process initialisation, except that the SAA AS2805 Interface closes all open files after they have been successfully reopened. If an error is encountered, the old environment is restored.
- 9501 – Results in the same processing as a warmboot command.
- 9502 – Cutover command to bump the Product Specific Posting Date.
- 9510 – Mark a station down. If no further send stations are available then mark the link down.
- 9511 – Mark a station up and set a wait for traffic timer providing we are not logged off and this is a send station. If we are logged off then insert a logon timer.
- 9512 – Suspend a station. Results in the same processing as a 9510 command.
- 9513 – Reinstate a station. Results in the same processing as a 9511 command.
- 9516 – Not being polled. Results in the same processing as a 9510 command.
- 9517 – Being Polled. Results in the same processing as a 9511 command.
- 9518 – X25 Station now marked up. Results in the same processing as a 9511 command.
- 9519 – X25 Station now marked down. Results in the same processing as a 9510 command.

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Appendix B

Message Mapping

Messages received from BASE24 destined for the co-network are mapped from an internal STM/PSTM message into an external AS2805 message before they are sent to the co-network. Similarly, message received from the co-network destined for BASE24 are mapped from the external AS2805 message into the internal STM/PSTM message before they are sent to BASE24.

Following is the mapping used for this conversion;

Internal ATM STM to External AS2805 Conversion

The following describes how external AS2805 fields are built from the internal STM fields.

AS2805 Fields

TYP	Fill with STM.TYP
PBIT-MAP/SBIT-MAP	Fill with values from EMF file
PAN.DATA-LGTH, PAN.DATA-VALUE	Use the utility HISWUTIL^ EXTRACT^TRACK2^INFO to extract these fields from STM.RQST.TRACK2. If the card was swiped, zero fill these values
PROC-CDE.TYP, PROC-CDE.SRC-ACCT-TYP, PROC-CDE.SRC-ACCT-SUB-TYP PROC-CDE.DEST-ACCT-TYP, PROC-CDE.DEST-ACCT-SUB-TYP	Fill as follows;

STM (ATM)	SEM (AS2805)
TRAN-CDE	PROC-CDE
TY FA TA	TY SR DE
0200 10	0200 01
11 00	10 00
01 00	20 00
31 00	30 00
0200 20	0200 21
00 11	10 00
00 01	20 00
00 31	30 00
0200 30	0200 31
11 00	10 00
01 00	20 00
31 00	30 00
0200 40	0200 40
11 01	10 20
01 11	20 10
31 11	30 10
31 01	30 20
0200 51	0200 40
11 31	10 30
01 31	20 30

TXN-AMT	Fill with RQST.AMT-1
TRAN-DAT-TIM	Fill with Guardian Timestamp
SYS-AUD-NUM	Fill with SEQ-NUM if the parameter

	in the LCONF indicates non-unique STANs. Otherwise use the next STAN from the PCT.
LOCAL-TXN-TIM	Fill with TRAN-TIM
LOCAL-TXN-DAT	Fill with TRAN-DAT
SETL-DAT	Fill with POST-DAT
PT-SRV-ENTRY-MODE	Fill with "021"
CRD-SEQ-NUM	Fill with RQST.MBR-NUM
PT-SRV-COND-CDE	Fill with "41"
ACQ-INST-ID-CDE	Fill with RQST.ACQ-INST-ID-NUM
FWD-INST-ID-CDE.DATA-LGTH	Fill with length of FRWD-INST-ID-NUM
FWD-INST-ID-CDE.DATA-VALUE	Fill with FRWD-INST-ID-NUM
TRACK2.DATA-LGTH	Fill with length of RQST.TRACK2, excluding start sentinel, end sentinel and LRC. If the card was manually entered, zero fill this field
TRACK2.DATA-VALUE	Fill with RQST.TRACK2, excluding start sentinel, end sentinel and LRC. If the card was manually entered, zero fill this field
AUTH-ID-RESP	N/A
RESP-CDE	Use Response Code Mapping table and RQST.RESP to fill
CRD-ACCEPT-TERM-ID	Fill with first eight bytes of TERM-ID
CRD-ACCEPT-ID-CDE	Fill with RETL-ID
CRD-ACCEPT-NAM-LOC	Fill with concatenation of RQST.TERM-NAM-LOC RQST.TERM-CITY RQST.TERM-ST-X
PIN-DATA	Use conversion routines for PIN processing. For hardware support a translation takes place, between RQST.PIN-KEY and current outbound PIN session key
SEC-DATA	Fill with PCT.KEY.OUTBOUND-CURR-INDX in last three bytes
LEDG-BAL	Fill from RQST.AMT-2
ACCT-BAL-CLR-FNDS	Fill from RQST.AMT-3
ORIG-DATA-ELTS	If this is a reversal, fill from original request in ILF. ORIG-TYP, ORIG-SYS-AUD-NUM,

	ORIG-ACQ-INST-ID-CDE, ORIG-FWD-INST-ID-CDE ORIG-TRAN-DAT-TIM - fill with exit time.
ACCT-ID-1	Use RQST.FROM-ACCT or RQST.TO-ACCT as defined for PROC-CDE
ACCT-ID-2	Use RQST.FROM-ACCT or RQST.TO-ACCT as defined for PROC-CDE

Internal POS PSTM to External AS2805 Conversion

AS2805 Fields

TYP	See below for TYP conversion
PBIT-MAP/SBIT-MAP	Fill with values from EMF file
PAN.DATA-LGTH, PAN.DATA-VALUE	Use the utility HISWUTIL^ EXTRACT^TRACK2^INFO to fill these fields from TRAN.TRACK2. If the card was swiped, zero fill these fields
PROC-CDE.TYP, PROC-CDE.SRC-ACCT-TYP, PROC-CDE.SRC-ACCT-SUB-TYP PROC-CDE.DEST-ACCT-TYP, PROC-CDE.DEST-ACCT-SUB-TYP	POS only processes "FROM" accounts

PSTM (POS) TRAN-CDE TC T AA C	SEM (AS2805) PROC-CDE 12 34 56
<u>0200 10</u>	<u>0200 00</u>
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00
<u>0200 11</u>	<u>0100 00</u>
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00
<u>0200 14</u>	<u>0200 20</u>
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00
<u>0200 15</u>	<u>0200 01</u>
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00
<u>0200 17</u>	<u>0200 31</u>
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00

0200 18	0200 09
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00

0210	0110

If Response is to 0100.

0210	0210

If Response is to 0200.

The transaction code fields for a Financial Response message are built from the details held for the matching Financial Request message.

0220 12	0220 00
x 00 x	00 00
x 31 x	30 00
x 01 x	20 00
x 11 x	10 00

0420	0420

The transaction code fields for a Financial Reversal message are built from the details held for the matching Financial Request message (except 220, Pre-Authorisation Completion).

TXN-AMT	Fill with TRAN.AMT-1
TRAN-DAT-TIM	Fill with Guardian timestamp
SYS-AUD-NUM	Fill with SEQ-NUM if the LCONF parameter UNIQUE-STAN indicates no to use unique STANs. Otherwise, use the next STAN from the PCT
LOCAL-TXN-TIM	Fill with TRAN-TIM
LOCAL-TXN-DAT	Fill with TRAN-DAT
EXPIRY-DAT	fill with TRAN.EXP-DAT. If the card was swiped, fill with zeroes
SETL-DAT	Fill with POST-DAT
MERCH-TYP	Fill with RETL-SIC-CDE
PT-SRV-ENTRY-MODE	If TRACK2.BYTE[0] = "M" Fill BYTE [0:1] with "01" else Fill BYTE [0:1] with "02".

	If PIN-LENGTH = 0 Fill BYTE [2] with "2" else Fill BYTE [2] with "1"
CRD-SEQ-NUM	Fill with TRAN.MBR-NUM
PT-SRV-COND-CDE	Fill with PT-SRV-COND-CDE
ACQ-INST-ID-CDE.DATA-LGTH	Fill with length of ACQ-INST-ID-NUM
ACQ-INST-ID-CDE.DATA-VALUE	Fill with ACQ-INST-ID-NUM
FWD-INST-ID-CDE.DATA-LGTH	Fill with length of FRWD-INST-ID-NUM
FWD-INST-ID-CDE.DATA-VALUE	Fill with FRWD-INST-ID-NUM
TRACK2.DATA-LGTH, TRACK2.DATA-VALUE	Fill with length of TRAN.TRACK2, excluding start sentinel, end sentinel and LRC. If the card was manually entered, fill this field with zeroes
AUTH-ID-RESP	Fill with TRAN.APPRV-CDE
RESP-CDE	Use Response Code Mapping table and TRAN.RESP-CDE to fill
CRD-ACCEPT-TERM-ID	Fill with first eight bytes of TERM-ID
CRD-ACCEPT-ID-CDE	Fill with RETL-ID
CRD-ACCEPT-NAM-LOC	Fill with concatenation of TERM-NAM-LOC TERM-CITY TERM-ST-X
PIN-DATA	Use conversion routines for PIN processing. For hardware support a translation takes place, between PSTM.PIN-KEY and current outbound PIN session key
SEC-DATA	Fill with details from KEYF file and place key index in last 3 bytes
CASH-AMT	Fill with TRAN.AMT-2 if it is a cashback purchase
LEDG-BAL	Fill from TRAN.AMT-1
ACCT-BAL-CLR-FNDS	Fill from TRAN.AMT-2
ORIG-DATA-ELTS	If this is a reversal, fill from original request in ILF. ORIG-TYP, ORIG-SYS-AUD-NUM, ORIG-ACQ-INST-ID-CDE, ORIG-FWD-INST-ID-CDE
ORIG-TRAN-DAT-TIM	fill with exit time
ACCT-ID-1	Fill from TRAN.ACCT

External AS2805 to Internal ATM STM Conversion

AS2805 Fields

TYP	Fill with TYP from SEM. Convert an AS2805 0100,0110 or 0120 to 0200,0210 or 0220
PROD-ID	Fill with "01"(ATM)
DPC-NUM	Fill with 0
TRAN-CDE	As for "STM TO SEM" but in reverse
FROM-ACCT-TYP	As for "STM TO SEM" but in reverse
TO-ACCT-TYP	As for "STM TO SEM" but in reverse
RTE.STAT	Fill with zeroes
RTE.NOT-ON-US-FLG	Fill with zeroes
RTE.ORIG-PRO-NAME	Fill with the symbolic name of this process, obtained from startup message from XPNET (NET.MYNAME)
RTE.HLD-ORIG-PRO-NAME	Fill with zeroes
RTE.ORIGINATOR	If this is a "0200" request, Fill with "7" (switch originated)
RTE.RESPONDER	If this is a "0210" response, Fill with "7" (switch originated)
RTE.EXCPT-FLG	Fill with zeroes
CRD-LN	Fill with zeroes
RCV-INST-ID-NUM	Fill with zeroes
CRD-FIID	Fill with zeroes
SHRG-GRP	If this is a request, fill with PCT.ATM.SHARING
TRAN-DAT	Fill with LOCAL-TXN-DAT
TRAN-TIM	Fill with LOCAL-TXN-TIM
TIM-OFST	Fill with zeroes
POST-DAT	Fill with PCT.SETL.ATM-DAT
ACQ-ICHG-SETL-DAT	If this is a request fill with SETL-DAT
ISS-ICHG-SETL-DAT	If this is a response fill with SETL-DAT

TERM-OWNER-FIID	Fill with PCT.FIID
TERM-ID	Fill from CRD-ACCEPT-TERM-ID
TERM-LN	Fill with PCT.LN
BRCH-ID	Fill with zeroes
REGN-ID	Fill with zeroes
SEQ-NUM	Fill from SYS-AUD-NUM
RQST.TRACK2	If PT-SRV-ENTRY-MODE = "02", indicating "mag stripe" Fill with ";" (start sentinel), TRACK2.DATA-VALUE and "?" (end sentinel). Else Fill with ";" (start sentinel), PAN.DATA-VALUE, "=", EXPIRY-DAT and "?" (end sentinel).
RQST.PIN-OFST	Fill with zeroes
RQST.MBR-NUM	Fill with CRD-SEQ-NUM
RQST.TERM-TYP	Fill with zeroes
RQST.ORIG-CRNCY-CDE	Fill with PCT.CRNCY-CDE
RQST.ACQ-INST-ID-NUM	Fill with ACQ-INST-ID-CDE
RQST.RVSL-CDE	Fill with zeroes
RQST.FROM-ACCT	Fill from ACCT-ID-1 or ACCT-ID-2 followings rules as defined in "STM TO SEM"
RQST.TO-ACCT	Fill from ACCT-ID-1 or ACCT-ID-2 followings rules as defined in "STM TO SEM"
RQST.PIN-TRIES	If this is a response and RESP-CDE = "55", incorrect PIN, increment this number by one, otherwise fill with "Z"
RQST.CUST-BAL-INFO	If this is a response, fill with "4"
RQST.AMT-1	If this is a request, fill with TXN-AMT
RQST.AMT-2	If this is a request, fill with zeroes. Otherwise fill with LEDG-BAL
RQST.AMT-3	If this is a request, fill with zeroes. Otherwise fill with ACCT-BAL-CLR-FNDS
RQST.DEP-BAL-CR	Fill with zero
RQST.RESP	Use RESP-CDE and Response Code

	Mapping Table to fill
RQST.TERM-NAME-LOC	Fill with CRD-ACCEPT-NAM-LOC (First 25 bytes)
RQST.TERM-OWNER-NAME	Fill with blanks
RQST.TERM-CITY	Fill with CRD-ACCEPT-NAM-LOC[26] for 13 bytes
RQST.TERM-ST-X	Fill with CRD-ACCEPT-NAM-LOC[39] for 2 bytes
RQST.TERM-TRAN-ALLOWED	Fill with PCT.ATM.NOT-ON-US
RQST.TRACE-NUM	Fill with zeroes
RQST.TERM-ST	Fill with zeroes
RQST.AUTH-DEST	Fill with zeroes
RQST.RTE-GRP	Fill with PCT.ATM.RTG-GRP
ENTRY-TIM,EXIT-TIM,RE-ENTRY-TIM	On a request, fill ENTRY-TIM with JULIAN TIMESTAMP. On a response, fill RE-ENTRY-TIM with JULIAN TIMESTAMP.
FRWD-INST-ID-NUM	Fill with FWD-INST-ID-CDE
CRD-ISS-ID-NUM	Fill with zeroes
PIN, PIN-SIZE, PIN-KEY, PIN-FRMT, PINPAD-CHAR, ANSI-OFST	Use conversion routines for PIN processing. For hardware support no translation takes place, current PIN session key is placed in STM.PIN-KEY

External AS2805 to Internal POS PSTM Conversion

AS2805 Fields	
TYP	Convert from TYP as defined in PSTM to SEM
PROD-ID	Fill with PCT.POS.PROD-ID
DPC-NUM	Fill with 0
RTE.STAT	Fill with 0
ORIG-PRO-NAME	Fill with the symbolic name of this process, obtained from startup message from XPNET (NET.MYNAME)
AST-RTN-PRO-NAME	If this is a "0200" request, Fill with PCT.POS-DEST
ORIGINATOR	If this is a request Fill with "7" (switch originated)
RESPONDER	If this is a response, Fill with "7" (switch originated)
POST-DAT	Fill with PCT.SETL.POS-DAT
TERM-LN	Fill with PCT.LN
TERM-FIID	Fill with PCT.FIID
TERM-ID	Fill from CRD-ACCEPT-TERM-ID
TERM-NAME-LOC	Fill with CRD-ACCEPT-NAM-LOC (First 25 bytes)
TERM-OWNER-NAME	If this is a request, fill with spaces
TERM-CITY	Fill with CRD-ACCEPT-NAM-LOC[26] for 13 bytes
TERM-ST	Fill with CRD-ACCEPT-NAM-LOC[39] for 2 bytes
ORIG-CRNCY-CDE	Fill with PCT.CRNCY-CDE
AUTH-IND2	Fill with zeroes
SEQ-NUM	Fill with SYS-AUD-NUM
PRE-AUTH-SEQ-NUM	If this is a pre authorisation request, Fill with SYS-AUD-NUM Else if this is a "0220" force post Fill with AUTH-ID-RESP
ACQ-INST-ID-NUM	If this is a request fill with ACQ-INST-ID-CDE

RCV-INST-ID-NUM	Fill with zeroes
RETL-ID	Fill with CRD-ACCEPT-ID-CDE
RETL-GRP	Fill with MERCH-TYP
RETL-SIC-CDE	Fill with MERCH-TYP
TRAN.TRAN-CDE	Convert from PROC-CDE as defined in PSTM TO SEM
TRAN.ORIG	If this is a request fill with "SAA "
TRAN.DEST	If this is a request fill with "B24 "
TRAN.CRD-LN	Fill with zeroes
TRAN.CRD-FIID	Fill with zeroes
TRAN.ACCT	Fill with ACCT-ID-1
TRAN.RESP-CDE	Use RESP-CDE and Response Code Mapping Table to fill
TRAN.TRACK2	If PT-SRV-ENTRY-MODE NOT = "01", "manual", Fill with ";" (start sentinel), TRACK2.DATA-VALUE and "?" (end sentinel). Else Fill with "M" (start sentinel), PAN.DATA-VALUE, "=", EXPIRY-DAT and "?" (end sentinel)
TRAN.MBR-NUM	Fill from CRD-SEQ-NUM
TRAN.EXP-DAT	Fill with EXPIRY-DAT
TRAN.AMT-1	If this is a request, fill with TXN-AMT. If this is a response to a pre-auth or bal inq, fill with LEDG-BAL
TRAN.AMT-2	If this is a request, fill with CASH-AMT. Otherwise if this is a response to a pre authorisation or balance inquiry fill with ACCT-BAL-CLR-FNDS
TRAN.PRE-AUTH-HLD	If this is a pre auth request fill with PCT.PRE-AUTH-HLD
TRAN.APPRV-CDE-LGTH	If this is a request fill with PCT.APPRV-CDE-LGTH
TRAN.APPRV-CDE	If this is a response, fill with AUTH-ID-RESP
TRAN.ICHG-RESP	If this is a response, fill with RESP-CDE left justified and blank filled
TRAN-DAT	Fill with LOCAL-TXN-DAT
TRAN-TIM	Fill with LOCAL-TXN-TIM

ACQ-ICHG-SETL-DAT	If this is a request fill with SETL-DAT
ISS-ICHG-SETL-DAT	If this is a response fill with SETL-DAT
RVSL-CDE	Fill with zeroes
PT-SRV-COND-CDE	Fill with PT-SRV-COND-CDE
PT-SRV-ENTRY-MDE	Fill with PT-SRV-ENTRY-MODE
PIN, PIN-SIZE, PIN-KEY, PIN-FRMT, PINPAD-CHAR, ANSI-OFST	Use conversion routines for PIN processing. For hardware support no translation takes place, current PIN session key is placed in PSTM.PIN-KEY
PIN-TRIES	If this is a response and RESP-CDE = "201", incorrect PIN, increment this number by one, otherwise fill with "Z"
HOST-TRACE-NUM	Fill with zeroes
ENTRY-TIM,EXIT-TIM,RE-ENTRY-TIM	If this is a request, fill ENTRY-TIM with JULIAN TIMESTAMP. If this is a response, fill RE-ENTRY-TIM with JULIAN TIMESTAMP
FRWD-INST-ID-NUM	Fill with FWD-INST-ID-CDE
CRD-ACCEPT-ID-NUM	Fill with CARD-ACCEPT-ID-CDE
CRD-ISS-ID-NUM	Fill with zeroes
NUM-SERVICES	If this is a request, fill with PCT.POS.NUM-SRVCS
SRVCS	If this is a request, fill with PCT.POS-ALLOWED-SRVCS for the number of occurrences in PCT.POS.NUM-SRVCS

APPENDIX C

Log Messages

Log messages are routed to certain destinations, based on the routing code that appears in the log messages. The SAA AS2805 Interface process routes all of its log messages with a standard routing code of zero.

The SAA Interface process generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

SAA AS2805 INTERCHANGE INTERFACE PROCESS LOG MESSAGE MANUAL

This appendix contains documentation for the log messages that can be generated by the SAA AS2805 Interface process. The product module ID (PMID) for the SAA AS2805 Interface process is SWSAA. Log messages are documented, in numeric order, by log message number.

NUM: Log Message Number

SEV: Severity code of the log message. The meaning of these codes are:

0 = Informational

1 = Warning

2 = Error

3 = Critical

TXT: `Text of the log message`

VAR: Definition of the variables, if any, in the log message

CAUSE: Cause of the message

REMEDY: The remedy gives instructions, when appropriate, about what actions could be taken to correct the situation that caused the log message to be generated. When the action requires additional support, the remedy instructs the network operator to contact their software support staff, their system manager or HELP24, depending on the level of support required and the seriousness of the situation. Some log messages do not require any action and, when this is the case, the remedy states this.

Note: The same log message can be logged under different situations (though the log message number should be unique). Since this manual describes all messages that can be logged, some duplication might become apparent.

NUM: 0005

SEV: 2

TXT: `Interprocess communication with \F, error \E`
`VAR:..\F = Filename`
`VAR:..\E = Guardian error number`

CAUSE: The SAA AS2805 Interface process received a communication failure other than a timeout.

REMEDY: If this error continues to occur, contact your software support staff.

NUM: 0010

SEV: 2

TXT: `Invalid message received from \S`
`VAR:  \S = Station name`

CAUSE: The SAA AS2805 Interface process received a message from the indicated station that could not be processed because the interface could not determine the message type. This message is followed by another message which is a dump of the message found to be invalid.

REMEDY: Contact your software support staff to determine the origin of the message, its contents and its intended destination. If this error continues contact your operations manager.

NUM: 0011

SEV: 2

TXT: `<text>`
`VAR:  <text> =  BASE24-pos Internal Message`

CAUSE: The SAA AS2805 Interface process received a message from the indicated station that could not be processed because the interface could not determine the message type. The message is dumped to the log after Log Message Number 0010.

REMEDY: Determine the contents and intended destination of the message. If this error continues to occur, contact your software support staff.

NUM: 0012

SEV: 0

TXT: `Key Exchange Type must be Dynamic Keks for this Command`

CAUSE: The SAA AS2805 Interface process received an 0800 Network Management KEKCHANGE request, but the TYPE OF KEY EXCHANGE field in ICF Screen 12 was not set to 4 (Dynamic KEK exchange will be used).

REMEDY: No action is required. Command is ignored.

NUM: 0013
SEV: 0
TXT: `Unable to perform Kekchange - Link is DOWN`
CAUSE: The SAA AS2805 Interface process received an 0800 Network Management KEKCHANGE request, but the link was down.
REMEDY: No action is required. Command is ignored. Bring the link to an UP status to allow KEKCHANGE requests to be processed.

NUM: 0015
SEV: 2
TXT: `Invalid message received from \F`
`VAR:  \F = Process name`
CAUSE: The SAA AS2805 Interface process received a message, from a process other than the XPNET Node, that it did not recognise as a valid message. The message was sent to the log without being processed further. This message is followed by another message, which is a dump of the message found to be invalid.

REMEDY: Contact your software support staff to determine the origin of the message, its contents, and its intended destination. If this error continues to occur, contact your operations manager.

NUM: 0016
SEV: 2
TXT: `<text>`
`VAR:  <text> =  BASE24-pos Internal Message`
CAUSE: The SAA AS2805 Interface process received a message, from a process other than the XPNET Node, that it did not recognise as a valid message. The message is dumped to the log after Log Message Number 0015.
REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0020
SEV: 2
TXT: `Invalid message received from \F`
`VAR:  \F = Process name`
CAUSE: The SAA AS2805 Interface process received a message that it did not recognise as a valid message. The message is sent to the log without being processed further. This message is followed by another message, which is a dump of the message found to be invalid.

REMEDY: Contact your software support staff to determine the origin of the message, its contents, and its intended destination.

NUM: 0021

SEV: 2

TXT: `<text>`

`VAR: <text> = BASE24-pos Internal Message`

CAUSE: The SAA AS2805 Interface process received a message that it did not recognise as a valid message. The message is dumped to the log after Log Message Number 0020.

REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0025

SEV: 3

TXT: `NETWORK initialization failed`

CAUSE: The SAA AS2805 Interface process abended because the call to NETINIT failed.

REMEDY: Contact your software support staff.

NUM: 0030

SEV: 3

TXT: `Not all necessary ASSIGNS and PARAMS defined`

CAUSE: During initialisation or warmboot, the SAA AS2805 Interface process successfully opened and read the Logical Network Configuration File (LCONF) but determined that required assigns and parameters were missing, or invalid.

REMEDY: Check the log for related error messages indicating the assign(s) and/or parameter(s) that were missing or invalid. The SAA Interface process must be restarted after the required changes to the LCONF have been made.

NUM: 0035

SEV: 3

TXT: `LOGGER initialization failed`

CAUSE: The SAA AS2805 Interface process abended because a call to the SKEL log procedure LOG^INIT failed. During LOG^INIT the process attempts to open the log file. If the log cannot be opened, the process attempts to open an alternate log device (default $0). If the alternate log device cannot be opened, the process abends.

REMEDY: Check the log for related error messages indicating the specific reason(s) the SAA AS2805 Interface process abended. Correct the specified errors and attempt to initialise the process again. For additional assistance, contact your software support staff.

NUM: 0040

SEV: 0

TXT: `Copyrighted by ACI Worldwide, Inc.  1988`

CAUSE: During process initialisation, the SAA AS2805 Interchange Interface logs an informational message regarding the copyright of the process.

REMEDY: No action is required.

NUM: 0043

SEV: 1

TXT: `ATM record not found in the TKN file - all STM Tokens will be Logged to the ILF`

CAUSE: An ATM record detailing which tokens should not be logged to the BASE24 Transaction Log File could not be located.

REMEDY: Add a valid ATM Transaction Log File record to the TKN for the SAA AS2805 Interchange Interface.

NUM: 0044

SEV: 1

TXT: `POS record not found in the TKN file - all PSTM Tokens will be Logged to the ILF`

CAUSE: A POS record detailing which tokens should not be logged to the BASE24 POS Transaction Log File could not be located.

REMEDY: Add a valid POS Transaction Log File record to the TKN for the SAA AS2805 Interchange Interface.

NUM: 0045

SEV: 0

TXT: `No reports will be run by this process`

CAUSE: During initialisation or warmboot, the SAA AS2805 Interface process did not find the assigns and parameters in the Logical Network Configuration File (LCONF) required to generate interchange reports. The process assumes no reports are to be generated.

REMEDY: No action is required. However, if reports should be generated for the SAA AS2805 Interface process, review the LCONF to ensure the following assigns and parameter are present, and valid, for this specific SAA AS2805 Interface process:

Assigns: SW-RPT-ERROR-FILE,

SW-RPT01-PROG-NAME,

SW-RPT09-PROG-NAME.

Param: SW-RPT-STARTUP

NUM: 0050

SEV: 0

TXT: `PROCESS initialized`

CAUSE: The SAA AS2805 Interface process successfully completed initialisation.

REMEDY: No action is required.

NUM: 0055

SEV: 3

TXT: `OPEN failed, file \F, error \E`
`VAR: \F = LCONF filename`
`VAR: \E = Guardian error number`

CAUSE: The SAA AS2805 Interface process abended because a call to the LCONF utility procedure LCONF^INIT failed. During process initialisation, LCONF^INIT must open the Logical Network Configuration File so that all appropriate assigns and params can be read by the SAA AS2805 Interchange Interface. If the LCONF cannot be opened the process abends.

REMEDY: Obtain the Guardian Error Number from the log message in order to determine the appropriate course of action. Correct the specified error and attempt to initialise the process again. For additional assistance, contact your software support staff.

NUM: 0056

SEV: 3

TXT: `LMCF assign not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Log Message Configuration File (LMCF) assign in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid LMCF assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0057

SEV: 3

TXT: `ICF assign not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Interchange Configuration File (ICF) assign in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid ICF assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0058
SEV: 3
TXT: `KEYF assign not found in \F`
`VAR:  \F = LCONF filename`
CAUSE: The SAA AS2805 Interface process could not find the Key File (KEYF) assign in the Logical Network Configuration File (LCONF).
REMEDY: Add a valid KEYF assign to the LCONF and restart the SAA AS2805 Interface process.

NUM: 0059
SEV: 3
TXT: `ILF assign not found in \F`
`VAR:  \F = LCONF filename`
CAUSE: The SAA AS2805 Interface process could not find the Interchange Log File (ILF) assign in the Logical Network Configuration File (LCONF).
REMEDY: Add a valid ILF assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0060
SEV: 3
TXT: `SAF assign not found in \F`
`VAR:  \F = LCONF filename`
CAUSE: The SAA AS2805 Interface process could not find the Store and Forward File (SAF) assign, for this process, in the Logical Network Configuration File (LCONF).
REMEDY: Add a valid SAF assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0061
SEV: 3
TXT: `EMF assign not found in \F`
`VAR:  \F = LCONF filename`
CAUSE: The SAA AS2805 Interface process could not find the External Message File (EMF) assign in the Logical Network Configuration File (LCONF).
REMEDY: Add a valid EMF assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0062

SEV: 3

TXT: `RPT assign not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Reports File (RPT) assign in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid RPT assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0063

SEV: 3

TXT: `RPT-ERROR-FILE assign not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Report Error File assign, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid RPT-ERROR-FILE assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0064

SEV: 1

TXT: `SW-RPT-OBEY-FILE assign not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Report Obey File assign, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SW-RPT-OBEY-FILE assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0065

SEV: 1

TXT: `SW-RPT01-PROG-NAME assign not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Interchange Clearings Statement Report assign, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SW-RPT01-PROG-NAME assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0066

SEV: 1

TXT: `SW-RPT09-PROG-NAME assign not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Interchange Transaction Detail Report assign, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SW-RPT09-PROG-NAME assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0067

SEV: 1

TXT: `SW-ILF-FILE-RETENTION param not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Interchange Log File retention period assign, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SW-ILF-FILE-RETENTION assign to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0068

SEV: 1

TXT: `SW-RPT-STARTUP param not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Report Obey File Start Time parameter, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SW-RPT-STARTUP parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0069

SEV: 1

TXT: `SPAN-FLAG param not found in \F`
`VAR: \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SPAN flag parameter, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SPAN-FLAG parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0070

SEV: 1

TXT: `DATE-DISPLAY-TYPE param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Date Display Type parameter, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid DATE-DISPLAY-TYPE parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0071

SEV: 1

TXT: `ACCT-NUM-TYP param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Account Number Type parameter, for this process, in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid ACCT-NUM-TYP parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0072

SEV: 1

TXT: `SAA-ILF-NOTIFY-INTERVAL param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA Interchange Log File (ILF) Notify Interval parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SAA-ILF-NOTIFY-INTERVAL parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0073

SEV: 1

TXT: `SAA-ILF-NOTIFY-PERCENTAGE param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA Interchange Log File (ILF) Notify Percentage parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SAA-ILF-NOTIFY-PERCENTAGE parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0074

SEV: 1

TXT: `SAA-SAF-NOTIFY-INTERVAL param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA Store And Forward (SAF) Notify Interval parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SAA-SAF-NOTIFY-INTERVAL parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0075

SEV: 1

TXT: `SAA-SAF-NOTIFY-PERCENTAGE param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA Store And Forward (SAF) Notify Percentage parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid SAA-SAF-NOTIFY-PERCENTAGE parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0076

SEV: 1

TXT: `FRWD-INST-ID-CDE param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Forwarding Institution Identification Number parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid FRWD-INST-ID-CDE parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0077

SEV: 1

TXT: `RCV-INST-ID-CDE param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Receiving Institution Identification Number parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid RCV-INST-ID-CDE parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0078

SEV: 1

TXT: `SAA-PROTOCOL-TYP set to default of NO ETX`

CAUSE: The SAA AS2805 Interface process could not find the SAA Protocol Type parameter in the Logical Network Configuration File (LCONF).

REMEDY: No action is necessary if special protocol processing is not required. However, if an ETX character should be appended to all messages, then add a valid SAA-PROTOCOL-TYP parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0079

SEV: 2

TXT: `Received invalid response code \\ to KEK change`
`VAR: \\ = invalid response code`

CAUSE: The SAA AS2805 Interface process received an 0810 Network Management KEKCHANGE acknowledgment and the response code was not approved.

REMEDY: Use the operator log to determine the value of the Response Code. Contact the Co-network partner to determine the reason for sending the specific Response Code. For further assistance, contact your software support staff.

NUM: 0080

SEV: 1

TXT: `Token assign not found in \F`

CAUSE: The Token File (TKN) assign could not be located in the LCONF and the SAA AS2805 Interchange Interface is operating in a Release 5.0 environment.

REMEDY: If the SAA AS2805 Interchange Interface should be operating in a Release 5.0 or 5.1 environment, add a valid TKN assign to the LCONF, else change the BASE24-REL parameter to 40 instead of 50.

NUM: 0085

SEV: 2

TXT: `SW-ILF-FILE-RETENTION param contains invalid data`

CAUSE: The SAA AS2805 Interface process successfully read the Interchange Log File (ILF) Retention parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SW-ILF-FILE-RETENTION parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0086
SEV: 2
TXT: `ATM-LN-CUTOVER param contains invalid data`
CAUSE: The SAA AS2805 Interface process successfully read the ATM Logical Network Cutover parameter and determined that it contained invalid data.
REMEDY: Correct the data in the ATM-LN-CUTOVER parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0089
SEV: 2
TXT: `POS-LN-CUTOVER param contains invalid data`
CAUSE: The SAA AS2805 Interface process successfully read the POS Logical Network Cutover parameter and determined that it contained invalid data.
REMEDY: Correct the data in the POS-LN-CUTOVER parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0090
SEV: 2
TXT: `SW-RPT-STARTUP param contains invalid data`
CAUSE: The SAA AS2805 Interface process successfully read the Report Obey File Start Time parameter and determined that it contained invalid data.
REMEDY: Correct the data in the SW-RPT-STARTUP parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0091
SEV: 1
TXT: `SPAN-FLAG param contains invalid data`
CAUSE: The SAA AS2805 Interface process successfully read the Span Flag parameter and determined that it contained invalid data.
REMEDY: Correct the data in the SPAN-FLAG parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0092
SEV: 1
TXT: `DATE-DISPLAY-TYPE param contains invalid data`
CAUSE: The SAA AS2805 Interface process successfully read the Date Display Type parameter and determined that it contained invalid data.
REMEDY: Correct the data in the DATE-DISPLAY-TYPE parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0093
SEV:	1
TXT:	`ACCT-NUM-TYP param contains invalid data`
CAUSE:	The SAA AS2805 Interface process successfully read the Account Number Type parameter and determined that it contained invalid data.
REMEDY:	Correct the data in the ACCT-NUM-TYP parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0094
SEV:	2
TXT:	`AFS-LN-CUTOVER param contains invalid data`
CAUSE:	The SAA AS2805 Interface process successfully read the AFS Logical Network Cutover parameter and determined that it contained invalid data.
REMEDY:	Correct the data in the AFS-LN-CUTOVER parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0095
SEV:	1
TXT:	`SAA-ILF-NOTIFY-INTERVAL param contains invalid data`
CAUSE:	The SAA AS2805 Interface process successfully read the SAA Interchange Log File (ILF) Notify Interval parameter and determined that it contained invalid data.
REMEDY:	Correct the data in the SAA-ILF-NOTIFY-INTERVAL parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0096
SEV:	1
TXT:	`SAA-ILF-NOTIFY-PERCENTAGE param contains invalid data`
CAUSE:	The SAA AS2805 Interface process successfully read the SAA Interchange Log File (ILF) Notify Percentage parameter and determined that it contained invalid data.
REMEDY:	Correct the data in the SAA-ILF-NOTIFY-PERCENTAGE parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0097
SEV:	1
TXT:	`SAA-SAF-NOTIFY-INTERVAL param contains invalid data`
CAUSE:	The SAA AS2805 Interface process successfully read the SAA Store And Forward (SAF) Notify Interval parameter and determined that it contained invalid data.
REMEDY:	Correct the data in the SAA-SAF-NOTIFY-INTERVAL parameter and restart the SAA AS2805 Interchange Interface.

NUM:	0098

SEV: 1

TXT: `SAA-SAF-NOTIFY-PERCENTAGE param contains invalid data`

CAUSE: The SAA AS2805 Interface process successfully read the SAA Store And Forward (SAF) Notify Percentage parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA-SAF-NOTIFY-PERCENTAGE parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0100

SEV: 2

TXT: `Kekchange received with invalid length for field 48 =#`

`VAR:  # = KEK Key (SEM field P-48)`

CAUSE: The SAA AS2805 Interface process could not extract the KEK Key (encrypted under the public key) and its corresponding KVC from field 48 of the SEM due to invalid KEK and KVC length received from the Co-network partner.

REMEDY: Contact your software support staff.

NUM: 0101

SEV: 1

TXT: `SAA-PROTOCOL-TYP param contains invalid data`

CAUSE: The SAA AS2805 Interface process successfully read the SAA Protocol Type parameter and determined that it contained invalid data.

REMEDY: No action is necessary if special protocol processing is not required. However, if an ETX character should be appended to all messages, then correct the data in the SAA-PROTOCOL-TYP parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0102

SEV: 1

TXT: `SAA-DEFAULT-CRD-LN PARAM LENGTH TOO LONG`

CAUSE: The SAA AS2805 Interface process successfully read the SAA Default Card Logical Network parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA-DEFAULT-CRD-LN parameter and restart the SAA AS2805 Interchange Interface.

NUM: 0103

SEV: 1

TXT: `SAA-LOGON-ACTION param invalid - must be 0, 1, or 2`

CAUSE: The SAA AS2805 Interface process successfully read the SAA logon action parameter and determined that it contained invalid data.

REMEDY: Change the value in the SAA-LOGON-ACTION parameter to "0", "1" or "2" and restart the SAA AS2805 Interchange Interface.

NUM: 0104

SEV: 1

TXT: `CVV-FLAG param contains invalid data`

CAUSE: The SAA AS2805 Interface process successfully read the SAA Card Verification Value Flag parameter and determined that it contained invalid data.

REMEDY: Change the value in the CVV-FLAG parameter to "Y" or "N" and restart the SAA AS2805 Interchange Interface.

NUM: 0105

SEV: 0

TXT: `Function NODEKEKTRANS failed decrypting kek keys`

CAUSE: The SAA AS2805 Interface process cannot perform a NODEKEKTRANS function because of error reported by secutils.

REMEDY: Ensure NODEKEKTRANS environment is performing as required. For further assistance, contact your software support staff.

NUM: 0106

SEV: 0

TXT: `Outbound KEK key has been updated for this I/C`

CAUSE: The SAA AS2805 Interchange Interface process received an 0810 Network Management KEKCHANGE acknowledgment indicating a successful KEKCHANGE.

REMEDY: Acknowledgment that the KEKCHANGE request was successful. No action required.

NUM: 0110

SEV: 1

TXT: `ATM-LN-CUTOVER param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the ATM Logical Network Cutover parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid ATM-LN-CUTOVER parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0111

SEV: 1

TXT: `POS-LN-CUTOVER param not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the POS Logical Network Cutover parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid POS-LN-CUTOVER parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0112

SEV: 1

TXT: `ATM-LN-CUTOVER param not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the ATM Logical Network Cutover parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid ATM-LN-CUTOVER parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0113

SEV: 1

TXT: `POS-LN-CUTOVER param not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the POS Logical Network Cutover parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid POS-LN-CUTOVER parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0114

SEV: 1

TXT: `AFS-LN-CUTOVER param not found in \F`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the AFS Logical Network Cutover parameter in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid AFS-LN-CUTOVER parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 0115

SEV: 0

TXT: `Inbound KEK key has been updated for this I/C`

CAUSE: The SAA AS2805 Interchange Interface process received an 0800 Network Management KEKCHANGE request and sent an 0810 indicating success.

REMEDY: KEKCHANGE request processed successfully. No action required.

NUM: 0116

SEV: 0

TXT: `<text>`

VAR: `<text>` = Error description returned by FORMAT^SEG^LOG^MSG

CAUSE: During initialisation, the SAA AS2805 Interface process could not disassemble its Interchange Configuration File (ICF) record.

REMEDY: Take the appropriate action based on the error description. For further assistance, contact your software support staff.

NUM: 0120

SEV: 3

TXT: `Extended memory allocation error #, <text>`

`VAR: # =  Error returned from HISWMEM^INIT`

`VAR: <text> = Description of reason for error`

CAUSE: During initialisation, the SAA AS2805 Interface process could not allocate the required amount of Extended Memory because of the indicated HISWMEM^INIT error and description.

REMEDY: Take the appropriate action (based on the reported HISWMEM^INIT error and description) and restart the process. For additional assistance, contact your software support staff.

NUM: 0125

SEV: 3

TXT: `<text>`

`VAR: <text> = Error description returned from TMPL^CREATE^LIKE`

CAUSE: During initialisation, the SAA AS2805 Interface process failed to create its Store-and-Forward file (SAF). A non-Guardian error occurred on a call to TMPL^CREATE^LIKE.

REMEDY: Determine appropriate action to be taken from the error description returned by TMPL^CREATE^LIKE. Ensure the Logical Network Configuration File contains a valid assign for the Store-and-Forward file (SAF) and its Template. For additional assistance contact your software support staff.

NUM: 0129

SEV: 3

TXT: `TKEY date \****** does not match ICF date`

`VAR:  \*'s = Tape Key File's creation date`

CAUSE: During initialisation, the SAA AS2805 Interface process opens and reads the first record in the Tape Key File. If the tape creation date in the Tape Key File's header record does not equal the tape date on ICF screen 12 the wrong TKEY file is being used.

REMEDY: Use the correct Tape Key File, or if the date on ICF screen 12 is incorrect - amend it, and warmboot the SAA AS2805 Interface process.

NUM: 0130

SEV: 3

TXT: TKEY date ****** does not match ICF date
VAR: *'s = Tape Key File's creation date

CAUSE: During initialisation, the SAA AS2805 Interface process opens and reads the first record in the Tape Key File. If the tape creation date in the Tape Key File's header record does not equal the tape date on ICF screen 12 the wrong TKEY file is being used.

REMEDY: Use the correct Tape Key File, or if the date on ICF screen 12 is incorrect - amend it and warmboot the SAA AS2805 Interface process.

NUM: 0131

SEV: 3

TXT: Extended memory allocation error #, <text>
VAR: # = Error returned from USER^ALLOCATESEGMENT
VAR: <text> = Description of reason for error

CAUSE: During initialisation, the SAA AS2805 Interface process could not allocate an Extended Memory Segment (for the Process Control Table (PCT), the Key File (KEYF) record and the External Message File (EMF) record), because of the indicated USER^ALLOCATESEGMENT Error and description.

REMEDY: Take appropriate action (based on the reported USER^ALLOCATESEGMENT error and description) and restart the process. For additional assistance, contact your software support staff.

NUM: 0135

SEV: 1

TXT: MESSAGE FAILURE # FROM \S
VAR: # = Message Failure Code
VAR: \S = Station Name

CAUSE: The SAA AS2805 Interface process received an internal message, returned by the XPNET Node, because it could not be delivered to its intended destination.

REMEDY: Contact your software support staff to determine the message contents and intended destination.

NUM: 0136

SEV: 2

TXT: <text>
VAR: <text> = BASE24-atm Internal Message

CAUSE: The SAA AS2805 Interface process received a failed BASE24-atm Standard Internal Message. The message is dumped to the log after Log Message Number 135.

REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0137

SEV: 2

TXT: `<text>`

`VAR:  <text> =  BASE24-pos Internal Message`

CAUSE: The SAA AS2805 Interface process received a failed BASE24-pos Standard Internal Message. The message is dumped to the log after Log Message Number 135.

REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0138

SEV: 1

TXT: `Message failure # from \S`

`VAR:  # = Message Failure Code`

`VAR:  \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an external message, returned by the XPNET Node, because it could not be delivered to its intended destination.

REMEDY: Contact your software support staff to determine the message contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0139

SEV: 2

TXT: `Error converting Iso to Sem field : ###`

`VAR:  ### = External Message Field ID.`

CAUSE: The SAA AS2805 Interface process received a failed external message that could not be successfully expanded into a BASE24 SEM.

REMEDY: Contact your software support staff to determine the message contents and bitmap make-up. If this error continues to occur, contact your software support staff.

NUM: 0140

SEV: 2

TXT: `<text>`

`VAR:   <text> = External (ISO) Message`

CAUSE: The SAA AS2805 Interface process received a failed (ISO) External Message that could not be expanded into a BASE24 SEM. The message is dumped to the log after Log Message Number 139.

REMEDY: Determine the message's contents and bitmap make-up. If this error continues to occur, contact your software support staff.

NUM: 0145
SEV: 2
TXT: `<text>`
`VAR: <text> = BASE24-atm Internal (reversal) Message`
CAUSE: When processing a failed internal response, a reversal could not be formatted successfully and is dumped to the log.
REMEDY: Use the operator log to determine which field in the BASE24-atm Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0146
SEV: 2
TXT: `<text>`
`VAR: <text> = BASE24-atm Internal (reversal) Message`
CAUSE: When processing a failed internal response, a reversal could not be formatted successfully and is dumped to the log.
REMEDY: Use the operator log to determine which field in the BASE24-atm Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0150
SEV: 2
TXT: `<text>`
`VAR: <text> = BASE24-pos Internal (reversal) Message`
CAUSE: When processing a failed internal response, a reversal could not be formatted successfully and is dumped to the log.
REMEDY: Use the operator log to determine which field in the BASE24-pos Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0151
SEV: 2
TXT: `<text>`
`VAR: <text> = BASE24-pos Internal (reversal) Message`
CAUSE: When processing a failed internal response, a reversal could not be formatted successfully and is dumped to the log.
REMEDY: Use the operator log to determine which field in the BASE24-pos Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0155

SEV: 2

TXT: `<text>`

VAR: `<text> = BASE24-atm Internal (reversal) Message`

CAUSE: When processing a failed external request, a reversal could not be formatted successfully and is dumped to the log.

REMEDY: Use the operator log to determine which field in the BASE24-atm Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0160

SEV: 2

TXT: `<text>`

VAR: `<text> = BASE24-pos Internal (reversal) Message`

CAUSE: When processing a failed external request, a reversal could not be formatted successfully and is dumped to the log.

REMEDY: Use the operator log to determine which field in the BASE24-pos Standard Internal Message was in error and contact your software support staff to take the appropriate action to correct it.

NUM: 0165

SEV: 2

TXT: `Error converting iso to sem field ###`

VAR: `### = External Message Field ID.`

CAUSE: The SAA AS2805 Interface process received an external (ISO) message from the Co-network but failed to expand it into a BASE24 SEM. The first field to fail expansion is specified in the message.

REMEDY: Compare the message's contents with its bitmap make-up. Ensure all the data sent corresponds to that specified by the message's bitmap. Modify the contents and/or bitmap as necessary.

NUM: 0166

SEV: 0

TXT: `BASE24 switch station is now UP (STATION: \S)`

VAR: `\S = Station name`

CAUSE: The SAA AS2805 Interface process checked the connection between the specific station and the interchange.

REMEDY: Confirmation of successful connection. No action required.

NUM: 0170

SEV: 0

TXT: `ATM Completion repsonse received with bad response`

CAUSE: A Completion Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0171

SEV: 0

TXT: `<text>`

`VAR: <text> = BASE24-atm SEM message`

CAUSE: A Completion Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log after Log Message Number 170.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0174

SEV: 0

TXT: `POS Completion response received with bad response`

CAUSE: Completion Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0175

SEV: 0

TXT: `POS 0130 Auth Ack received with bad response`

CAUSE: An Authorisation Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0176

SEV: 0

TXT: `<text>`
`VAR:  <text> = BASE24-pos SEM message`

CAUSE: A Completion Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log after Log Message Number 174.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0177

SEV: 0

TXT: `<text>`
`VAR:  <text> = BASE24-pos SEM message`

CAUSE: An Authorisation Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log after Log Message Number 175.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0180

SEV: 0

TXT: `ATM reversal response received with bad response`

CAUSE: A Reversal Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0181

SEV: 0

TXT: `<text>`
`VAR:  <text> = BASE24-atm SEM message`

CAUSE: A Reversal Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log after Log Message Number 180.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0185

SEV: 0

TXT: `POS reversal response received with bad response`

CAUSE: A Reversal Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0186

SEV: 0

TXT: `<text>`
`VAR: <text> = BASE24-pos SEM message`

CAUSE: A Reversal Acknowledgment was received from the Co-network but the response was not approved. The message is dumped to the log after Log Message Number 185.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0190

SEV: 2

TXT: `Duplicate ATM reconciliation advice received for \\\\`
`VAR: \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the current Settlement cutover date, in the Process Control Table (PCT), indicates that a cutover message has already been received for today's date.

REMEDY: Contact the Interchange partner as well as your software support staff and inform them that a cutover message has been received by this SAA AS2805 Interchange Interface after the process has already cutover.

NUM: 0191

SEV: 2

TXT: `Bad date received in ATM reconciliation advice \\\\`
`VAR: \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the Settlement Date, in the request, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).

REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0195

SEV: 2

TXT: `Duplicate POS reconciliation advice received for \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the current Settlement cutover date, in the Process Control Table (PCT), indicates that a cutover message has already been received for today's date.

REMEDY: Contact the Interchange partner as well as your software support staff and inform them that a cutover message has been received by this SAA AS2805 Interface after the process has already cutover.

NUM: 0196

SEV: 2

TXT: `Bad date received in POS reconciliation advice \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the Settlement Date, in the request, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).

REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0197

SEV: 2

TXT: `Duplicate reconciliation advice received for \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the current Settlement cutover date, in the Process Control Table (PCT), indicates that a cutover message has already been received for today's date.

REMEDY: Contact the Interchange partner as well as your software support staff and inform them that a cutover message has been received by this SAA AS2805 Interchange Interface after the process has already cutover.

NUM: 0198

SEV: 2

TXT: `Bad date received in reconciliation advice \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation request was received from the Co-network but the Settlement Date, in the request, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).

REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0200

SEV: 2

TXT: `Bad date received in ATM reconciliation ack \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation acknowledgment was received from the Co-network but the Settlement Date, in the response, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).

REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0201

SEV: 0

TXT: `ATM reconciliation Ack received for \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation acknowledgment was received from the Co-network.

REMEDY: Informational message. No action required.

NUM: 0202

SEV: 1

TXT: `No timer for reconciliation Ack : key \?`
`VAR:  \? = Primary key identifying the suspended message`

CAUSE: A reconciliation acknowledgment was received from the Co-network but neither a Reconciliation nor a SAF timer could be found for the Reconciliation Response. The message is dropped.

REMEDY: Contact the Co-network partner to determine its reason for sending the specified message. For further assistance, contact your software support group.

NUM: 0203

SEV: 0

TXT: `Mismatch occurred during reconciliation for ATM acquirer totals`

CAUSE: A reconciliation acknowledgment was received from the Co-network but it's settlement code indicates that a reconciliation mismatch has occurred.

REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0204

SEV: 1

TXT: `ATM reconciliation not approved \\`
`VAR: \\ = Response Code`

CAUSE: A reconciliation acknowledgment was received from the Co-network but its response code was not approved.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0210

SEV: 2

TXT: `Bad date received in POS reconciliation ack \\\\`
`VAR: \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation acknowledgment was received from the Co-network but the Settlement Date, in the response, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).

REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0211

SEV: 0

TXT: `POS reconciliation Ack received for \\\\`
`VAR: \\\\ = Settlement Date (MMDD)`

CAUSE: A reconciliation acknowledgment was received from the Co-network.

REMEDY: Informational message. No action required.

NUM: 0213
SEV: 0
TXT: `Mismatch occurred during reconciliation for POS acquirer totals`
CAUSE: A reconciliation acknowledgment was received from the Co-network but its settlement code indicates that a reconciliation mismatch has occurred.
REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0214
SEV: 1
TXT: `POS Reconciliation not approved \\`
`VAR:  \\ = Response Code`
CAUSE: A reconciliation acknowledgment was received from the Co-network but its response code was not approved.
REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0215
SEV: 2
TXT: `Bad date received in reconciliation ack \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`
CAUSE: A reconciliation acknowledgment was received from the Co-network but the Settlement Date, in the request, is not equal to the Settlement Date in the Interface's Process Control Table (PCT).
REMEDY: Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine why the Settlement Dates are not synchronised. Correct the appropriate partner's Interchange Configuration File (ICF) record and warmboot that process.

NUM: 0216
SEV: 0
TXT: `Reconciliation Ack received for \\\\`
`VAR:  \\\\ = Settlement Date (MMDD)`
CAUSE: A reconciliation acknowledgment was received from the Co-network.
REMEDY: Informational message. No action required.

NUM: 0217

SEV: 1

TXT: `No timer for Reconciliation Ack : key \?`

`VAR:  \? = Primary key identifying the suspended message`

CAUSE: A reconciliation acknowledgment was received from the Co-network but neither a Reconciliation nor a SAF timer could be found for the Reconciliation Request. The message is dropped.

REMEDY:

NUM: 0218

SEV: 0

TXT: `Mismatch occurred during reconciliation for acquirer totals`

CAUSE: A reconciliation acknowledgment was received from the Co-network but its settlement code indicates that a reconciliation mismatch has occurred.

REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0219

SEV: 1

TXT: `Reconciliation not approved \\`

`VAR:  \\ = Response Code`

CAUSE: A reconciliation acknowledgment was received from the Co-network but its response code was not approved.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0220

SEV: 1

TXT: `Received invalid 0800 NMM-INFO-CODE \\\; ignored.`

`VAR:  \\\ = Info Code (P-70) from SEM`

CAUSE: The SAA AS2805 Interface process received an 0800 Network Management Message with an invalid NMM Code (ie. neither an echo test nor a logon).

REMEDY: Use the operator log to determine the value of the invalid NMM Code. Contact the Co-network Partner to determine the reason for sending the invalid Code. For further assistance, contact your software support staff.

NUM: 225

SEV: 2

TXT: `Echo test received when station # not logged on`
`VAR:  # = Index into PCT for appropriate station`

CAUSE: The SAA AS2805 Interface process received an 0800 Network Management Echo Test Request for a station that is known but not logged on.

REMEDY: Determine which station the log message is referring to and either log the station on, or contact the Co-network partner to disable the echo test messages.

NUM: 0231

SEV: 0

TXT: `BASE24 switch station is now UP (STATION: \S)`
`VAR:  \S = Station name`

CAUSE: The SAA AS2805 Interface process received an 0800 Network Management logon request for a known station that is currently logged off.

REMEDY: Confirmation of a successful logon. No action required.

NUM: 0232

SEV: 0

TXT: `BASE24 switch link is now UP (STATION: \S)`
`VAR:  \S = Station name`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the specified station and the Interchange.

REMEDY: Confirmation of a successful link connection. No action required.

NUM: 0233

SEV: 0

TXT: `Logon received while Link logged on`

CAUSE: The SAA AS2805 Interface process received an 0800 Network Management logon request for a known station that is already logged on.

REMEDY: Contact the Co-network Partner to determine the reason for sending a logon request.

NUM: 0234

SEV: 2

TXT: `SCM Function NODERESP failed with error ####`
`VAR:  ####  = Error returned from secutils`

CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.

REMEDY: Ensure SCM environment is performing as required.

NUM: 0235

SEV: 1

TXT: `Received invalid 0810 NMM-INFO-CODE \\\; ignored.`
`VAR:  \\\ = Info Code (P-70) from SEM`

CAUSE: The SAA AS2805 Interface process received an 0810 Network Management Message with an invalid NMM Code (ie. neither an echo test nor a logon).

REMEDY: Use the operator log to determine the value of the invalid NMM Code. Contact the Co-network Partner to determine the reason for sending the invalid Code. For further assistance, contact your software support staff.

NUM: 0240

SEV: 2

TXT: `Received invalid response code \\ to echo test`
`VAR:  \\ = Response Code`

CAUSE: The SAA AS2805 Interface process received an 0810 Network Management Echo Test Acknowledgment and the response code was not approved.

REMEDY: Use the operator log to determine the value of the Response Code. Contact the Co-network Partner to determine the reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0245

SEV: 2

TXT: `Invalid response code received to logon : \\`
`VAR:  \\ = Response Code`

CAUSE: The SAA AS2805 Interface process received an 0810 Network Management logon acknowledgment and the response code was not approved.

REMEDY: Use the operator log to determine the value of the Response Code. Contact the Co-network Partner to determine the reason for sending the specified Response Code. For further assistance, contact your software support staff.

NUM: 0246

SEV: 0

TXT: `BASE24 switch station is now UP (STATION: \S)`
`VAR:  \S = Station name`

CAUSE: The SAA AS2805 Interface process received an 0810 Network Management logon acknowledgment for a known station which is currently logged off.

REMEDY: Confirmation of a successful station logon. No action required.

NUM: 0247

SEV: 0

TXT: `BASE24 switch link is now UP`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the specified station and the Interchange.

REMEDY: Confirmation of a successful link connection. No action required.

NUM: 0248

SEV: 3

TXT: `No Proof of Endpoint data`

CAUSE: The SAA AS2805 Interface process could not perform Proof of Endpoint because no data was provided by partner.

REMEDY: Ensure this interchange should be performing Proof of Endpoint and if so contact interchange the partner to ensure they are sending correct data. If problem persists contact your software support staff.

NUM: 0249

SEV: 3

TXT: `Proof of Endpoint Failure`

CAUSE: The SAA AS2805 Interface process failed to verify Proof of Endpoint because data was incorrect.

REMEDY: This error message might happen because a number of logon responses have been queued to the interchange process and the interchange partner has processed old logon requests. The SAA AS2805 Interface process will send a logoff to the interchange partner to try to get to a known state. It will then try the logon sequence again. If problem persists contact your software support staff.

NUM: 0250

SEV: 1

TXT:
```
Received invalid 0820 NMM-INFO-CODE \\\; ignored.
VAR:  \\\ = Info Code (P-70) from SEM
```

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management Message with an invalid NMM Code (ie. neither a logoff nor a key change).

REMEDY: Use the operator log to determine the value of the invalid NMM Code. Contact the Co-network Partner to determine the reason for sending the invalid Code. For further assistance, contact your software support staff.

NUM: 0255

SEV: 1

TXT: `BASE24 Link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management Logoff message.

REMEDY: Confirmation of logoff. No action required.

NUM: 0256

SEV: 1

TXT: `BASE24 Link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process sent an 0820 Network Management Logoff message.

REMEDY: Confirmation of logoff. No action required.

NUM: 0260

SEV: 0

TXT: `<text>`

`VAR:  <text> = Force Post Completion message`

CAUSE: The SAA AS2805 Interface process attempted to format a SEM 0220 completion message from a BASE24-atm 0220 message. The STM is dumped to the log.

REMEDY: Use the operator log to determine which field in the BASE24-atm Standard Internal Message was in error and contact your software support staff to take the appropriate action.

NUM: 0265

SEV: 2

TXT: `<text>`

`VAR:  <text> = Reversal message`

CAUSE: The SAA AS2805 Interface process attempted to format a SEM 0420 reversal message from a BASE24-atm Standard Internal Message (STM) 0420 message. The STM is dumped to the log.

REMEDY: Use the operator log to determine which field in the BASE24-atm Standard Internal Message was in error and contact your software support staff to take the appropriate action.

NUM: 0270

SEV: 0

TXT: `<text>`

`VAR:   <text> = Force Post Completion message`

CAUSE: The SAA AS2805 Interface process attempted to format a SEM 0220 completion message from a BASE24-pos Standard Internal Message 0220 message. The PSTM is dumped to the log.

REMEDY: Use the operator log to determine which field in the PSTM was in error and contact your software support staff to take the appropriate action.

NUM: 0275

SEV: 0

TXT: `<text>`

`VAR:   <text> = Reversal message`

CAUSE: The SAA AS2805 Interface process attempted to format a SEM 0420 reversal message from a BASE24-pos Standard Internal Message 0420 message. The PSTM is dumped to the log.

REMEDY: Use the operator log to determine which field in the PSTM was in error and contact your software support staff to take the appropriate action.

NUM: 0281

SEV: 0

TXT: `*** Logon initiated for all stations by operator ***`

CAUSE: The SAA AS2805 Interface process received a logon all command from the operator's Network Control Screen.

REMEDY: Confirmation that command was received and processed. No action required.

NUM: 0286

SEV: 0

TXT: `*** Link logged off by operator ***`

CAUSE: The SAA AS2805 Interface process received a LOGOFF ALL command from the operator's Network Control Screen.

REMEDY: Confirmation that command was received and processed. No action required.

NUM: 0290

SEV: 0

TXT: `Base24 logged \\\, Co-network logged \\\`

`VAR:  \\\ = Base24 logged`

`VAR:  \\\ = Co-network logged`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0291

SEV: 0

TXT: `ICF indicates no security keys being used`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0292

SEV: 0

TXT:
```
ICF indicates \\\\\\\\\\\\ session keys
VAR:  \\\\\\\\\\\\ = Static
VAR:  \\\\\\\\\\\\ = Tape File
VAR:  \\\\\\\\\\\\ = Dynamic
VAR:  \\\\\\\\\\\\ = Dynamic KEK
```

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0294

SEV: 0

TXT: `Base24 session keys      : NOT YET GENERATED`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0295

SEV: 0

TXT: `Base24 session keys     : OK`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0296

SEV: 0

TXT: `Co-network Session keys : NOT YET RECEIVED`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0297

SEV: 0

TXT: `Co-network Session keys : OK`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0300

SEV: 0

TXT: `Link is \\\\`
`VAR:  \\\\ = Text: " UP " or "DOWN"`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0301

SEV: 0

TXT: `BASE24-atm posting date: \\\\\\`
`VAR:  \\\\\\ = BASE24-atm Posting Date`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0302

SEV: 0

TXT: `BASE24-pos posting date: \\\\\\`
`VAR:  \\\\\\ = BASE24-pos Posting Date`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0303

SEV: 0

TXT: `Co-network posting date: \\\\\\`
`VAR:  \\\\\\ = Co-network Posting Date`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0304

SEV: 0

TXT: `# outbound requests pending of # maximum requests`
`VAR:  # = Number of outbound requests pending`
`VAR:  # = Maximum number of outbound requests allowed`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0305

SEV: 0

TXT: `# inbound requests pending of # maximum requests`
`VAR:  # = Number of inbound requests pending`
`VAR:  # = Maximum number of inbound requests allowed`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0306

SEV: 0

TXT: `# timers active of # maximum timers`
`VAR:  # = Number of timers currently running`
`VAR:  # = Maximum number of timers allowed to be running at once`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: o action required.

NUM: 0307

SEV: 0

TXT: `current ILF filename: \F`
`VAR:  \F = Institution Log File filename`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0308

SEV: 0

TXT: `# %full, notify interval (#), notify % (#)`
`VAR:  # = Percentage full of current ILF`
`VAR:  # = ILF Notify Interval`
`VAR:  # = ILF Notify Percentage`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0309

SEV: 0

TXT: `SAF filename: \F`
`VAR:  \F = Store and Forward filename`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0310
SEV: 0
TXT: # %full, notify interval (#), notify % (#)
VAR: # = Percentage full of SAF
VAR: # = SAF Notify Interval
VAR: # = SAF Notify Percentage
CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.
REMEDY: No action required.

NUM: 0311
SEV: 0
TXT: $ SAF records on file
VAR: $ = Number of records in the SAF
CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.
REMEDY: No action required.

NUM: 0312
SEV: 0
TXT: SAF cycle in progress with # retries
VAR: # = Number of SAF retries
CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.
REMEDY: No action required.

NUM: 0313
SEV: 0
TXT: Station \S is \\\\ Type: \\\\\\\\\\\\
VAR: \S = Station name
VAR: \\\\ = Text: " UP " or "DOWN"
VAR: \\\\\\\\\\\\ = TEXT : “SEND/RECEIVE” or “RECEIVE ONLY”
CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.
REMEDY: No action required.

NUM: 0314
SEV: 0
TXT: >>>>> Reconciliation recovery in progress <<<<<
CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.
REMEDY: No action required.

NUM: 0315

SEV: 0

TXT: `BASE24-atm Release \\\;  BASE24-pos Release \\\`

CAUSE: The SAA AS2805 Interface process received a detailed STATUS command.

REMEDY: No action required.

NUM: 0316

SEV: 0

TXT: `Link is \\\\`

`VAR:` \\\\ = Text "UP" or "DOWN"

CAUSE: The SAA AS2805 Interface process received an abbreviated STATUS command.

REMEDY: No action required.

NUM: 0317

SEV: 0

TXT: `Co-network posting date: \\\\\\`

`VAR:  \\\\\\ = Co-network Posting Date`

CAUSE: The SAA AS2805 Interface process received an abbreviated STATUS command.

REMEDY: No action required.

NUM: 0318

SEV: 0

TXT: `$ SAF records on file`

`VAR:  $ = Number of records in the SAF`

CAUSE: The SAA AS2805 Interface process received an abbreviated STATUS command.

REMEDY: No action required.

NUM: 0319

SEV: 0

TXT: `Station \S is \\\\ Type : \\\\\\\\\\\\`

`VAR:  \S = Station name`

`VAR:  \\\\ = Text: " UP " or "DOWN"`

`VAR:  \\\\\\\\\\\\ = TEXT : “SEND/RECEIVE” or “RECEIVE ONLY”`

CAUSE: The SAA AS2805 Interface process received an abbreviated STATUS command.

REMEDY: No action required.

NUM: 0320

SEV: 0

TXT: `*** \\\ Performance Statistics Summary ***`
`VAR:  \\\ = Text: "ATM"`

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0321

SEV: 0

TXT: `Minutes in period: #   Outbound rqst limit:  #`
`VAR:  # = Number of minutes in each performance period`
`VAR:  # = Maximum number of outbound requests allowed`

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0322

SEV: 0

TXT: `         REQUESTS - TIMEOUTS - PERCENT - AVERAGE`

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0323

SEV: 0

TXT: `PERIOD #   $$$$       $$$$       ###%     ##.##`
`VAR:  # = Period identifier`
`VAR:  $$$$ = Number of outbound requests generated in period`
`VAR:  $$$$ = Number of timeouts processed in period`
`VAR:  ###% = Timeouts as percent of outbound requests`
`VAR:  ##.## = Average response time (in seconds)`

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0324

SEV: 0

TXT: `Minutes in period: #   Inbound  rqst limit:  #`
`VAR:  # = Number of minutes in period`
`VAR:  # = Maximum number of inbound requests allowed`

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0325

SEV: 0

TXT: REQUESTS - TIMEOUTS - PERCENT - AVERAGE

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0326

SEV: 0

TXT:
```
PERIOD #   $$$$       $$$$       ###%     ##.##
VAR: # = Period identifier
VAR:$$$$=Number of inbound requests processed in period
VAR: $$$$ = Number of timeouts processed in period
VAR: ###% = Timeouts as percent of inbound requests
VAR: ##.## = Average response time (in seconds)
```

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0327

SEV: 0

TXT:
```
*** \\\ Performance Statistics Summary ***
VAR:  \\\ = Text: "POS"
```

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0328

SEV: 0

TXT:
```
Minutes in period: #   Outbound rqst limit:  #
VAR:  # = Number of minutes in period
VAR:  # = Maximum number of outbound requests allowed
```

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0329

SEV: 0

TXT: REQUESTS - TIMEOUTS - PERCENT - AVERAGE

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0330

SEV: 0

TXT:
```
PERIOD #   $$$$       $$$$       ###%     ##.##
```

VAR: # = Period identifier
VAR: $$$$ = Number of outbound requests generated in period
VAR: $$$$ = Number of timeouts processed in period
VAR: ###% = Timeouts as percent of outbound requests
VAR: ##.## = Average response time (in seconds)

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0331

SEV: 0

TXT: Minutes in period: #   Inbound rqst limit: #
VAR: # = Number of minutes in period
VAR: # = Maximum number of inbound requests allowed

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0332

SEV: 0

TXT: REQUESTS - TIMEOUTS - PERCENT - AVERAGE

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0333

SEV: 0

TXT: PERIOD # $$$$ $$$$ ###%   ##.##
VAR: # = Period identifier
VAR: $$$$=Number of inbound requests received in period
VAR: $$$$ = Number of timeouts processed in period
VAR: ###% = Timeouts as percent of inbound requests
VAR: ##.## = Average response time (in seconds)

CAUSE: The SAA AS2805 Interface process received a PERFORMANCE command.

REMEDY: No action required.

NUM: 0340
SEV: 2
TXT: `Warmboot failed - previous status retained`
CAUSE: During warmboot, the SAA AS2805 Interface process fails to successfully re-process the LCONF.
REMEDY: Look for further error messages indicating specific reasons why the warmboot failed. Contact your software support staff to have the initialisation parameters verified and corrected.

NUM: 0341
SEV: 2
TXT: `Warmboot failed - previous status retained`
CAUSE: During warmboot, the SAA AS2805 Interface process fails to successfully re-read the ICF, KEYF or EMF, or re-initialisation does not complete successfully.
REMEDY: Look for further error messages indicating specific reasons why the warmboot failed. Contact your software support staff to have the initialisation parameters verified and corrected.

NUM: 0342
SEV: 0
TXT: `Warmboot successful - new status implemented`
CAUSE: The SAA AS2805 Interface process received a warmboot, or 9501, command from the operator's Network Control Screen.
REMEDY: Confirmation that the command was received and processed successfully. No action required.

NUM: 0343
SEV: 1
TXT: `Warmboot could not access TKN file`
CAUSE: During warmboot, the SAA AS2805 Interface process could not access the TKN file.
REMEDY: No action required.

NUM: 0345

SEV: 0

TXT:
```
Station \S is now \*******************
VAR:  \S = Station name
VAR:  \*'s = Text   (depends on command):
 "MARKED DOWN" - 9510
 "MARKED UP"   - 9511
 "SUSPENDED"   - 9512
 "REINSTATED"  - 9513
 "NOT BEING POLLED"  - 9516
 "BEING POLLED"- 9517
 "MARKED UP"   - 9518
 "MARKED DOWN" - 9519
 "IN ABNORMAL STATE" - (catch all)
```

CAUSE: The SAA AS2805 Interface process received a 951x station status message from the operator's Network Control Screen.

REMEDY: Confirmation that the message was received and processed. No action required.

NUM: 0350

SEV: 0

TXT: `*** Request timed out to Base24 ***`

CAUSE: A timer (type: 0, subtype: 1) expired whilst waiting for BASE24 to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process.

NUM: 0351

SEV: 1

TXT: `ATM record not found in the TKN file - all STM Tokens will be Logged to the ILF`

CAUSE: An ATM record detailing which tokens should not be logged to the BASE24 Transaction Log File could not be located during warmboot.

REMEDY: Add a valid ATM Transaction Log File record to the TKN for the SAA AS2805 Interchange Interface.

NUM: 0352

SEV: 1

TXT: `POS record not found in the TKN file - all PSTM Tokens will be Logged to the ILF`

CAUSE: A POS record detailing which tokens should not be logged to the BASE24 POS Transaction Log File could not be located during warmboot.

REMEDY: Add a valid POS Transaction Log File record to the TKN for the SAA AS2805 Interchange Interface.

NUM: 0355

SEV: 0

TXT: `*** Request timed out to Base24 ***`

CAUSE: A timer (type: 0, subtype: 2) expired whilst waiting for BASE24 to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process.

NUM: 0360

SEV: 0

TXT: `*** Request timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`

CAUSE: A timer (type: 1, subtype: 1) expired whilst waiting for the Co- network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0365

SEV: 0

TXT: `*** Request timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`

CAUSE: A timer (type: 1, subtype: 2) expired whilst waiting for the Co- network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0370

SEV: 0

TXT: `*** Reconciliation timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`

CAUSE: A Reconciliation timer (type: 4) expired whilst waiting for the Co-network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0375
SEV: 0
TXT: `ATM Reconciliation record sent for \\\\\\`
`VAR:  \\\\\\ = Settlement Date`
CAUSE: The ATM Settlement Interval timer (type: 5, subtype: 1) expired.
REMEDY: Confirmation that Settlement has been initiated. No action required.

NUM: 0380
SEV: 0
TXT: `POS Reconciliation record sent for \\\\\\`
`VAR:  \\\\\\ = Settlement Date`
CAUSE: The POS Settlement Interval timer (type: 5, subtype: 2) expired.
REMEDY: Confirmation that Settlement has been initiated. No action required.

NUM: 0382
SEV: 0
TXT: `Reconciliation record sent for \\\\\\`
`VAR:  \\\\\\ = Settlement Date`
CAUSE: The Settlement Interval timer (type: 5) expired and the Reconciliation Flag (on ICF screen 12) is set to "ATM AND POS".
REMEDY: Confirmation that Settlement has been initiated. No action required.

NUM: 0390
SEV: 0
TXT: `*** Completion timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`
CAUSE: A timer (type: 10, subtype: 1) expired whilst waiting for the Co-network.
REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on. If further assistance is required, contact HELP24.

NUM: 0395
SEV: 0
TXT: `*** Completion timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`
CAUSE: A timer (type: 10, subtype: 2) expired whilst waiting for the Co- network.
REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on. If further assistance is required, contact HELP24.

NUM:	0400
SEV:	2

TXT:	`Failure on OPEN^FILE, file \F, error \E`
	`VAR:  \F = Report Obey File filename`
	`VAR:  \E = Error returned by OPEN^FILE`

CAUSE:	The SAA AS2805 Interface process' attempt at creating and opening the Report Obey file failed, with the specified error.
REMEDY:	Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM:	0401
SEV:	1
TXT:	`Unable to run report obey file \F` `VAR:  \F = Report obey file filename`
CAUSE:	The SAA AS2805 Interface process attempted to run its report obey file and failed.
REMEDY:	Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM:	0402
SEV:	2

TXT:	`Failure on WRITE^FILE, file \F, error \E`
	`VAR:  \F = Report Obey File filename`
	`VAR:  \E = Error returned by WRITE^FILE`

CAUSE:	The SAA AS2805 Interface process' attempt at writing to the new Report obey file failed, with the specified error.
REMEDY:	Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM:	0403
SEV:	1
TXT:	`Unable to run report obey file \F` `VAR:  \F = Report obey file filename`
CAUSE:	The SAA AS2805 Interface process attempted to run its report obey file and failed.
REMEDY:	Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0404

SEV: 2

TXT:
```
Failure on WRITE^FILE, file \F, error \E
VAR:  \F = Report Obey File filename
VAR:  \E = Error returned by WRITE^FILE
```

CAUSE: The SAA AS2805 Interface process' attempt at writing to the new Report obey file failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0405

SEV: 1

TXT:
```
Unable to run report obey file \F
VAR:  \F = Report obey file filename
```

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0406

SEV: 2

TXT:
```
Failure on WRITE^FILE, file \F, error \E
VAR:  \F = Report Obey File filename
VAR:  \E = Error returned by WRITE^FILE
```

CAUSE: The SAA AS2805 Interface process' attempt at writing to the new Report obey file failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0407

SEV: 1

TXT:
```
Unable to run report obey file \F
VAR:  \F = Report obey file filename
```

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0408
SEV: 2
TXT: `Failure on WRITE^FILE, file \F, error \E`
`VAR:  \F = Report Obey File filename`
`VAR:  \E = Error returned by WRITE^FILE`
CAUSE: The SAA AS2805 Interface process' attempt at writing to the new Report obey file failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0409
SEV: 1
TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`
CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.
REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0410
SEV: 2
TXT: `Failure on WRITE^FILE, file \F, error \E`
`VAR:  \F = Report Obey File filename`
`VAR:  \E = Error returned by WRITE^FILE`
CAUSE: The SAA AS2805 Interface process' attempt at writing to the new Report obey file failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0411
SEV: 1
TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`
CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.
REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0412

SEV: 2

TXT: `Failure on CLOSE^FILE, file \F, error \E`
`VAR:  \F = Report Obey File filename`
`VAR:  \E = Error returned by WRITE^FILE`

CAUSE: The SAA AS2805 Interface process' attempt at closing the new Report obey file failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of closing the specified file.

NUM: 0413

SEV: 1

TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0414

SEV: 2

TXT: `REPORT PROCESS \F, NEWPROCESS ERROR \P`
`VAR:  \F = Process ID`
`VAR:  \P = Error returned by NEWPROCESS`

CAUSE: The SAA AS2805 Interface process' attempt at starting a new process, to run the new report obey file, failed with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of creating the Report process.

NUM: 0415

SEV: 1

TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0416

SEV: 2

TXT: `Failure on OPEN, file \F, error \E`
`VAR:  \F = New Process ID`
`VAR:  \E = Error returned by HISWFILE^OPEN`

CAUSE: The SAA AS2805 Interface process' attempt at opening the new process (to run the Report Obey File) failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 0417

SEV: 1

TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0418

SEV: 2

TXT: `Failure on WRITE, file \F, error \E`
`VAR:  \F = New Process ID`
`VAR:  \E = Error returned by HISWFILE^WRITE`

CAUSE: The SAA AS2805 Interface process' attempt at sending the new process (to run its Report Obey File) its startup message failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0419

SEV: 1

TXT: `Unable to run report obey file \F`
`VAR:  \F = Report obey file filename`

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0420

SEV: 2

TXT: `Failure on CLOSE, file \F, error \E`

`VAR: \F = New Process ID`

`VAR: \E = Error returned by HISWFILE^CLOSE`

CAUSE: The SAA AS2805 Interface process' attempt at closing the new process (to run the Report Obey File) failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of closing the specified file.

NUM: 0421

SEV: 1

TXT: `Unable to run report obey file \F`

`VAR: \F = Report obey file filename`

CAUSE: The SAA AS2805 Interface process attempted to run its report obey file and failed.

REMEDY: Examine the log for specific error message relating to the task the SAA AS2805 Interchange Interface was performing at the time of error. Take whatever action is required, based on this action. If further assistance is required, contact your software support staff.

NUM: 0422

SEV: 0

TXT: `REPORT PROCESS \D STARTING REPORT OBEY FILE \F`

`VAR: \D = Process name (running report obey file)`

`VAR: \F = Report obey file filename`

CAUSE: The reports interval timer (type: 11) expired and the SAA AS2805 Interface process successfully started to run its new Report Obey File.

REMEDY: If necessary, the reports can be run again using the specified obey file. No action is required.

NUM: 0425

SEV: 0

TXT: `*** Reversal timed out to \\\\ ***`

`VAR: \\\\ = Logical Network`

CAUSE: A timer (type: 14, subtype: 1) expired whilst waiting for the Co- network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0426

SEV: 1

TXT: `Error # updating token '\\' in PSTM`

`VAR: # = Token error number`

`VAR: \\ = Token identifier`

CAUSE: An attempt to update an existing token in the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0427

SEV: 1

TXT: `Error # updating token '\\' in STM`

`VAR: # = Token error number`

`VAR: \\ = Token identifier`

CAUSE: An attempt to update an existing token in the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0428

SEV: 1

TXT: `Error # updating token '\\' in PSTM`

`VAR: # = Token error number`

`VAR: \\ = Token identifier`

CAUSE: An attempt to update an existing token in the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0429

SEV: 1

TXT: `Error # appending token '\\' to STM`

`VAR: # = Token error number`

`VAR: \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0430

SEV: 0

TXT:
```
*** Reversal timed out to \\\\ ***
VAR:  \\\\ = Logical Network
```

CAUSE: A timer (type: 14, subtype: 2) expired whilst waiting for the Co-network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0431

SEV: 1

TXT:
```
Error # appending token '\\' to STM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0432

SEV: 1

TXT:
```
Error # appending token '\\' to STM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0433

SEV: 1

TXT:
```
Error # appending token '\\' to STM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0434

SEV: 1

TXT: `Error # appending token '\\' to STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0435

SEV: 0

TXT: `RCV^REPLY error \E`
`VAR: \E = Error returned by RCV^REPLY`

CAUSE: The SAA AS2805 Interface process attempted to send a formatted PCS response message to the Health Monitor Server, and failed. This error should only occur if your institution has purchased the BASE24-pcs Network Product.

REMEDY: This error may cause a programmatic dump. The operator must take the appropriate action to correct the error that was encountered. For further assistance, contact your software support staff.

NUM: 0436

SEV: 1

TXT: `Error # appending token '\\' to STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0437

SEV: 1

TXT: `Error # appending token '\\' to PSTM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0438

SEV: 1

TXT: `Error # appending token '\\' to PSTM`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0439

SEV: 1

TXT: `Error # appending token '\\' to PSTM`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0440

SEV: 2

TXT: `AJ - Error converting current time to integer`

CAUSE: The SAA AS2805 Interface process failed to convert the current system time from ASCII to integer format, when checking the posting date.

REMEDY: Contact your software support staff.

NUM: 0441

SEV: 1

TXT: `Error # updating token '\\' in STM`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt to update an existing token in the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0442

SEV: 1

TXT: `Error # updating token '\\' in STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to update an existing token in the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0443

SEV: 1

TXT: `Error # retrieving token '\\' from PSTM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0444

SEV: 1

TXT: `Error # retrieving token '\\' from STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0445

SEV: 1

TXT: `Error converting Stm -> Sem : bad pin format`

CAUSE: The SAA AS2805 Interface process failed to format the PIN from an internal to an external format, when converting an 0200 STM to an ATM SEM request.

REMEDY: Contact your software support staff.

NUM: 0446

SEV: 1

TXT: `Error converting Stm -> Sem: track2 too long`

CAUSE: The SAA AS2805 Interface process failed to format thetrack2 from an internal to an external format, when converting an 0200 STM to an ATM STM request.

REMEDY: Contact your software support staff.

NUM: 0447

SEV: 1

TXT: `Error converting Pstm -> Sem: track2 too long`

CAUSE: The SAA AS2805 Interface process failed to format the track2 from an internal to an external format, when converting an 0200 PSTM to POS SEM message.

REMEDY: Contact your software support staff.

NUM: 0448

SEV: 1

TXT: `Error # retrieving token '\\' from PSTM`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0449

SEV: 1

TXT: `Error # retrieving token '\\' from PSTM`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0450

SEV: 1

TXT: `Error converting Pstm -> Sem : bad pin format`

CAUSE: The SAA AS2805 Interface process failed to format the PIN from an internal to an external format, when converting an 0200 PSTM to a POS SEM request.

REMEDY: Contact your software support staff.

NUM: 0451

SEV: 1

TXT: `Error # retrieving token '\\' from PSTM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0452

SEV: 1

TXT: `Error # retrieving token '\\' from STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0453

SEV: 1

TXT: `Error # retrieving token '\\' from STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0454

SEV: 1

TXT: `Error # retrieving token '\\' from STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`

CAUSE: An attempt to retrieve the specified token from the BASE24-atm Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0455

SEV: 1

TXT: `Error converting Sem -> Pstm : BAD TRAN-CDE`

CAUSE: The SAA AS2805 Interface process failed to format the Processing Code from an external to an internal format, when converting an POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0456

SEV: 1

TXT: `Error converting Sem -> Stm : bad pin frmt`

CAUSE: The SAA AS2805 Interface process failed to format the PIN from an external to an internal format, when converting an ATM SEM request to a 0200 STM message.

REMEDY: Contact your software support staff.

NUM: 0457

SEV: 1

TXT: `Error converting Sem -> Stm : bad amount`

CAUSE: The SAA AS2805 Interface process failed to format the Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM request to a 0200 STM message.

REMEDY: Contact your software support staff.

NUM: 0458

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Ledger Balance Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM request to a 0200 STM message.

REMEDY: Contact your software support staff.

NUM: 0459

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Clear Funds Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM request to a 0200 STM message.

REMEDY: Contact your software support staff.

NUM: 0460
SEV: 1
TXT: `Error # updating token '\\' in STM`
`VAR: # = Token error number`
`VAR: \\ = Token identifier`
CAUSE: An attempt to update an existing token in the BASE24-atm Standard Internal Message failed with the specified token-specific error.
REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0461
SEV: 1
TXT: `Error converting Sem -> Pstm : bad tran-cde`
CAUSE: The SAA AS2805 Interface process failed to format the Transaction Code from an external to an internal format, when converting an POS SEM request to a 0200 PSTM message.
REMEDY: Contact your software support staff.

NUM: 0462
SEV: 1
TXT: `Error converting Sem -> Pstm : bad tran-cde`
CAUSE: The SAA AS2805 Interface process failed to format the Transaction Code from an external to an internal format, when converting an POS SEM request to a 0200 PSTM message.
REMEDY: Contact your software support staff.

NUM: 0463
SEV: 1
TXT: `Error converting Sem -> Stm : bad proc-cde`
CAUSE: The SAA AS2805 Interface process failed to format the Processing Code from an external to an internal format, when converting an ATM SEM request to a 0200 STM message.
REMEDY: Contact your software support staff.

NUM: 0465
SEV: 1
TXT: `Error converting Sem -> Stm : bad amount`
CAUSE: The SAA AS2805 Interface process failed to format the Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM response to an 0210 STM message.
REMEDY: Contact your software support staff.

NUM: 0466

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Ledger Balance Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM response to a 0210 STM message.

REMEDY: Contact your software support staff.

NUM: 0467

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Clear Funds Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM response to a 0210 STM message.

REMEDY: Contact your software support staff.

NUM: 0468

SEV: 1

TXT:
```
Error # updating token '\\' in PSTM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to update an existing token in the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0469

SEV: 1

TXT:
```
Error # updating token '\\' in PSTM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to update an existing token in the BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0470

SEV: 1

TXT: `Error converting Sem -> Stm : bad proc-cde`

CAUSE: The SAA AS2805 Interface process failed to format the Processing Code from an external format to an internal format, when converting an ATM SEM 0220 to a 0220 STM message.

REMEDY: Contact your software support staff.

NUM: 0471

SEV: 1

TXT: `Error converting Sem -> Stm : bad amount`

CAUSE: The SAA AS2805 Interface process failed to format the Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM 0220 to a 0220 STM message.

REMEDY: Contact your software support staff.

NUM: 0472

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Ledger Balance Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM 0220 to a 0220 STM message.

REMEDY: Contact your software support staff.

NUM: 0473

SEV: 1

TXT: `Error converting Sem -> Stm : bad balance`

CAUSE: The SAA AS2805 Interface process failed to format the Clear Funds Amount from an external ASCII representation to an internal integer representation when converting an ATM SEM 0220 to a 0220 STM message.

REMEDY: Contact your software support staff.

NUM: 0474

SEV: 1

TXT:
```
Error # appending token '\\' to PSTM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0475

SEV: 1

TXT: `Error converting Sem -> Pstm : bad amount`

CAUSE: The SAA AS2805 Interface process failed to format the Amount from an external ASCII representation to an internal integer representation when converting a POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0476

SEV: 1

TXT: `Error converting Sem -> Pstm : bad cash amt`

CAUSE: The SAA AS2805 Interface process failed to format the Cash Amount from an external ASCII representation to an internal integer representation when converting a POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0477

SEV: 1

TXT: `Error converting Sem -> Pstm : bad pin frmt`

CAUSE: The SAA AS2805 Interface process failed to format the PIN from an external format to an internal format, when converting a POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0478

SEV: 0

TXT: `Error converting Sem -> Pstm : bad track2`

CAUSE: The SAA AS2805 Interface process failed to convert the Track 2 length from an external ASCII representation to an integer value when converting a POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0479

SEV: 1

TXT: `Error converting Sem -> Pstm : bad pan`

CAUSE: The SAA AS2805 Interface process failed to convert the PAN length from an external ASCII representation to an integer value when converting a POS SEM request to a 0200 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0481

SEV: 1

TXT: `Error converting Sem -> Pstm : bad amount`

CAUSE: The SAA AS2805 Interface process failed to format the Amount from an external ASCII representation to an internal integer representation when converting a POS SEM response to a 0210 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0482

SEV: 1

TXT: `Error converting Sem -> Pstm : bad cash amt`

CAUSE: The SAA AS2805 Interface process failed to format the Cash Amount from an external ASCII representation to an internal integer representation when converting a POS SEM response to a 0210 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0483

SEV: 0

TXT: `Error converting Sem -> Pstm : bad track2`

CAUSE: The SAA AS2805 Interface process failed to convert the Track 2 length from an ASCII format to an integer value, when converting a POS SEM response to a 0220 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0484

SEV: 1

TXT: `Error converting Sem -> Pstm : bad pan`

CAUSE: The SAA AS2805 Interface process failed to convert the PAN length from an ASCII format to an integer value, when converting a POS SEM response to a 0220 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0485

SEV: 1

TXT: `Error converting Sem -> Pstm : bad amount`

CAUSE: The SAA AS2805 Interface process failed to convert the Amount from an external ASCII representation to an internal integer representation, when converting a POS SEM response to a 0220 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0486

SEV: 1

TXT: `Error converting Sem -> Pstm : bad cash amt`

CAUSE: The SAA AS2805 Interface process failed to convert the Cash Amount from an external ASCII representation to an internal integer representation, when converting a POS SEM response to a 220 PSTM message.

REMEDY: Contact your software support staff.

NUM: 0487

SEV: 1

TXT:
```
Error # appending token '\\' to PSTM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0488

SEV: 1

TXT:
```
Error # appending token '\\' to PSTM
VAR:  # = Token error number
VAR:  \\ = Token identifier
```

CAUSE: An attempt to add the specified token to the end of the specified BASE24-pos Standard Internal Message failed with the specified token-specific error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 0489

SEV: 2

TXT: `SETTLEMENT AMOUNT MISMATCH`

CAUSE: The SAA AS2805 Interface process found a difference between the Net Amount sent in the 0520 Settlement Advice and the Net Amount maintained by this interface, when formatting the 0530 Settlement Acknowledgment.

REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0490

SEV: 2

TXT: `SETTLEMENT AMOUNT MISMATCH`

CAUSE: The SAA AS2805 Interface process found a difference between the Net Amount sent in the 0520 Settlement Advice and the Net Amount maintained by this interface, when formatting the 0530 Settlement Acknowledgment.

REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0491

SEV: 2

TXT: `SETTLEMENT AMOUNT MISMATCH`

CAUSE: The SAA AS2805 Interface process found a difference between the Net Amount sent in the 0520 Settlement Advice and the Net Amount maintained by this interface, when formatting the 0530 Settlement Acknowledgment.

REMEDY: Contact your software support staff. Inform your Co-network reconciliation department that a mismatch has occurred.

NUM: 0492

SEV: 0

TXT: `<text>`

`VAR:  <text> = BASE24 SEM message`

CAUSE: The SAA AS2805 Interface process found an error when verifying a SEM's Message Authentication Code and so dumped the SEM to the log.

REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your software support staff.

NUM: 0493

SEV: 1

TXT: `Number of bad Mac's exceeds, specified count`

CAUSE: The number of allowed MAC errors has reached the permitted maximum. The switch link will be marked down.

REMEDY: If this message is a one-off occurrence, no action is required since the interface will log itself back on again. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0495

SEV: 2

TXT: `Unable to write to NET process`

CAUSE: The SAA AS2805 Interface process attempted to send a BASE24 Internal message (STM) to the network, but failed.

REMEDY: Examine the log for additional message relating to the action the SAA AS2805 Interchange Interface took on the failed message. If further assistance is required, contact your software support staff.

NUM: 0496

SEV: 2

TXT: `Unable to process msg with MAC err (msg type:` \\\\)

`VAR:` \\\\ = SEM Message Type

CAUSE: The SAA AS2805 Interface process found an error when verifying a SEM's Message Authentication Code.

REMEDY: Determine the message's contents and intended destination. If this error continues to occur, contact your operations manager.

NUM: 0500

SEV: 2

TXT: `Unable to write to NET process`

CAUSE: The SAA AS2805 Interface process attempted to send a BASE24 POS Standard Internal message (PSTM) to the network, but failed.

REMEDY: Examine the log for additional message relating to the action the SAA AS2805 Interchange Interface took on the failed message. If further assistance is required, contact your software support staff.

NUM: 0505

SEV: 2

TXT: `Error converting sem to iso field ###`

`VAR: ### = BASE24 SEM field ID`

CAUSE: The SAA AS2805 Interface process attempted to compress an expanded SAA SEM message into an ISO SEM message, but failed at the specified field.

REMEDY: Contact your software support staff.

NUM: 0506

SEV: 2

TXT: `Unable to write sem via NETWRITE`

CAUSE: The SAA AS2805 Interface process attempted to send an ISO SEM message to the Co-network, but failed.

REMEDY: Examine the log for additional message relating to the action the SAA AS2805 Interface process took on the failed message. If this error continues to occur contact your software support staff because the logical link will be marked down.

NUM: 0507

SEV: 2

TXT: `Unable to find emf entry for sem typ: \\\\`

`VAR:  \\\\ = BASE24 SEM Message type`

CAUSE: The SAA AS2805 Interface process attempted to locate the correct entry in the extended memory EMF table for the specified message type, but failed.

REMEDY: Ensure a valid entry exists in the External Message file (EMF) for this message type and warmboot the process. If this error continues to occur, contact your sofware support staff.

NUM: 0508

SEV: 2

TXT: `Unable to find emf entry for sem typ: \\\\`

`VAR:  \\\\ = BASE24 SEM Message Type`

CAUSE: The SAA AS2805 Interface process attempted to locate the correct entry in the extended memory EMF table for the specified message type, but failed.

REMEDY: Ensure a valid entry exists in the External Message file (EMF) for this message type and warmboot the process. If this error continues to occur, contact your software support staff.

NUM: 0510

SEV: 2

TXT: `Invalid acquirer date (date: \\\\)`

`VAR:  \\\\ = Acquirer Date from request message`

CAUSE: The SAA AS2805 Interface process compared the date sent from the Acquirer to that in its Process Control Table and found a difference. The request will be denied.

REMEDY: Contact your software support staff to determine why the acquirer sent the bad date.

NUM: 0515

SEV: 2

TXT: `Unable to add ILF record : Invalid date (\\\\)`

`VAR:  \\\\ = Interchange Log File (ILF) date`

CAUSE: The SAA AS2805 Interface process could not determine which Interchange Log File (ie. current or next) to add a transaction to.

REMEDY: Contact your software support staff.

NUM: 0516
SEV: 2
TXT: `ILF record:`
CAUSE: The SAA AS2805 Interface process could not add a transaction to the appropriate Interchange Log File, so the record is dumped to the log.
REMEDY: This message simply indicates that an ILF record follows. No action required.

NUM: 0517
SEV: 0
TXT: `<text>`
`VAR: <text> = ILF record`
CAUSE: After a transaction has been successfully processed, the SAA AS2805 Interface process could not add the transaction to the appropriate Interchange Log File, so the record is dumped to the log.
REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0518
SEV: 2
TXT: `ILF record:`
CAUSE: The SAA AS2805 Interface process could not add a transaction to the appropriate Interchange Log File, so the record is dumped to the log.
REMEDY: This message simply indicates that an ILF record follows. No action required.

NUM: 0519
SEV: 0
TXT: `<text>`
`VAR: <text> = ILF record`
CAUSE: After a transaction has been successfully processed, the SAA AS2805 Interface process could not add the transaction to the appropriate Interchange Log File, so the record is dumped to the log.
REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0525
SEV: 2
TXT: `Unable to save totals on ILF`
CAUSE: The SAA AS2805 Interface process could not position to the first record in the Interchange Log File (ILF).
REMEDY: Contact your software support staff.

NUM: 0526
SEV: 2
TXT: `Unable to save totals on ILF`
CAUSE: The SAA AS2805 Interface process could not obtain the current Interchange Log File (ILF) record address.
REMEDY: Contact your software support staff.

NUM: 0527
SEV: 2
TXT: `Unable to save totals on ILF`
CAUSE: The SAA AS2805 Interface process could not update the first record in the Interchange Log File with the current totals.
REMEDY: Contact your software support staff.

NUM: 0530
SEV: 2
TXT: `ILF record:`
CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File.
REMEDY: This message simply indicates that an ILF record follows. No action required.

NUM: 0531
SEV: 2
TXT: `<text>`
`VAR: <text> = ILF record`
CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File and the record is dumped to the log.
REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0532
SEV: 2
TXT: `<text>`
`VAR: <text> = ILF record`
CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File and the record is dumped to the log.
REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0533

SEV: 1

TXT: `\?, prikey: \?, file: \F`

`VAR:  \? = Error text describing error`

`VAR:  \? = Interchange Log File primary key`

`VAR:  \F = Interchange Log File filename`

CAUSE: The SAA AS2805 Interface process could not keyposition on the appropriate Interchange Log File (ILF) record.

REMEDY: Examine the log to determine why the process could not position on the required record and take the appropriate action based on the reason.

NUM: 0534

SEV: 3

TXT: `<text>`

`VAR:   <text>` = ILF record

CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File and the record is dumped to the log.

REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0535

SEV: 2

TXT: `Unable to add SAF record, file \F, error \E`

`VAR:  \F = Store and Forward filename`

`VAR:  \E = Error returned by HISWSAF^ADD`

CAUSE: The SAA AS2805 Interface process could not add the record to the Store and Forward (SAF) File.

REMEDY: Examine the log for related error messages indicating why the record could not be added to the SAF and take the appropriate action based on that message.

NUM: 0536

SEV: 0

TXT: `<text>`

`VAR:  <text> = Store and Forward record`

CAUSE: The SAA AS2805 Interface process could not add the record to the Store and Forward (SAF) File and the record is dumped to the log.

REMEDY: Examine the log for related error messages indicating why the record could not be added to the SAF and take the appropriate action based on that message.

NUM: 0537

SEV: 3

TXT: `SAF record reached maximum number of retries SAF RECORD PURGED :`

CAUSE: The SAA AS2805 Interface process determines that the number of SAF retries has reached the maximum allowed.

REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0538

SEV: 0

TXT: `<text>`

`VAR: <text> = Store and Forward record`

CAUSE: The SAA AS2805 Interface process determines that the number of SAF retries has reached the maximum allowed and the record is dumped to the log.

REMEDY: Examine the log for related error messages indicating why the record could not be added to the SAF and take the appropriate action based on that message.

NUM: 0539

SEV: 2

TXT: `ILF record:`

CAUSE: The SAA AS2805 Interface process could not read an existing record in the Interchange Log file.

REMEDY: Contact your software support staff.

NUM: 0540

SEV: 3

TXT: `Failure on % FULL CALCULATION, File \F, Error \E`

`VAR: \F = Filename of file used in calculation`

`VAR: \E = Error returned by CALC^PERCENTAGE^FULL`

CAUSE: The SAA AS2805 Interface process detected an error when calculating the percentage full of the specified file.

REMEDY: Take whatever action is required, based on the reported error and the action of calculating the percent full of the specified file. If further assistance is required, contact your software support staff.

NUM: 0541

SEV: 1

TXT: `*** Warning *** File \F is # percent full`

`VAR:  \F = Filename of the file used in calculation`

`VAR:  # = Percentage full of the specified file`

CAUSE: The SAA AS2805 Interface process received a STATUS command from the operator's Network Control Screen and the specified file has reached the size specified by the LCONF parameter SAA-fff-NOTIFY-PERCENTAGE, where fff is the File Identifier.

REMEDY: Contact your software support staff. The specified file must be carefully monitored from now on to prevent it from becoming completely full.

NUM: 0542

SEV: 2

TXT: `ILF record:`

CAUSE: The SAA AS2805 Interface process could not read an existing record in the Interchange Log file.

REMEDY: Contact your software support staff.

NUM: 0543

SEV: 2

TXT: `ILF record:`

CAUSE: The SAA AS2805 Interface process could not read an existing record in the Interchange Log file.

REMEDY: Contact your software support staff.

NUM: 0544

SEV: 2

TXT: `<text>`

`VAR:  <text> = ILF record`

CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File and the record is dumped to the log.

REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0545

SEV: 1

TXT: `Co-network switch link is DOWN`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the Interchange and BASE24.

REMEDY: Confirmation of a successful link disconnection. No action required.

NUM: 0546

SEV: 2

TXT: `ILF record:`

CAUSE: The SAA AS2805 Interface process could not read an existing record in the Interchange Log file.

REMEDY: Contact your software support staff.

NUM: 0547

SEV: 1

TXT:
```
\?, prikey: \?, file: \F
VAR:  \? = Error text describing error
VAR:  \? = Interchange Log File primary key
VAR:  \F = Interchange Log File filename
```

CAUSE: The SAA AS2805 Interface process could not read the appropriate Interchange Log File (ILF) record.

REMEDY: Examine the log to determine why the process could not read the required record and take the appropriate action based on the reason.

NUM: 0548

SEV: 1

TXT:
```
\?, prikey: \?, file: \F
VAR:  \? = Error text describing error
VAR:  \? = Interchange Log File primary key
VAR:  \F = Interchange Log File filename
```

CAUSE: The Interchange Log File (ILF) record read by the SAA AS2805 Interchange Interface did not have the requested primary key.

REMEDY: Examine the log to determine why the process could not read the record with the required primary key and take the appropriate action based on the reason.

NUM: 0549

SEV:

TXT:
```
text>
VAR:  <text> = ILF record
```

CAUSE: The SAA AS2805 Interface process could not update an existing record in the Interchange Log File and the record is dumped to the log.

REMEDY: Contact your software support staff. This record must be processed manually.

NUM: 0550

SEV:

TXT: `Unable to update ICF with new posting date`

CAUSE: The SAA AS2805 Interface process could not update the Interchange Configuration File (ICF) with the new Posting Date at Cutover.

REMEDY: Examine the log for related error messages indicating why the ICF record could not be updated and take the appropriate action based on that message.

NUM: 0551

SEV:

TXT: `CUTOVER COMPLETE - NEW POST DATE - \\\\\\`
`VAR:  \\\\\\ = Settlement date`

CAUSE: The SAA AS2805 Interface process completed Cutover to the new Posting Date successfully.

REMEDY: Confirmation that Cutover has completed. No action required.

NUM: 0552

SEV:

TXT: `Co-network switch link is DOWN`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the Interchange and BASE24.

REMEDY: Confirmation of a successful link disconnection. No action required.

NUM: 0554

SEV:

TXT: `Failure on OPEN, file \F, error \E`
`VAR:  \F = Interchange Log File filename`
`VAR:  \E = Error returned by HISWFILE^OPEN`

CAUSE: The SAA AS2805 Interface process failed to open the specified Interchange Log File (ILF).

REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 0555

SEV:

TXT: `<text>`

`VAR: <text> = Error description returned by TMPL^CREATE^LIKE`

CAUSE: The SAA AS2805 Interface process failed to create the new Interchange Log File (ILF).

REMEDY: Determine from the log the reason for the error message. Take the appropriate action based on that reason.

NUM: 0557

SEV: 3

TXT: `Failure on WRITE, file \F, error \E`

`VAR: \F = Interchange Log File filename`

`VAR: \E = Error returned by HISWFILE^WRITE`

CAUSE: The SAA AS2805 Interface process failed to write the totals record to the specified Interchange Log File (ILF).

REMEDY: Take whatever action is required, based on the reported error and the action of writing to the specified file.

NUM: 0558

SEV: 3

TXT: `Failure on UPDATE, file \F, error \E`

`VAR: \F = Interchange Log File filename`

`VAR: \E = Error returned by HISWFILE^UPDATE`

CAUSE: The SAA AS2805 Interface process failed to update the totals record of the specified Interchange Log File (ILF) with the record's address within the file.

REMEDY: Take whatever action is required, based on the reported error and the action of updating the specified file.

NUM: 0559

SEV: 3

TXT: `Failure on READ, file \F, error \E`

`VAR: \F = Interchange Log File filename`

`VAR: \E = Error returned by HISWFILE^READ`

CAUSE: The SAA AS2805 Interface process failed to read the totals record of the specified Interchange Log File (ILF).

REMEDY: Take whatever action is required, based on the reported error and the action of reading the specified file.

NUM: 0560

SEV: 2

TXT: `Mandatory bit ### not set on incoming message - Type: \\\\`

`VAR:  ### = SEM field ID`

`VAR:  \\\\ = SEM Message Type`

CAUSE: The SAA AS2805 Interface process detected the absence of the specified bit setting in the bitmap of the specified message type. The specified bit must be set.

REMEDY: Contact your software support staff to verify the message's contents and bitmap make-up. If this error continues to occur, contact your operations manager.

NUM: 0561

SEV: 0

TXT: `Unexpected bit ### set on incoming message - Type: \\\\`

`VAR:  ### = SEM field ID`

`VAR:  \\\\ = SEM Message Type`

CAUSE: The SAA AS2805 Interface process detected the presence of the specified bit setting in the bitmap of the specified message type. The specified bit should not be set.

REMEDY: Contact your software support staff to verify the message's contents and bitmap make-up. If this error continues to occur, contact your operations manager.

NUM: 0620

SEV: 0

TXT: `Unknown ATM sem reponse code: \\`

`VAR:  \\ = ATM SEM Response Code`

CAUSE: The SAA AS2805 Interface process could not convert the ATM SEM response code from an ASCII format into an integer format.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine their reason for sending the specified Response Code.

NUM: 0621

SEV: 0

TXT: `Unknown ATM sem reponse code: \\`

`VAR:  \\ = ATM SEM Response Code`

CAUSE: The SAA AS2805 Interface process could not match the ATM SEM response code to a valid internal BASE24 response code. The internal response code defaults to "050".

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0624

SEV: 0

TXT: `Unknown POS sem reponse code: \\`
`VAR:  \\ = POS SEM Response Code`

CAUSE: The SAA AS2805 Interface process could not convert the POS SEM response code from an ASCII format into an integer format.

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0625

SEV: 0

TXT: `Unknown POS sem reponse code: \\`
`VAR:  \\ = POS SEM Response Code`

CAUSE: The SAA AS2805 Interface process could not match the POS SEM response code to a valid internal BASE24 response code. The internal response code defaults to "100".

REMEDY: Use the operator log to determine the value of the Response Code. Retrieve the transaction from the Interchange Log File (ILF) and contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0800

SEV: 1

TXT: `SAA-RACAL-TERM-MASTER param not 16 hex characters`

CAUSE: The SAA AS2805 Interface process successfully read the SAA RACAL Terminal Master Key parameter and determined that it did not contain 16 valid HEX characters.

REMEDY: Change the data in the SAA-RACAL-TERM-MASTER parameter to contain 16 valid HEX characters, and restart the SAA AS2805 Interface process.

NUM: 0801

SEV: 0

TXT: `Keychange advice only required for dynamic link`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request when the Interface's key management is not Dynamic.

REMEDY: Modify the Key Exchange Type field on screen 12 of the Interchange Configuration File (ICF) to contain the same value for both partners in the co-network. Warmboot both process(es).

NUM: 0802

SEV: 0

TXT: `Keychange advice received, base24 not logged on`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request when the Interface is not logged on to the Co-network.

REMEDY: Either take the required action to logon to the Co-network, or contact the Co-network partner to inform them of the Interface's status.

NUM: 0803

SEV: 0

TXT: `Keychange advice received, co-net not logged on`

cAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request when the Co-network is not logged on to the Interface.

REMEDY: Contact the Co-network partner to determine its status and why it has not logged on.

NUM: 0804

SEV: 2

TXT: `Keychange advice received, invalid field 53 value : #`
`VAR:  # = In Key Index (SEM field P-53)`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with a value other than 1 or 2 in field P-53.

REMEDY: Determine from the log what the invalid data is. Contact the Co- network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0805

SEV: 2

TXT: `Keychange advice received, invalid field 48 length : #`
`VAR:  # = Session Keys (SEM field P-48)`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with an invalid length value in field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0806

SEV: 2

TXT: `Keychange advice received, bad session keys`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with an invalid MAC session key in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0807

SEV: 2

TXT: `Keychange advice received, bad session keys`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with an invalid PIN session key in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0808

SEV: 2

TXT: `Keychange advice received, bad session keys`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0809

SEV: 0

TXT: `BASE24 switch link is now UP`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the specified station and the Interchange.

REMEDY: Confirmation of a successful link connection. No action required.

NUM: 0810

SEV: 1

TXT: `Received invalid 0830 NMM-INFO-CODE \\\; ignored.`
`VAR:  \\\ = Info Code (P-70) from SEM`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management Message with an invalid NMM Code (ie. not a logoff nor a key change).

REMEDY: Use the operator log to determine the value of the invalid NMM Code. Contact the Co-network Partner to determine the reason for sending the invalid Code. For further assistance, contact your software support staff.

NUM: 0811

SEV: 0

TXT: `Keychange response only required for dynamic link`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response when the Interface's key management is not Dynamic.

REMEDY: Modify the Key Exchange Type field on screen 12 of the Interchange Configuration File (ICF) to contain the same value for both partners in the co-network. Warmboot the appropriate process(es).

NUM: 0812

SEV: 2

TXT: `Keychange response received, bad response code : \\`
`VAR:  \\ = Response Code`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management response with an invalid response code.

REMEDY: Use the operator log to determine the value of the Response Code. Contact the Co-network Partner to determine its reason for sending the specified Response Code.

NUM: 0813

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid response code. The appropriate station is marked down.

REMEDY: Confirmation that the station has been logged off. No action required.

NUM: 0814

SEV: 2

TXT: `Keychange response received, invalid field 48 length:#`
`VAR:  # = Length attribute of SEM field P-48`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0815

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0816

SEV: 2

TXT: `Keychange response received, invalid kvcs`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Code (KVC) in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0817

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Code (KVC) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0818

SEV: 0

TXT: `BASE24 switch link is now UP`

CAUSE: The SAA AS2805 Interface process checked the status of the link between the specified station and the Interchange.

REMEDY: Confirmation of a successful link connection. No action required.

NUM: 0819

SEV: 0

TXT: `*** Keychange timed out to \\\\ ***`
`VAR:  \\\\ = Logical Network`

CAUSE: A timer (type: 3) expired whilst waiting for the Co-network to respond.

REMEDY: If this message is a one-off occurrence, no action is required. If however, this problem persists, contact your software support staff to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0820

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process checked the status of the link between BASE24 and the Interchange.

REMEDY: Confirmation of a successful link disconnection. No action required.

NUM: 0821

SEV: 0

TXT: `Keychange inbound accepted`

CAUSE: The SAA AS2805 Interface process received a valid 0820 Network Management key change request message.

REMEDY: Confirmation of acceptance of key change request. No action required.

NUM: 0822
SEV: 0
TXT: `Keychange outbound accepted`
CAUSE: The SAA AS2805 Interface process received a valid 0830 Network Management key change response message.
REMEDY: Confirmation of acceptance of key change response. No action required.

NUM: 0823
SEV: 2
TXT: `SCM Function RTMK failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0824
SEV: 2
TXT: `SCM Function KVCREQSIN failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0825
SEV: 2
TXT: `SCM Function NODEPROOF failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0826
SEV: 2
TXT: `SCM Function NODEKEKGEN failed with error ####`
`VAR:   ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interchange Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0827
SEV: 2
TXT: `SCM Function KVCREQSIN failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0828
SEV: 1
TXT: `Logoff received while in logoff state - msg ignored`
CAUSE: The SAA AS2805 Interface process has received a logoff message while in a logoff state. This message is ignored.
REMEDY: This message is used to highlight that the SAA AS2805 Interface process will not reset any timers or take any logoff action as it might interfere with it's current processing. This message can occur when there is a number of queued logon requests/responses which cause the interchange processes to try to reach a known state i.e. by sending a logoff. As logon processing should only take typically 30 - 60 seconds, if this message persists, contact local support staff.

NUM: 0829
SEV: 2
TXT: `SCM Function NODEKEYGEN failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 0830
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0831
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0832
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0833
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0834
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0835
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0836
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0837

SEV: 2

TXT: `Keychange advice received, bad session keys`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0838

SEV: 2

TXT: `Keychange advice received, bad session keys`

CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid session keys in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0839

SEV: 2

TXT: `KVC processing not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support Key Verification Codes (KVCs), but the SAA AS2805 Interface process expected to find KVCs included in field P-48 of the received 0820 Network Management key change request.

REMEDY: Check that the field KVC PROCESSING in ICF Screen 12 for this Interchange is allowed to be set to Y for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0840

SEV: 2

TXT: `Keychange not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support the keychange function, but the SAA AS2805 Interface process received an 0820 Network Management key change request.

REMEDY: Check that the field TYPE OF KEY EXCHANGE in ICF Screen 12 for this Interchange is allowed to be set to 3 (Dynamic key exchange) for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0841

SEV: 2

TXT: `Keychange response received,invalid field 48 length : #`
`VAR:  # = Length attribute of SEM field P-48`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0842

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION:\S)`

VAR: \S = Station name

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0843

SEV: 2

TXT: `Keychange response received, invalid KVSS`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0844

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION:\S)`
`VAR: \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0845

SEV: 2

TXT: `Keychange response received, invalid field 48 length: #`
`VAR:  # = Length attribute of SEM field P-48`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0846

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR: \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0847

SEV: 2

TXT: `Keychange response received, invalid kvcs`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0848

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR: \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0849

SEV: 2

TXT: `Keychange response received, invalid field 48 length: #`
`VAR:  # = Length attribute of SEM field P-48`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 0850

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR: \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid length for field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0851

SEV: 2

TXT: `Keychange response received, invalid KVCs`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0852

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0853

SEV: 2

TXT: `Keychange response received, invalid kvcs`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48.

REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0854

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR: \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with invalid Key Verification Codes (KVCs) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0855

SEV: 1

TXT: `Can only compare 4 hex chars of KVC for Racal.`

CAUSE: The Racal Security Module and/or the security utilities software being used, does not support KVCs with a length of 6 hexadecimal characters (i.e 3 bytes or 24 bits). Only the first 4 hexadecimal characters (i.e 2 bytes or 16 bits) of the KVC will be compared until the next process start or warmboot, when the SAA AS2805 Interface process receives an 0830 Network Management key change response containing Key Verification Codes (KVCs) included in field P-48.

REMEDY: No action required, unless 24 bit KVC processing to full AS2805.6.3 standards is mandatory on this Interchange. Contact the Co-network partner to determine what KVC data was sent. Contact your software support staff to upgrade the Racal Security Module and/or security utilities software, if appropriate.

NUM: 0856

SEV: 2

TXT: `KVC processing not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support Key Verification Codes (KVCs), but the SAA AS2805 Interface process expected to find KVCs included in field P-48 of the received 0830 Network Management key change response.

REMEDY: Check that the field KVC PROCESSING in ICF Screen 12 for this Interchange is allowed to be set to Y for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0857

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR:  \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with unexpected Key Verification Codes (KVCs) in field P-48. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0858

SEV: 2

TXT: `Keychange not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support the keychange function, but the SAA AS2805 Interface process received an 0830 Network Management key change response.

REMEDY: Check that the field TYPE OF KEY EXCHANGE in ICF Screen 12 for this Interchange is allowed to be set to 3 (Dynamic key exchange) for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0859

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF (STATION: \S)`
`VAR:  \S = Station Name`

CAUSE: The SAA AS2805 Interface process received an unexpected 0830 Network Management key change response. The station is subsequently logged off.

REMEDY: Confirmation of station being logged off. No action required.

NUM: 0860

SEV: 2

TXT: `Keychange not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support the keychange function, but the SAA AS2805 Interface process attempted to format an 0820 Network Management key change request.

REMEDY: Check that the field TYPE OF KEY EXCHANGE in ICF Screen 12 for this Interchange is allowed to be set to 3 (Dynamic key exchange) for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0861

SEV: 2

TXT: `KVC processing not supported for SECURE-DEV-TYP`

CAUSE: The Security Module being used does not support Key Verification Codes (KVCs), but the SAA AS2805 Interface process expected to find KVCs included in field P-48 of the received 0830 Network Management key change response.

REMEDY: Check that the field KVC PROCESSING in ICF Screen 12 for this Interchange is allowed to be set to Y for the Security Module type specified in the LCONF parameter, SECURE-DEV-TYP. Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 0862

SEV: 1
TXT: ERROR FORMATTING SEM ADVICE
CAUSE: The SAA AS2805 Interchange Interface process could not format the standard external message.
REMEDY: Check that the security box is functioning and if problem persists, contact your software support staff to verify the formatting procedures.

NUM: 0863
SEV: 0
TXT: *** KEK CHANGE TIMED OUT TO \\\\ ***
VAR: \\\\ = Logical Network
CAUSE: A timer (type: 3) expired whilst waiting for the Co-network to respond.
REMEDY: If this message is a one-off occurrence, no action is required. If however this problem persists, contact your software support group to review the timer values in the Interchange Configuration File (ICF) record for this process, and to ensure the Interchange partner is up and logged on.

NUM: 0864
SEV: 1
TXT: BASE24 SWITCH LINK IS LOGGED OFF
CAUSE: The SAA AS2805 Interchange Interface process checked the status of the link between BASE24 and the Interchange.
REMEDY:
Confirmation of a successful link disconnection. No action required.

NUM: 0901
SEV: 3

TXT: `Failure on OPEN, file \F, error \E`

`Failure on KEYPOSITION, file \F, error \E`

`Failure on READ, file \F, error \E`

```
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0902

SEV: 3

TXT: `Failure on OPEN, file \F, error \E`

`Failure on KEYPOSITION, file \F, error \E`

`Failure on READ, file \F, error \E`

```
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0903

SEV: 3

TXT: `Failure on OPEN, file \F, error \E`
`Failure on KEYPOSITION, file \F, error \E`
`Failure on READ, file \F, error \E`
`VAR: \F = Interchange Configuration File (ICF) filename`
`or Key File (KEYF) filename`
`or Store-and-Forward Filename`
`or External Message File (EMF) filename`
`or Tape Key (TKEY) filename`
`VAR: \E = Guardian error number`

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0904

SEV: 3

TXT: `Failure on OPEN, file \F, error \E`
`Failure on KEYPOSITION, file \F, error \E`
`Failure on READ, file \F, error \E`
`VAR: \F = Interchange Configuration File (ICF) filename`
`or Key File (KEYF) filename`
`or Store-and-Forward Filename`
`or External Message File (EMF) filename`
`or Tape Key (TKEY) filename`
`VAR: \E = Guardian error number`

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0905

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0906

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0907

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 908

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0909

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0910

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0911

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0912

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0913

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0914

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0915

SEV: 3

TXT:
```
Failure on OPEN, file \F, error \E
Failure on KEYPOSITION, file \F, error \E
Failure on READ, file \F, error \E
VAR:  \F =  Interchange Configuration File (ICF)
filename
   or Key File (KEYF) filename
   or Store-and-Forward Filename
   or External Message File (EMF) filename
   or Tape Key (TKEY) filename
VAR:  \E = Guardian error number
```

CAUSE: During initialisation, the SAA AS2805 Interface process could not perform file operation on specified file because of the indicated Guardian file error.

REMEDY: Take the appropriate action (based on the reported Guardian error) and restart the process.

NUM: 0990

SEV: 3

TXT:
```
Failure on OPEN, File \F, Error \E
VAR:  \F = Interchange Log File (ILF) filename
VAR:  \E = Error returned by HISWFILE^OPEN
```

CAUSE: The SAA AS2805 Interface process could not open the specified Interchange Log File (ILF) because of the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 0995

SEV: 3

TXT:
```
Failure on CLOSE, File \F, Error \E
VAR:  \F = Interchange Log File (ILF) filename
VAR:  \E = Error returned by HISWFILE^CLOSE
```

CAUSE: The SAA AS2805 Interface process could not close the specified Interchange Log File (ILF) because of the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of closing the specified file.

NUM: 1000
SEV: 3
TXT: `Failure on CLOSE, File \F, Error \E`
`VAR:  \F = Interchange Log File (ILF) filename`
`VAR:  \E = Error returned by HISWFILE^CLOSE`
CAUSE: The SAA AS2805 Interface process could not close the specified Interchange Log File (ILF) because of the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of closing the specified file.

NUM: 1005
SEV: 3
TXT: `Failure on PURGE, File \F, Error \E`
`VAR:  \F = Interchange Log File (ILF) filename`
`VAR:  \E = Error returned by HISWFILE^PURGE`
CAUSE: The SAA AS2805 Interface process could not purge the specified Interchange Log File (ILF) because of the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of purging the specified file.

NUM: 1006
SEV: 0
TXT: `File \F, purged. Age exceeds retention period.`
`VAR:  \F = Interchange Log File (ILF) filename`
CAUSE: The SAA AS2805 Interface process successfully purged the specified Interchange Log File (ILF) because its age exceeded the retention period in LCONF parameter SW-ILF-FILE-RETENTION.
REMEDY: Unless a subsequent error 1010 is logged, the SAA AS2805 Interface process also successfully purged the associated Interchange Log Alternate Key Files (ILF0, etc).

NUM: 1010
SEV: 3
TXT: `Failure on PURGE, File \F, Error \E`
`VAR:  \F = Interchange Log File Alternate Key (ILF0) filename`
`VAR:  \E = Error returned by HISWFILE^PURGE`
CAUSE: The SAA AS2805 Interface process could not purge one of the specified Interchange Log Alternate Key Files (ILF0, etc) because of the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of purging the specified file.

NUM: 1015

SEV: 3

TXT: `Failure on OPEN, File \F, Error \E`

`VAR: \F = Interchange Report Obey File filename`

`VAR: \E = Error returned by HISWFILE^OPEN`

CAUSE: The SAA AS2805 Interface process could not open the specified Interchange Report Obey File because of the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 1020

SEV: 3

TXT: `Failure on CLOSE, File \F, Error \E`

`VAR: \F = Interchange Report Obey File filename`

`VAR: \E = Error returned by HISWFILE^CLOSE`

CAUSE: The SAA AS2805 Interface process could not close the specified Interchange Report Obey File because of the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of closing the specified file.

NUM: 1025

SEV: 3

TXT: `Failure on PURGE File \F, Error \E`

`VAR: \F = Interchange Report Obey File filename`

`VAR: \E = Error returned by HISWFILE^PURGE`

CAUSE: The SAA AS2805 Interface process could not purge the specified Interchange Report Obey File because of the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of purging the specified file.

NUM: 1026

SEV: 0

TXT: `File \F, purged. Age exceeds retention period.`

`VAR: \F = Interchange Report Obey File filename`

CAUSE: The SAA AS2805 Interface process successfully purged the specified Interchange Report Obey File because its age exceeded the retention period in LCONF parameter SW-ILF-FILE-RETENTION.

REMEDY: No Action Required

NUM: 1100

SEV: 1

TXT: `Unable to add Token '\\', no room available`
`VAR:  \\ = Token identifier`

CAUSE: An attempt was made to add the specified token to the Standard Internal Message (STM) when there was not enough room available to accommodate it.

REMEDY: Review the size on the STM to determine why there is not enough room available for the specified token. If the condition persists contact your software support staff.

NUM: 1105

SEV: 2

TXT: `Error # while attempting to add token '\\'`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt was made to add the specified token to the Standard Internal Message (STM) and failed with the specified error.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 1110

SEV: 2

TXT: `Error # while attempting to get token '\\'`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`

CAUSE: An attempt was made to retrieve the specified token from the Standard Internal Message (STM) and failed with the specified error. (The token exists in the STM).

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 1115

SEV: 2

TXT: `Error # while converting token '\\'. (Token before \?, Token after \?)`
`VAR:  # = Token error number`
`VAR:  \\ = Token identifier`
`VAR:  \? = Token data`
`VAR:  \? = Token data`

CAUSE: An attempt was made to convert the specified token from one format to another. The token images before and after the conversion attempt are specified.

REMEDY: Take the appropriate action based on the reported error. If the condition persists contact your software support staff.

NUM: 1120

SEV: 2

TXT: `SAA-UNIQUE-STAN-FLAG param not found in \F; defaulting to Y`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA unique stan flag parameter in the logical network configuration file.

REMEDY: Add the valid SAA unique stan flag parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 1121

SEV: 2

TXT: `SAA-REVERSE-BAL-INQ param not found in \F; defaulting to N`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA reverse balance enquiry parameter in the logical network configuration file.

REMEDY: Add a valid SAA reverse balance enquiry parameter in the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 1122

SEV: 2

TXT: `SAA-LOGON-REQUIRED param not found in \F; defaulting to 'Y'`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA logon required parameter in the logical network configuration file.

REMEDY: Add a valid SAA logon required parameter in the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 1123

SEV: 2

TXT: `SAA-LOGON-ACTION param not found in \F; defaulting to 1`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA logon action parameter in the logical network configuration file.

REMEDY: Add the valid SAA logon action parameter in the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 1124

SEV: 2

TXT: `SAA-EBCDIC-CONVERSION param not found in \F; defaulting to 'N'`

`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the SAA ebcdic conversion parameter in the logical network configuration file.

REMEDY: Add a valid SAA ebcdic conversion parameter to the LCONF and restart the SAA AS2805 Interchange Interface.

NUM: 1125

SEV: 2

TXT: `SAA-UNIQUE-STAN-FLAG param contains invalid data - must be 'Y' or 'N'; defaulting to 'Y'`

CAUSE: The SAA AS2805 Interface process successfully read the SAA unique stan flag parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA unique stan flag parameter to 'Y' or 'N' and restart the SAA AS2805 Interchange Interface.

NUM: 1126

SEV: 2

TXT: `SAA-LOG-REQUIRED param contains invalid data - must be 'Y' or 'N'; defaulting to 'Y'`

CAUSE: The SAA AS2805 Interface process successfully read the SAA log required parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA log required parameter to 'Y' or 'N' and restart the SAA AS2805 Interchange Interface.

NUM: 1127

SEV: 2

TXT: `SAA-EBCDIC-CONVERSION PARAM INVALID - must be 'Y' or 'N'; defaulting to 'N'`

CAUSE: The SAA AS2805 Interface process successfully read the SAA ebcdic conversion parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA ebcdic conversion parameter to 'Y' or 'N' and restart the SAA AS2805 Interchange Interface.

NUM: 1128

SEV: 2

TXT: `SAA-REVERSE-BAL-INQ param contains invalid data - must be 'Y' or 'N'; defaulting to 'N'`

CAUSE: The SAA AS2805 Interface process successfully read the SAA reverse balance enquiry parameter and determined that it contained invalid data.

REMEDY: Correct the data in the SAA reverse balance enquiry parameter to 'Y' or 'N' and restart the SAA AS2805 Interchange Interface.

NUM: 1129

SEV: 2

TXT: `Unable to retrieve suspense record for timed out request (addr: $)`

`VAR:  $ = address of suspense record`

CAUSE: The SAA AS2805 Interface process was unable to retrieve the suspense record for deletion.

REMEDY: The transaction has been timed out therefore no action is required. If the problem persists, please contact your local support group.

NUM: 1130

SEV: 2

TXT: `Unable to retrieve suspense record for timed out request (addr: $)`

`VAR:  $ = address of suspense record`

CAUSE: The SAA AS2805 Interface process was unable to retrieve the suspense record for deletion.

REMEDY: The transaction has been timed out therefore no action is required. If the problem persists, please contact your local support group.

NUM: 1131

SEV: 2

TXT: `Unable to retrieve suspense record for timed out request (addr: $)`

`VAR:  $ = address of suspense record`

CAUSE: The SAA AS2805 Interface process was unable to retrieve the suspense record for deletion.

REMEDY: The transaction has been timed out therefore no action is required. If the problem persists, please contact your local support group.

NUM: 1132

SEV: 0

TXT: Failure to restore old posting date on ICF - Manual update is required

CAUSE: The SAA AS2805 Interface process could not restore the ICF posting date after a warmboot failed.

REMEDY: Reset the ICF posting date manually.

NUM: 1133

SEV: 0

TXT: ATM Reconciliation logged to ILF for \\\\\\
`VAR:  \\\\\\ = Posting Date`

CAUSE: The SAA AS2805 Interface process.logged the cutover of the current atm posting date when ICF atm settlement timer expired

REMEDY: No action required on this information message.

NUM: 1134

SEV: 0

TXT: POS Reconciliation logged to ILF for \\\\\\
`VAR:  \\\\\\ = Posting Date`

CAUSE: The SAA AS2805 Interface process.logged the cutover of the current atm posting date when ICF pos settlement timer expired

REMEDY: No action required on this information message.

NUM: 1135

SEV: 0

TXT: Reconciliation logged to ILF for \\\\\\
`VAR:  \\\\\\ = Posting Date`

CAUSE: The SAA AS2805 Interface process.logged the cutover of the current atm and pos posting date when ICF settlement timer expired

REMEDY: No action required on this information message.

NUM: 1136

SEV: 0

TXT: Restored ICF Post Date - \\\\\\
`VAR:  \\\\\\ = Posting Date`

CAUSE: The SAA AS2805 Interface process restored the ICF posting date after a warmboot failed.

REMEDY: No action required on this message, check other.log messages for reason warmboot failed.

NUM: 1137

SEV: 3

TXT: `KEY6 assign not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Key 6 File (KEY6) assign in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid KEY6 assign to the LCONF and restart the SAA AS2805 Interface process.

NUM: 1138

SEV: 3

TXT: `KEYT assign not found in \F`
`VAR:  \F = LCONF filename`

CAUSE: The SAA AS2805 Interface process could not find the Key T File (KEYT) assign in the Logical Network Configuration File (LCONF).

REMEDY: Add a valid KEYT assign to the LCONF and restart the SAA AS2805 Interface process.

NUM: 1139

SEV: 2

TXT: `The DES-TRIPLE-SINGLE param contains invalid data. Triple to single PIN translations will not be supported by this interface process`

CAUSE: The SAA AS2805 Interface process successfully read the Interchange Log File (ILF) TDES parameter and determined that it contained invalid data. Must be set to “Y” or “N”.

REMEDY: Correct the data in the `DES-TRIPLE-SINGLE` parameter and restart the SAA AS2805 Interchange Interface.

NUM: 1140

SEV: 2

TXT: `Failure on OPEN, file \F, error \E`
`VAR:  \F = KEY6 filename`
`VAR:  \E = Error returned by hiswfile^open`

CAUSE: The SAA AS2805 Interface process' attempt at opening the KEY6 file failed, with the specified error.

REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 1141
SEV: 2
TXT: `Failure on KEYPOSITION, file \F, error \E`
`VAR: \F = KEY6 filename`
`VAR: \E = Error returned by hiswfile^keyposition`
CAUSE: The SAA AS2805 Interface process' attempt at positioning the KEY6 file at the record required failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of positioning the specified file.

NUM: 1142
SEV: 2
TXT: `Failure on READ, file \F, error \E`
`VAR: \F = KEY6 filename`
`VAR: \E = Error returned by hiswfile^read`
CAUSE: The SAA AS2805 Interface process' attempt at reading the KEY6 file at the record required failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of reading the specified file.

NUM: 1143
SEV: 2
TXT: `Failure on OPEN, file \F, error \E`
`VAR: \F = KEYT filename`
`VAR: \E = Error returned by hiswfile^open`
CAUSE: The SAA AS2805 Interface process' attempt at opening the KEYT file failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of opening the specified file.

NUM: 1144
SEV: 2
TXT: `Failure on KEYPOSITION, file \F, error \E`
`VAR: \F = KEYT filename`
`VAR: \E = Error returned by hiswfile^keyposition`
CAUSE: The SAA AS2805 Interface process' attempt at positioning the KEYT file at the record required failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of positioning the specified file.

NUM: 1145
SEV: 2
TXT: `Failure on READ, file \F, error \E`
`VAR:  \F = KEYT filename`
`VAR:  \E = Error returned by hiswfile^read`
CAUSE: The SAA AS2805 Interface process' attempt at reading the KEYT file at the record required failed, with the specified error.
REMEDY: Take whatever action is required, based on the reported error and the action of reading the specified file.

NUM: 1146
SEV: 2
TXT: `SCM Function NODERESP failed with error ####`
`VAR:  ####  = Error returned from secutils`
CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.
REMEDY: Ensure SCM environment is performing as required.

NUM: 1147
SEV: 0
TXT: `Function NODEKEKTRANS failed decrypting kek keys`
CAUSE: The SAA AS2805 Interface process cannot perform a NODEKEKTRANS function because of error reported by secutils.
REMEDY: Ensure NODEKEKTRANS environment is performing as required. For further assistance, contact your software support staff.

NUM: 1148
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid MAC session key in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 1149
SEV: 2
TXT: `Keychange advice received, bad session keys`
CAUSE: The SAA AS2805 Interface process received an 0820 Network Management key change request with invalid PIN session key in field P-48.
REMEDY: Contact the Co-network partner to determine what data was sent. Contact your software support staff to verify and correct the bad data.

NUM: 1150

SEV: 2

TXT: `Keychange advice received, invalid field 48 length : #`
`VAR:  # = Session Keys (SEM field P-48)`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change acknowledgement with an invalid length value in field P-48.

REMEDY: Determine from the log what the invalid data is. Contact the Co-network partner to establish why it sent the bad data. If further assistance is required, contact your software support staff.

NUM: 1151

SEV: 1

TXT: `BASE24 switch link is LOGGED OFF`

CAUSE: The SAA AS2805 Interface process received an 0830 Network Management key change response with an invalid field P-48 length. The appropriate station is marked down.

REMEDY: Confirmation that the station has been logged off. No action required.

NUM: 1154

SEV: 2

TXT: `SCM Function NODEPROOF failed with error ####`
`VAR:  ####  = Error returned from secutils`

CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.

REMEDY: Ensure SCM environment is performing as required.

NUM: 1155

SEV: 2

TXT: `SCM Function NODEKEKGEN failed with error ####`
`VAR:  ####  = Error returned from secutils`

CAUSE: The SAA AS2805 Interface process cannot perform an SCM function because of error reported by secutils.

REMEDY: Ensure SCM environment is performing as required.

NUM: 1156

SEV: 0

TXT: `Session key generation failed`

CAUSE: The SAA AS2805 Interface process MAC session key generation failed formulating an 0820 Network Management key change request.

REMEDY: Refer to previous specific error log message from secutils.

NUM: 1157

SEV: 0

TXT: `Session key generation failed`

CAUSE: The SAA AS2805 Interface process PIN session key generation failed formulating an 0820 Network Management key change request.

REMEDY: Refer to previous specific error log message from secutils.

NUM: 1158

SEV: 3

TXT: `Proof of Endpoint Failure`

CAUSE: The SAA AS2805 Interface process failed to verify Proof of Endpoint because data was incorrect.

REMEDY: This error message might happen because a number of logon responses have been queued to the interchange process and the interchange partner has processed old logon requests. The SAA AS2805 Interface process will send a logoff to the interchange partner to try to get to a known state. It will then try the logon sequence again. If problem persists contact your software support staff.

NUM: 1159

SEV: 0

TXT: `Failure on generation of new key`

CAUSE: The SAA AS2805 Interface process PIN transport key generation failed formulating an 0820 Network Management key change request.

REMEDY: Refer to previous specific error log message from secutils.

NUM: 1160

SEV: 3

TXT: `ICFE assign not found in \F.`
`VAR:  \F     = ICFE file name`

CAUSE: The SAA AS2805 Interface process could not find the specified assign in the Logical Network Configuration File (LCONF).

REMEDY: Add the specified assign to the LCONF and restart the process.

NUM: 1161

SEV: 3

TXT: `APCFEMT assign not found in \F.`
`VAR:  \F     = APCFEMT file name`

CAUSE: The SAA AS2805 Interface process could not find the specified assign in the Logical Network Configuration File (LCONF).

REMEDY: Add the specified assign to the LCONF and restart the process.

NUM: 1162

SEV: 3

TXT: `IPCFEMT assign not found in \F.`
`VAR:  \F     = IPCFEMT file name`

CAUSE: The SAA AS2805 Interface process could not find the specified assign in the Logical Network Configuration File (LCONF).

REMEDY: Add the specified assign to the LCONF and restart the process.

NUM: 1163

SEV: 0

TXT: `Proof of endpoint not supported for settings of this SECURE-DEV-TYP.  Defaulting to 'N'`

CAUSE: The SAA AS2805 Interface process ICF PROOF ENDPOINT FLAG set to “Y” is only valid for an SCM security device

REMEDY: Information only, this flag setting is overridden.

NUM: 1165

SEV: 1

TXT: `Invalid EMT name \'s specified in warmboot command`
`VAR:  \? = EMT name from the EMT specific warmboot command`

CAUSE: The warmboot command contains an Extended Memory Table (EMT) name not equal to APCFEMT or IPCFEMT. The processing continued with the current EMT.

REMEDY: Add the specified assign to the LCONF and restart the process.

NUM: 1166

SEV: 1

TXT: `Open error, File \F, Error \E during warmboot of the EMT.`
`VAR:  \F - LCONF name`
`VAR:  \E - Error`

CAUSE: An error occurred during an attempt to open the LCONF during a warmboot of an Extended Memory Table (EMT). This interface continued processing without performing the warmboot.

REMEDY: Correct the problem according to the specified NSK error number. If the error is hardware-related, contact Tandem. If the error is software related, contact HELP24.

NUM: 1167

SEV: 1

TXT: `Assign \? not found in LCONF during warmboot of the EMT`
`VAR: \? - EMT name from the EMT specific warmboot command`

CAUSE: The indicated file assign could not be found in the LCONF. The warmboot failed and the interface continued processing with the current Extended Memory Table (EMT) segment id.

REMEDY: Make corrections in the LCONF and warmboot the interface process.

NUM: 1168

SEV: 1

TXT: `Filename expand failure on \?, Invalid filename, during warmboot of the EMT.`
`VAR: \? = EMT file name`

CAUSE: The expand of the Extended Memory Table (EMT) filename is unsuccessful. The warmboot failed and the interface continued processing with the current EMT segment id.

REMEDY: Make corrections to the LCONF and warmboot Interface

NUM: 1169

SEV: 1

TXT: `\? warmboot aborted, previous status retained.`
`VAR: \? = EMT file name`

CAUSE: The specified Extended Memory Table (EMT) was not successfully warmbooted.

REMEDY: Check to see if the valid file exists. Make sure to deliver the the warmboot command in right order.

NUM: 1170

SEV: 0

TXT: `\? warmboot completed.`
`VAR: \? = EMT name`

CAUSE: The Extended Memory Table (EMT) has warmbooted successfully.

REMEDY: No action is required.

NUM: 1171

SEV: 2

TXT: `Authorization destination not specified in File \F. Transaction not logged to the BASE24-atm TLF. Message follows:`
`VAR:  \F = ICFE file name`

CAUSE: The transaction is configured to be sent to the destination specified in the Acquirer Processing Code File EMT (APCFEMT) for authorization. The APCFEMT is configured to send the response to the Authorization Process Name in the BASE24 product segment of the Enhanced Interchange Configuration File (ICFE) to be logged to the BASE24 product transaction log file. However, the ICFE Authorization Process Name is blanks. Therefore, the transaction is not sent to the Base24 authorization Process for logging to the TLF. The response was dumped to EMS.

REMEDY: Determine if a valid process name should be added to the ICFE or if the APCF is misconfigured. If the ICFE is updated, warmboot the interface. If the APCF is updated, rebuild the APCFEMT and send an APCFEMT warmboot command to the interface.

NUM: 1172

SEV: 2

TXT: `<text>`
`VAR:  <text> =  BASE24-atm Internal Message`

CAUSE: The message is dumped to the log after Log Message Number 1171. Refer this message for cause.

REMEDY: Refer Log Message Number 1171 for remedy.

NUM: 1173

SEV: 2

TXT: `Authorization destination not specified in File \F. Transaction not logged to the BASE24-pos PTLF. Message follows:`
`VAR:  \F = ICFE file name`

CAUSE: The transaction is configured to be sent to the destination specified in the Acquirer Processing Code File EMT (APCFEMT) for authorization. The APCFEMT is configured to send the response to the Authorization Process Name in the BASE24 product segment of the Enhanced Interchange Configuration File (ICFE) to be logged to the BASE24 product transaction log file. However, the ICFE Authorization Process Name is blanks. Therefore, the transaction is not sent to the Base24 authorization Process for logging to the PTLF. The response was dumped to EMS.

REMEDY: Determine if a valid process name should be added to the ICFE or if the APCF is misconfigured. If the ICFE is updated, warmboot the interface. If the APCF is updated, rebuild the APCFEMT and send an APCFEMT warmboot command to the interface.

NUM: 1174

SEV: 2

TXT: `<text>`

`VAR:  <text> =  BASE24-pos Internal Message`

CAUSE: The message is dumped to the log after Log Message Number 1173. Refer this message for cause.

REMEDY: Refer Log Message Number 1173 for remedy.

ACI Worldwide, Inc.

APPENDIX D

Token Documentation

The table below identifies the various tokens used by the SAA Interface along with their token IDs.

For more information on other BASE24 tokens please refer to the ***BASE24 Tokens Manual***.

ID	Token Name	Product
04	BASE24-atm Release 4.0 Token	POS
25	Surcharge Data Token	BASE
C0	BASE24-pos Release 5.1 Token	POS
C6	Train Stain XID Token	POS
CE	Authentication Data Token	POS
CH	POS Data1 Token	POS
Y0	Faulty Card Read Token	POS
Y1	Electronic Fallback Token	POS
Y2	Credit Eftpos Token	POS

SAA Tokens

This section describes the SAA message tokens. Tokens are described in alphanumeric order, according to token ID. The Header token and token header, which do not have token IDs, are described first. The table below identifies the SAA tokens and their corresponding token IDs. For tokens with ASCII formats, the ASCII formats follow the corresponding binary format.

For more information on other BASE24 tokens please refer to the ***BASE24 Tokens Manual***.

ID	Token Name	Product
X1	SAA External Data Token - ATM	BASE24-atm
X2	SAA External Data Token - POS	BASE24-pos
X3	Inventory Recharge Token	BASE24-atm
X4	Inventory Activation Number Token	BASE24-atm
X5	Inventory Stock Type Token	BASE24-atm

Token X1 SAA External Data Token - ATM

The SAA External Data Token is used to maintain certain data from the External message (either native ATM or POS message, or SEM from the Interchange). This external data can be used for a variety of purposes, particularly:

- Since the token can be logged to various Log Files, ENFORM reports can be written to extract the External data.
- SAA will not need to convert and re-convert data from external to internal format if it is kept.

Position	Level	Field Name and Description	Data Type
1-4	**02**	**MSG-TYP** The external request message type. The response message type is not maintained.	**PIC X (04)**
5-10	**02**	**PROC-CDE** The external message processing code.	**PIC X (06)**
11-12	**02**	**RESP-CDE** The external response code.	**PIC X (02)**

Token X2 SAA External Data Token - POS

The SAA External Data Token is used to maintain certain data from the External message (either native ATM or POS message, or SEM from the Interchange). This external data can be used for a variety of purposes, particularly:

- Since the token can be logged to various Log Files, ENFORM reports can be written to extract the External data.
- SAA will not need to convert and re-convert data from external to internal format if it is kept.

Position	Level	Field Name and Description	Data Type
1-4	**02**	**MSG-TYP** The external request message type. The response message type is not maintained.	**PIC X (04)**
5-10	**02**	**PROC-CDE** The external message processing code.	**PIC X (06)**
11-12	**02**	**RESP-CDE** The external response code.	**PIC X (02)**

Token X3 Inventory Recharge Token - Binary Format

The INV-RECHARGE-TKN contains information associated with the purchase of a pre-paid recharge voucher. This could be a mobile phone recharge or a phone coupon. It is formatted by the acquiring process (e.g. N50 Device Handler) and sent to the Inventory Supply system. The Inventory Supply system uses information in the token to determine the Stock Type purchased and returns information relevant to receipt printing in the same token.

Position	Level	Field Name and Description	Data Type
1-4	**02**	**STOCK-TYPE** Stock Type code as known by the Inventory Supply System. Formatted by the Delivery System.	**PIC X (04)**
5-12	**02**	**STOCK-VALUE** Denomination of the Voucher. In whole dollars. Formatted by the Delivery System.	**TYPE BINARY 64**
13-24	**02**	**DELIVERY-SEQ-NUM** The Sequence Number assigned to this transaction by the delivery system. Left justified, space filled. Formatted by the Delivery System.	**PIC X (12)**
25-40	**02**	**STOCK-NUMBER** Control or serial number associated with the stock item. Formatted by the Inventory Supply system on responses. Left justified, space filled. Available for printing on receipt.	**PIC X (16)**
41-48	**02**	**STOCK-EXP-DATE** Expiry date of the stock item. Format CCYYMMDD. Formatted by the Inventory Supply system on responses. Available for printing on receipt.	**PIC X (08)**
49	**02**	**ACTIVATION-CODE** Activation Code indicating the method of activation. "0". Activation number included in INV-ACTIVATION-NUMBER-TKN; "1" Account has been auto-activated (for future use).	**PIC X (01)**
50	**02**	**REVERSAL-TYP** Reversal flag. Formatted by Delivery System to indicate the type of reversal. The Inventory Supply system will use this to determine whether the stock item can be made available for resale. "0" - not a reversal; "1" - normal reversal, stock item can be made available for resale; "2" - doubtful reversal, stock item should NOT be made available for resale.	**PIC X (01)**

Token X3 Inventory Recharge Token - ASCII Format

The fields in the ASCII format inventory Recharge Token are described below.

Position	Level	Field Name and Description	Data Type
1-4	**02**	**STOCK-TYPE**	**PIC X (04)**
5-16	**02**	**STOCK-VALUE**	**PIC 9 (12)**
17-28	**02**	**DELIVERY-SEQ-NUM**	**PIC X (12)**
29-44	**02**	**STOCK-NUMBER**	**PIC X (16)**
45-52	**02**	**STOCK-EXP-DATE**	**PIC X (08)**
53	**02**	**ACTIVATION-CODE**	**PIC X (01)**
54	**02**	**REVERSAL-TYP**	**PIC X (01)**

Token X4 Inventory Activation Number Token

The INV-ACTIVATION-NUMBER-TKN contains the activation number associated with the purchase of a pre-paid recharge voucher. This is the number a customer needs to use when activating their voucher with the Telco. This could be a mobile phone recharge or a phone coupon. It is formatted by the Inventory Supply system and returned to the Delivery system for printing on a receipt.

Position	Level	Field Name and Description	Data Type
1-16	**02**	**ACTIVATION-NUMBER** Activation Number for the voucher purchased. Formatted by the Inventory Supply system. Left justified, space filled. Used by the Delivery System for receipt printing.	**PIC X (16)**

Token X5 Inventory Stock Type Token

This token is formatted by the N50 Device Handler and included in the internal message (STM) sent to the Inventory Transaction Manager.

Position	Level	Field Name and Description	Data Type
1-4	**02**	**ACQUIRER-STOCK-CODE**	**PIC X (04)**
		Acquirer Stock Code corresponding to the Voucher being purchased. Format is "VTnn".	
5-8	**02**	**ISSUER-STOCK-TYPE**	**PIC X (04).**
		Issuer Stock Type corresponding to the Voucher being purchased. The Stock Type identifies the Stock being purchased within the Inventory Issuer system.	

APPENDIX E

AS2805 commands supported by Transaction Security Services (TSS)

The A0-A3 and A6-A9 commands are supported by TSS. While A4 and A5 (Nodeproof and Noderesp) commands are supported by both TSS and the Base24 AS2805 Interchange Interface (SAA) in compliance to the Australian Payments Clearing Association's CECS (Consumer Electronic Clearing System) requirement.

COMMAND	**Command Description**
A0	AS2805 Gernerate Keys
A1	AS2805 Verify PPASN
A2	AS2805 Key Translate
A3	AS2805 Pin Translate
A4	AS2805 Generate Random Number
A5	AS2805 Decrypt Random Number
A6	AS2805 Pin Verify
A7	AS2805 Generate Interface Keys
A9	AS2805 Key Set

TSS Message A0 – AS2805 Generate Keys

This command will be used by AS2805 compliant POS device handlers and will result in one of the following ERACOM AMB commands to be initiated:

3000 – Terminal Key Generate 1 (Old command)
3500 – Terminal Key Generate 1 (New command)
3010 – Terminal Key Generate 2 (Old command)
3510 – Terminal Key Generate 2 (New command)

AS2805 Generate Keys Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A0’	04.4
Version	Alpha-numeric	2	2	’01’	04.4
Transaction Origin	Numeric	1	4	Valid Values: ‘0’ – device	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
Key Type	Numeric	2	21	‘01’ (PIN Key) (FUTURE USE) ‘02’ (Data key)(FUTURE USE) ‘03’ (MAC Key) (FUTURE Use) ‘30’ (PIN Encryption Key and MAC Key)	04.4
Key Generate Option	Alpha-numeric	1	23	Valid values: ‘1’ ‘2’ Blank – N/A	04.4
Total Length		24			

The following determines which Eracom command is formatted:

CSEC Key Length	Key Gen Option	HSM Command
S	1	3000
S	2	3010
D, T	1	3500
D, T	2	3510

AS2805 Generate Keys Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A0’	04.4
Version	Alpha-numeric	2	2	‘01’	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error	04.4

Parameter	**Data Type**	**Field Length**	**Offset**	**Value Returned By Security Services**	New for Release
				03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	
Key Type	Numeric	2	6	'01' (PIN Key) (FUTURE USE) '02' (Data key)(FUTURE USE) '03' (MAC Key) (FUTURE Use) '30' (PIN Encryption Key and MAC Key)	04.4
Key Length	Numeric	1	8	1 (single), 2 (double), 3 (triple)	04.4
MAC Key - Send	Hex	48	9	Space filled if MAC Key not requested	04.4
MAC Key - Receive	Hex	48	57	Space filled if MAC Key not requested	04.4
PIN Encryption Key	Hex	48	105	Space filled if PIN Encryption Key not requested	04.4
Data Encryption - Send	Hex	48	153	Space filled if Data Encryption Key not requested	04.4
Data Encryption - Receive	Hex	48	201	Space filled if Data Encryption Key not requested	04.4
Encryption Type	Alpha-numeric	2	249	Blank (unknown) '00 (unknown) '01' (ECB) '02' (TECB) '03' (CBC) '04' (TCBC) '05' (CFB) '06' (TCFB) '07' (OFB) '08' (TOFB)	04.4
Total Length		251			

TSS Message A1 – AS2805 Verify PPASN

This command will be used by AS2805 compliant POS device handlers or interfaces and will result in ERACOM AMB command E500 to be initiated:

AS2805 Verify PPASN Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A1'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Transaction Origin	Numeric	1	4	Valid Values: '0' – device '1' – interface	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
PPASN	Hex	18	21	Encrypted PPASN from device. Left hand 32 bits, Provided as ASCII HEX characters.	04.4
Total Length		39			

AS2805 Verify PPASN Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A1'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	04.4
Total Length		6			

TSS Message A2 – AS2805 Key Translate

This command will be used by AS2805 compliant interfaces and will result in ERACOM AMB command 4500 to be initiated:

AS2805 Key Translate Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A2’	04.4
Version	Alpha-numeric	2	2	’01’	04.4
Transaction Origin	Numeric	1	4	Valid Values: ‘0’ – device (not supported) ‘1’ – interface	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
MAC Key - Receive	Hex	48	21		04.4
PIN Encryption Key	Hex	48	69		04.4
Data Encryption - Receive	Hex	48	117		04.4
Key Index of New Key	Alpha-numeric	1	165	‘1’ or ‘2’	04.4
Total Length		166			

AS2805 Key Translate Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A2’	04.4
Version	Alpha-numeric	2	2	‘01’	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	04.4
Total Length		6			

TSS Message A3 – AS2805 Pin Translate

This command will be used by AS2805 compliant interfaces and will result in ERACOM AMB command 6610 to be initiated:

AS2805 PIN Translate Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A3'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Transaction Origin	Numeric	1	4	Valid Values: '0' – device '1' – interface (not supported)	04.4
Acquirer Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
Issuer Key Locator	Alpha-numeric	16	21	Key to ISEC record	04.4
Encrypted PIN	Hex	16	37		04.4
PAN Digits	Alpha-numeric	19	53		04.4
PAN Length	Numeric	2	72		04.4
STAN	Alpha-numeric	6	74		04.4
Amount	Numeric	12	80		04.4
Key Index of the Input PIN Encryption Key to use	Alpha-numeric	1	92	Space '1' or '2' (FUTURE USE)	04.4
Key Index of the Output PIN Encryption Key to use	Alpha-numeric	1	93	'1' or '2'	04.4
Total Length		94			

AS2805 PIN Translate Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A3'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error	04.4

Parameter	**Data Type**	**Field Length**	**Offset**	**Value Returned By Security Services**	**New for Release**
				08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	
Translated PIN	Binary	8	6		04.4
Key Index of the Output PIN Encryption Key used	Alpha-numeric	1	14	‘1’ or ‘2’	04.4
Total Length		15			

TSS Message A4 – AS2805 Generate Random Number

This command will be used by AS2805 compliant interfaces and will result in ERACOM AMB command E520 and ERACOM Mark II EE3033 command to be initiated:

AS2805 Generate Random Number Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A4’	04.4
Version	Alpha-numeric	2	2	’01’	04.4
Transaction Origin	Numeric	1	4	Valid Values: ‘0’ – device (not supported) ‘1’ – interface	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
Total Length		21			

AS2805 Generate Random Number Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A4’	04.4
Version	Alpha-numeric	2	2	‘01’	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	04.4
Random Number	Hex	48	6		04.4
Random Number Inverted	Hex	48	54		04.4
Total Length		102			

TSS Message A5 – AS2805 Decrypt Random Number

This command will be used by AS2805 compliant interfaces and will result in ERACOM AMB command E530 and ERACOM Mark II EE3034 command to be initiated:

AS2805 Decrypt Random Number Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A5'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Transaction Origin	Numeric	1	4	Valid Values: '0' – device (not supported) '1' – interface	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
Random Number	Hex	48	21		04.4
Total Length		69			

AS2805 Decrypt Random Number Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	'A5'	04.4
Version	Alpha-numeric	2	2	'01'	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	04.4
Random Number Inverted	Hex	48	6		04.4
Total Length		54			

TSS Message A6 – AS2805 Pin Verify

This command will be used by Auth and will result in one of the following ERACOM AMB commands to be initiated:

6510 – PIN Verify 6.4 – IBM3624
6511 – PIN Verify 6.4 – Visa

AS2805 PIN Verify Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A6’	04.4
Version	Alpha-numeric	2	2	’01’	04.4
PIN Verification Method	Alpha-numeric	1	4	Valid Values – **‘1’ – PIN Verify 6.4 – IBM3624** **‘2’ – PIN Verify 6.4 – Visa**	04.4
Transaction Origin	Numeric	1	5	Valid Values: ‘0’ – device ‘1’ – interface (not supported)	04.4
Acquirer Key Locator	Alpha-numeric	16	6	Key to CSEC/ISEC record	04.4
Auth Key Locator	Alpha-numeric	16	22	KEYA Group	04.4
Card Expiration Date	Alpha-numeric	4	38		04.4
Encrypted PIN	Hex	16	42		04.4
STAN	Alpha-numeric	6	58		04.4
Amount	Numeric	12	64		04.4
KPVV Index	Numeric	1	76		04.4
PAN Digits	Alpha-numeric	19	77		04.4
PAN Length	Numeric	2	96		04.4
PVV / Offset	Hex	12	98		04.4
PIN Offset Length	Numeric	2	110		04.4
ISEC PIN Key Index to be used	Alpha-numeric	1	112	Space ‘1’ or ‘2’ (FUTURE USE)	04.4
Total Length		113			

AS2805 PIN Verify Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A6’	04.4
Version	Alpha-numeric	2	2	‘01’	04.4
Response Status	Alpha-	2	4	Valid Values:	04.4

Parameter	**Data Type**	**Field Length**	**Offset**	**Value Returned By Security Services**	**New for Release**
	numeric			00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	
Total Length		6			

TSS Message A7 – AS2805 Generate Interface Keys

This command will be used by AS2805 compliant interfaces and will result in ERACOM AMB command 3A00 to be initiated:

AS2805 Generate Interface Keys Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A7’	04.4
Version	Alpha-numeric	2	2	’01’ or ‘02’	04.4
Transaction Origin	Numeric	1	4	Valid Values: ‘1’ – interface	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC/ISEC record	04.4
Key Type	Numeric	2	21	‘30’ (PIN Encryption Key and MAC Key)	04.4
Index Update	Alpha-numeric	1	23	"Y" (Yes, update the index); "N" (No, do not update the index)	04.4
Key Index to Generate	Alpha-numeric	1	24	1’ or ‘2’ or space	04.4
Total Length		25			

AS2805 Generate Interface Keys Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A7’	04.4
Version	Alpha-numeric	2	2	‘01’ or ‘02’	04.4
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command	04.4
Key Type	Numeric	2	6	‘30’ (PIN Encryption Key and MAC Key)	04.4
Key Length	Numeric	1	8	‘1’ (single), ‘2’ (double), ‘3’ (triple)	04.4
MAC Key - Send	Hex	48	9	Space filled if MAC Key not requested	04.4

Parameter	**Data Type**	**Field Length**	**Offset**	**Value Returned By Security Services**	**New for Release**
PIN Encryption Key	Hex	48	57	Space filled if PIN Encryption Key not requested	04.4
Data Encryption - Send	Hex	48	105	Space filled if Data Encryption Key not requested	04.4
Key Index of New Key	Alpha-numeric	1	153	‘1’ or ‘2’	04.4
MAC key check digits	Hex	6	154		09.2 only for version ‘2’ of the msg
PIN key check digits	Hex	6	160		09.2 only for version ‘2’ of the msg
Data key check digits	Hex	6	166		09.2 only for version ‘2’ of the msg
Total Length		172			

TSS Message A9 – AS2805 Key Set

This command will be used by AS2805 compliant POS device handlers and will result in the CSEC record being updated with new key values.

AS2805 Key Set Request

Parameter	Data Type	Field Length	Offset	Value Passed To Security Services	New for Release
Header	Standard				04.4
Command Type	Alpha-numeric	2	0	‘A9’	04.4
Version	Alpha-numeric	2	2	’01’	04.4
Transaction Origin	Numeric	1	4	Valid Values: ‘0’ – device ‘1’ – interface (not currently supported)	04.4
Key Locator	Alpha-numeric	16	5	Key to CSEC record	04.4
KEK1	Hex	52	21	2 byte format code, 2 byte KM Index, 48 byte key	04.4
KEK2	Hex	52	73	2 byte format code, 2 byte KM Index, 48 byte key	04.4
PPASN	Hex	18	125	2 byte KM Index, 16 byte PPASN	04.4
Total Length		143			

AS2805 Key Set Response

Parameter	Data Type	Field Length	Offset	Value Returned By Security Services
Header	Standard			
Command Type	Alpha-numeric	2	0	‘A9’
Version	Alpha-numeric	2	2	‘01’
Response Status	Alpha-numeric	2	4	Valid Values: 00 – no error 01 – validation failed 02 – hardware error 03 – operation failed 04 – value missing 05 – key sync error 06 – database error 08 – PIN length error 09 – record missing 10 – key missing 11 – duplicate key 12 – invalid serial number 99 – invalid command
Total Length		6		